

# **Exhibit 102**

**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

**EXHIBIT 102**

**Domestic Industry Claim Chart for Claims of U.S. Patent No. 12,373,366 (the “’366 Patent”) Based on Samsung Practicing ’366 Products**

As shown in the chart below, the Samsung Practicing ’366 Products, which include Samsung DDR5 RDIMMS and MRDIMMS, practice at least claim 1 of the ’366 patent. It should be understood that Netlist’s contentions are not limited to these products but instead encompass any and all reasonably similar products and components that practice the claims in the same or similar manner as discussed below. Netlist may supplement, modify or amend its contentions under the doctrine of equivalents based on any claim constructions, additional information obtained through formal discovery or other means, and/or based on other circumstances including any assertions by Respondents that the Samsung Practicing ’366 Products do not practice the ’366 patent as to any one or more of the limitations in the claim(s) below.

To the extent Respondents contend that any differences are alleged to exist between the asserted claims below and the Samsung Practicing ’366 Products, such differences are insubstantial and the Samsung Practicing ’366 Products perform substantially the same function, in substantially the same way, to yield substantially the same result, and therefore the Samsung Practicing ’366 Products practice the claims below under the doctrine of equivalents. Netlist may supplement, modify or amend its contentions under the doctrine of equivalents based on any claim constructions, additional information obtained through formal discovery or other means, and/or based on other circumstances including any assertions that the Samsung Practicing ’366 Products do not satisfy the technical domestic industry by Respondents as to any one or more of the limitations in the claims below.

## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>[1.pre] A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM via the set of data conduits, and address and control signals are delivered from the system board to the DIMM via the set of address and control conduits, the DIMM comprising:</p> | <p>The Accused Instrumentalities (such as a Samsung EC8 RDIMM or EC8 MRDIMM) are each a dual in-line memory module (DIMM). Analysis below uses Samsung’s RDIMM format as an example, but MRDIMMs would operate in materially the same way in all relevant parts. Ex. CC-145 (DDR5 MRDIMM: A Transformational Evolution For DDR5 DIMM) (“DDR5 MRDIMM is pin compatible with traditional DIMM, and it can be used in the same DIMM slots that are used for DDR5 RDIMMs”); Ex. CC-13 (JEDEC Press Release) (JEDEC noting that other planned features for MRDIMMs include “Utilization of standard DDR5 DIMM components including DRAM, DIMM Form Factor &amp; Pinout, SPD, PMIC, and TS for ease of adoption”; “Efficient I/O scaling using RCD/DB logic process capability”; and “Leverage existing LRDIMM ecosystem for design and test infrastructure”).</p> <p>Examples of Samsung DIMMs are shown below. <i>See, e.g.</i>, Ex. CC-116:</p>  <p>The image shows two Samsung DDR5 EC8 RDIMM memory modules. The top module is a full-length DIMM with a green PCB and gold-plated pins. The bottom module is a half-length DIMM with a white label. Both modules feature multiple black DRAM chips and a central white label with technical specifications and a QR code. The label on the bottom module reads: SAMSUNG, DDR5 EC8 RDIMM, 80CE04232645DAAD41, TOFK000, 64GB 2Rx4 PC5-4800B-RA0-1010-XT, Vietnam, M321R8GA0B80-CQKZJ, RE1MR6G. The label also includes a CE mark, a UKCA mark, and a QR code.</p> |

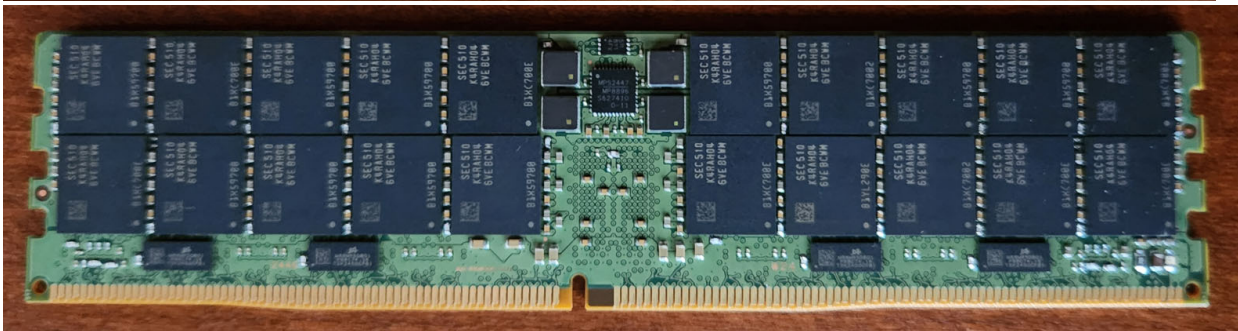
U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p data-bbox="573 310 1793 383">Samsung 64GB DDR5 4800MHz PC5-38400 ECC RDIMM 2Rx4 (EC8 10x4) Dual Rank 1.1V Registered DIMM 288-Pin Server RAM Memory M321R8GA0BB0-CQK.</p> <p data-bbox="573 435 1352 467"><i>See also, e.g., Ex. CC-117 (Samsung 5600MHz PC5-44800):</i></p>  <p data-bbox="573 805 1793 878">Samsung 64GB DDR5 5600MHz PC5-44800 ECC RDIMM 2Rx4 (EC8 10x4) Dual Rank 1.1V Registered DIMM 288-Pin Server RAM Memory M321R8GA0PB0-CWM.</p> <p data-bbox="573 930 1260 963"><i>See also images of Samsung DDR5 RDIMM products:</i></p> |

U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <div data-bbox="573 302 1896 751"><p>A photograph of a Samsung 64GB DDR5 PC5-4800 EC8 RDIMM 2Rx4 M321R8GA0BB0-CQK memory module. The module is green with gold-plated pins. It features a white label with the Samsung logo, part number 80CE03251213251911, and other specifications. The module is populated with 16 black memory chips, each labeled with 'SEC513' and 'K4D8G08'.</p></div> <p data-bbox="573 760 1770 792">Ex. CC-118 (Samsung 64GB DDR5 PC5-4800 EC8 RDIMM 2Rx4 M321R8GA0BB0-CQK).</p> <div data-bbox="573 841 1896 1243"><p>A photograph of a Samsung 64GB DDR5 PC5-6400 EC8 RDIMM 2Rx4 M321R8GA0PB2-CCP memory module. The module is green with gold-plated pins. It features a white label with the Samsung logo, part number 80CE01251501789891, and other specifications. The module is populated with 16 black memory chips, each labeled with 'SEC513' and 'K4D8G08'.</p></div> <p data-bbox="573 1252 1759 1284">Ex. CC-119 (Samsung 64GB DDR5 PC5-6400 EC8 RDIMM 2Rx4 M321R8GA0PB2-CCP).</p> |

U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

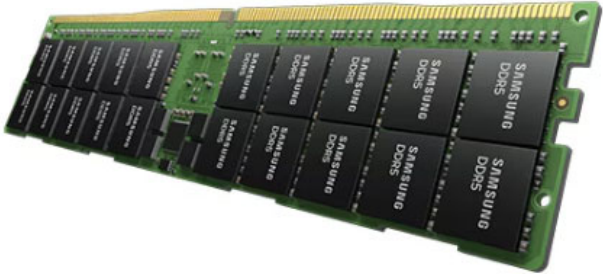
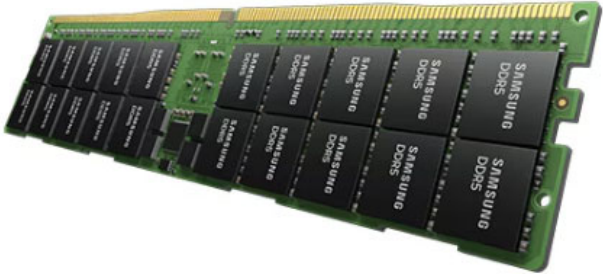
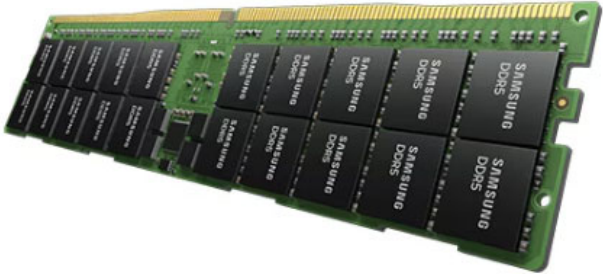
| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <div data-bbox="571 305 1780 651">A top-down view of a Samsung MRDIMM M327R8GA0EB0-CLVXB memory module. The module is green with gold-plated pins. It features a white label with the Samsung logo, part number M327R8GA0EB0-CLVXB, and other specifications. The module is populated with multiple black memory chips.</div> <div data-bbox="571 651 1801 980">A bottom-up view of the same Samsung MRDIMM M327R8GA0EB0-CLVXB memory module, showing the underside of the green PCB and the gold-plated pins.</div> <div data-bbox="571 980 1780 1279">A close-up view of the white label on the Samsung MRDIMM M327R8GA0EB0-CLVXB memory module. The label contains the Samsung logo, part number M327R8GA0EB0-CLVXB, and other specifications.</div> |

An image of a Samsung MRDIMM M327R8GA0EB0-CLVXB. See Exs. CC-111, CC-112, CC-113.

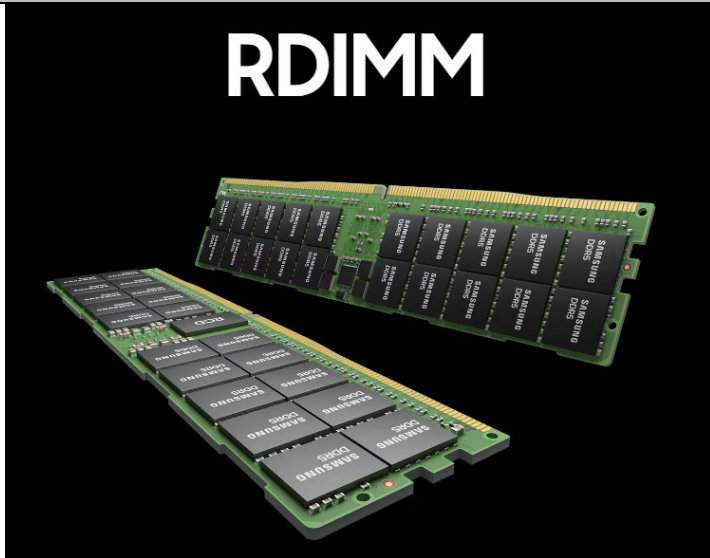
## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>Each accused Samsung DIMM is configured to fit into a memory slot connector of a system board of a computer system. For example, one Samsung web site illustrates that “DDR5 DIMM” is coupled to a “DIMM slot.” <i>See</i> Ex. CC-104 (Samsung Tech Blog, “Optimized DDR5 DIMM Solutions: Powering Leading-Edge Server Memory Modules” (“Optimized DDR5 DIMM Solutions”)); <i>see also, e.g.</i>, Ex. CC-145 (DDR5 MRDIMM: A Transformational Evolution For DDR5 DIMM) (“DDR5 MRDIMM is pin compatible with traditional DIMM, and it can be used in the same DIMM slots that are used for DDR5 RDIMMs”).</p> <div data-bbox="898 651 1512 1258" data-label="Diagram"> </div> <p>As another example, Samsung’s web site advertises various dual in-line memory modules “for a wide range of applications.” Ex. CC-95 (Samsung DRAM Modules). For instance, Samsung advertises registered DIMMs (“RDIMMs”), including ones with ECC capability, as suitable for</p> |

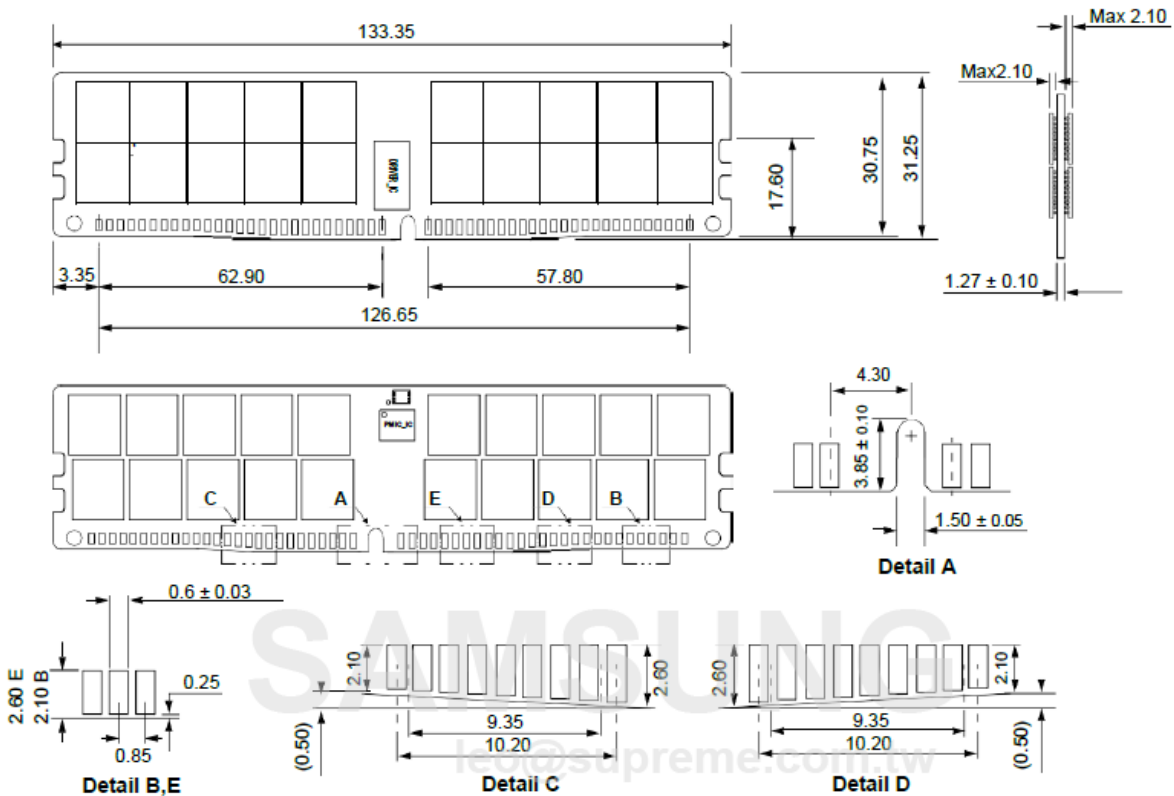
## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                                                                                                                                                                                                                                                                                                               | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                       |                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                       | <p>server applications. <i>Id.</i></p> <p>The RDIMM includes edge connections with gold fingers which are configured to fit into a memory slot connector of a server computer's motherboard (system board).</p> <table border="1" data-bbox="573 472 1892 829"><tr><td data-bbox="573 472 1234 829"><p><b>RDIMM</b></p><p><b>Registered DIMM</b></p><p>Include a register for enhancing the clock, command and control signals</p><p>Error correction is available by adding 8-bit parity signals</p><p>Supports x4 / x8 Organization / up to 2 ranks per DIMM and 3DPC configuration</p><p>Application : Server</p></td><td data-bbox="1234 472 1892 829"></td></tr></table> <p><i>See also</i> Ex. CC-144 (Samsung Module RDIMM):</p> | <p><b>RDIMM</b></p> <p><b>Registered DIMM</b></p> <p>Include a register for enhancing the clock, command and control signals</p> <p>Error correction is available by adding 8-bit parity signals</p> <p>Supports x4 / x8 Organization / up to 2 ranks per DIMM and 3DPC configuration</p> <p>Application : Server</p> |  |
| <p><b>RDIMM</b></p> <p><b>Registered DIMM</b></p> <p>Include a register for enhancing the clock, command and control signals</p> <p>Error correction is available by adding 8-bit parity signals</p> <p>Supports x4 / x8 Organization / up to 2 ranks per DIMM and 3DPC configuration</p> <p>Application : Server</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                       |                                                                                     |

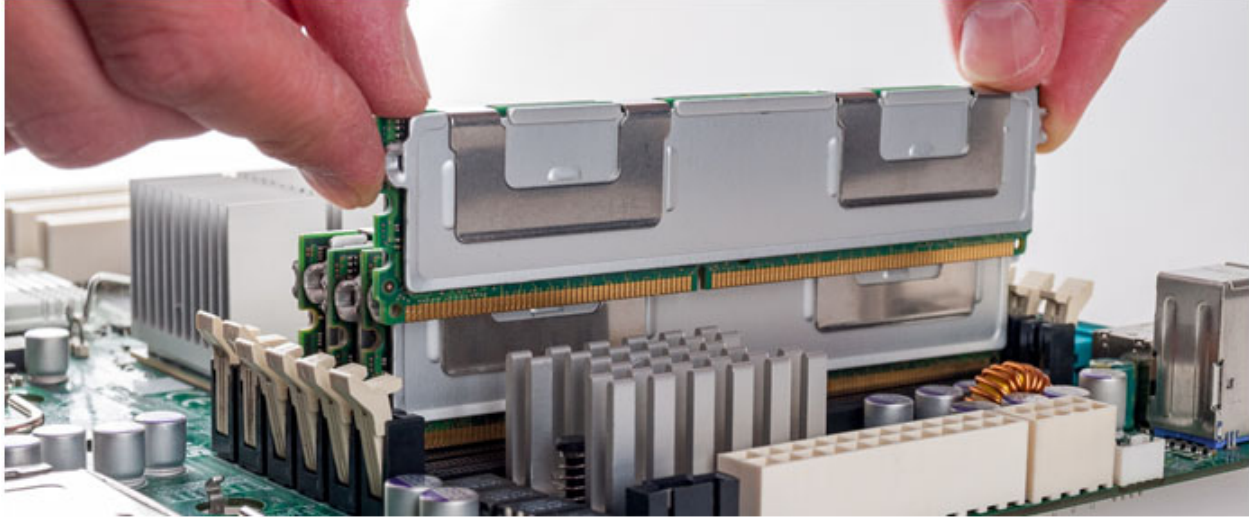
## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                                | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                      |                                 |                                      |                                 |                                        |                                             |                                        |                                             |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |        |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                   |      |        |       |          |       |                 |      |
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|                                        | <div><div><div>RDIMM</div><div></div></div><div><table><tr><th><input type="radio"/> Part Number ▾</th><th><input type="radio"/> DDR ▾</th><th><input type="radio"/> Applications ▾</th><th><input type="radio"/> Density ▾</th><th><input type="radio"/> Speed ▾</th><th><input type="radio"/> Voltage ▾</th><th><input type="radio"/> Product Status ▾</th><th><input type="radio"/> Rank x Organization ▾</th></tr><tr><td><input type="radio"/> M321R2GA3EB2-CWM</td><td>DDR5</td><td>Server</td><td>16 GB</td><td>5600 Mbps</td><td>1.1 V</td><td>Mass Production</td><td>1R x 8</td></tr><tr><td><input type="radio"/> M321R4GA0EB2-CWM</td><td>DDR5</td><td>Server</td><td>32 GB</td><td>5600 Mbps</td><td>1.1 V</td><td>Mass Production</td><td>1R x 4</td></tr><tr><td><input type="radio"/> M321R4GA3EB2-CWM</td><td>DDR5</td><td>Server</td><td>32 GB</td><td>5600 Mbps</td><td>1.1 V</td><td>Mass Production</td><td>2R x 8</td></tr><tr><td><input type="radio"/> M321R8GA0EB2-CWM</td><td>DDR5</td><td>Server</td><td>64 GB</td><td>5600 Mbps</td><td>1.1 V</td><td>Mass Production</td><td>2R x 4</td></tr><tr><td><input type="radio"/> M321RAJA0MB2-CWM</td><td>DDR5</td><td>Server</td><td>128 GB</td><td>5600 Mbps</td><td>1.1 V</td><td>Mass Production</td><td>2R x 4</td></tr><tr><td><input type="radio"/> M321RYGA0PB2-CWM</td><td>DDR5</td><td>Server</td><td>96 GB</td><td>5600 Mbps</td><td>1.1 V</td><td>Mass Production</td><td>2R x 4</td></tr><tr><td><input type="radio"/> MDRRW4QDBC2</td><td>DDR5</td><td>Server</td><td>96 GB</td><td>6400Mbps</td><td>1.1 V</td><td>Mass Production</td><td>2Rx4</td></tr></table></div></div> | <input type="radio"/> Part Number ▾  | <input type="radio"/> DDR ▾     | <input type="radio"/> Applications ▾ | <input type="radio"/> Density ▾ | <input type="radio"/> Speed ▾          | <input type="radio"/> Voltage ▾             | <input type="radio"/> Product Status ▾ | <input type="radio"/> Rank x Organization ▾ | <input type="radio"/> M321R2GA3EB2-CWM | DDR5 | Server | 16 GB | 5600 Mbps | 1.1 V | Mass Production | 1R x 8 | <input type="radio"/> M321R4GA0EB2-CWM | DDR5 | Server | 32 GB | 5600 Mbps | 1.1 V | Mass Production | 1R x 4 | <input type="radio"/> M321R4GA3EB2-CWM | DDR5 | Server | 32 GB | 5600 Mbps | 1.1 V | Mass Production | 2R x 8 | <input type="radio"/> M321R8GA0EB2-CWM | DDR5 | Server | 64 GB | 5600 Mbps | 1.1 V | Mass Production | 2R x 4 | <input type="radio"/> M321RAJA0MB2-CWM | DDR5 | Server | 128 GB | 5600 Mbps | 1.1 V | Mass Production | 2R x 4 | <input type="radio"/> M321RYGA0PB2-CWM | DDR5 | Server | 96 GB | 5600 Mbps | 1.1 V | Mass Production | 2R x 4 | <input type="radio"/> MDRRW4QDBC2 | DDR5 | Server | 96 GB | 6400Mbps | 1.1 V | Mass Production | 2Rx4 |
| <input type="radio"/> Part Number ▾    | <input type="radio"/> DDR ▾                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <input type="radio"/> Applications ▾ | <input type="radio"/> Density ▾ | <input type="radio"/> Speed ▾        | <input type="radio"/> Voltage ▾ | <input type="radio"/> Product Status ▾ | <input type="radio"/> Rank x Organization ▾ |                                        |                                             |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |        |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                   |      |        |       |          |       |                 |      |
| <input type="radio"/> M321R2GA3EB2-CWM | DDR5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Server                               | 16 GB                           | 5600 Mbps                            | 1.1 V                           | Mass Production                        | 1R x 8                                      |                                        |                                             |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |        |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                   |      |        |       |          |       |                 |      |
| <input type="radio"/> M321R4GA0EB2-CWM | DDR5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Server                               | 32 GB                           | 5600 Mbps                            | 1.1 V                           | Mass Production                        | 1R x 4                                      |                                        |                                             |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |        |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                   |      |        |       |          |       |                 |      |
| <input type="radio"/> M321R4GA3EB2-CWM | DDR5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Server                               | 32 GB                           | 5600 Mbps                            | 1.1 V                           | Mass Production                        | 2R x 8                                      |                                        |                                             |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |        |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                   |      |        |       |          |       |                 |      |
| <input type="radio"/> M321R8GA0EB2-CWM | DDR5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Server                               | 64 GB                           | 5600 Mbps                            | 1.1 V                           | Mass Production                        | 2R x 4                                      |                                        |                                             |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |        |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                   |      |        |       |          |       |                 |      |
| <input type="radio"/> M321RAJA0MB2-CWM | DDR5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Server                               | 128 GB                          | 5600 Mbps                            | 1.1 V                           | Mass Production                        | 2R x 4                                      |                                        |                                             |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |        |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                   |      |        |       |          |       |                 |      |
| <input type="radio"/> M321RYGA0PB2-CWM | DDR5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Server                               | 96 GB                           | 5600 Mbps                            | 1.1 V                           | Mass Production                        | 2R x 4                                      |                                        |                                             |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |        |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                   |      |        |       |          |       |                 |      |
| <input type="radio"/> MDRRW4QDBC2      | DDR5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Server                               | 96 GB                           | 6400Mbps                             | 1.1 V                           | Mass Production                        | 2Rx4                                        |                                        |                                             |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                        |      |        |        |           |       |                 |        |                                        |      |        |       |           |       |                 |        |                                   |      |        |       |          |       |                 |      |

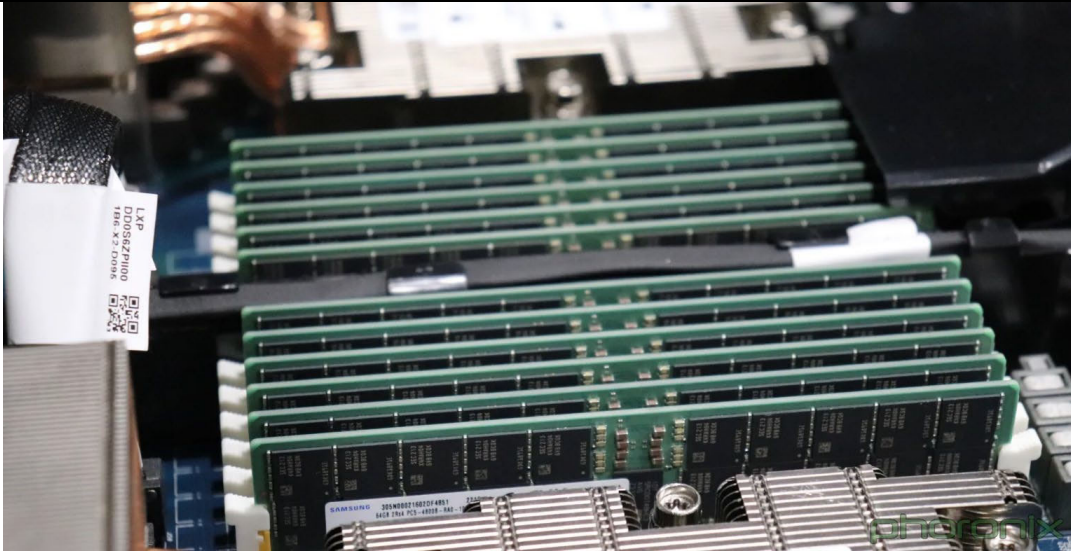
U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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|         | <p data-bbox="569 349 1392 381"><i>See also</i> Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 96:</p> <p data-bbox="604 440 1608 479"><b>19.4 4Gx4 based x80 Module (2 Ranks) - M321R8GA0BB0</b></p> <p data-bbox="1654 500 1759 524">Units : mm</p>  <p>The technical drawing illustrates the dimensions of a DDR5 RDIMM module. The top view shows a rectangular module with a total length of 133.35 mm and a total width of 31.25 mm. The module is divided into two ranks, each containing four memory chips. The bottom view shows the module with labels A, B, C, D, and E indicating specific components or features. Detailed views A, B, C, D, and E provide further dimensions for specific components, such as the height of the module (17.60 mm), the width of the module (30.75 mm), and the dimensions of the memory chips (e.g., 2.60 mm, 2.10 mm, 0.85 mm, 0.25 mm). The drawing also includes a large 'SAMSUNG' watermark and a URL 'le@supreme.com.tw'.</p> |

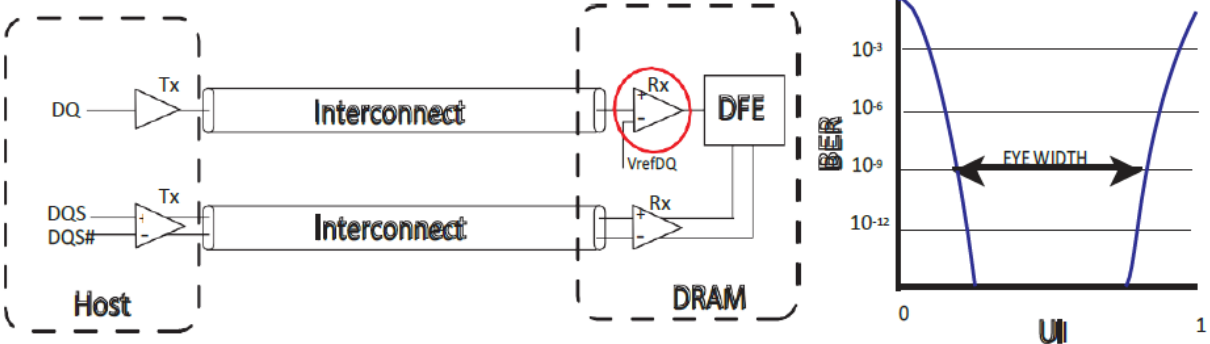
U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p data-bbox="573 310 846 337"><i>See also id.</i> at 93–95.</p> <p data-bbox="573 383 1854 456"><i>See also</i> Ex. CC-120 (Samsung, “Memory VR changes in the system by DDR5 adoption”) (showing a DIMM being inserted into a memory slot on a motherboard of a computer system).</p>  <p data-bbox="573 1036 1304 1063">Ex. CC-70 (DDR5 Memory Channel Scaling Performance):</p> |

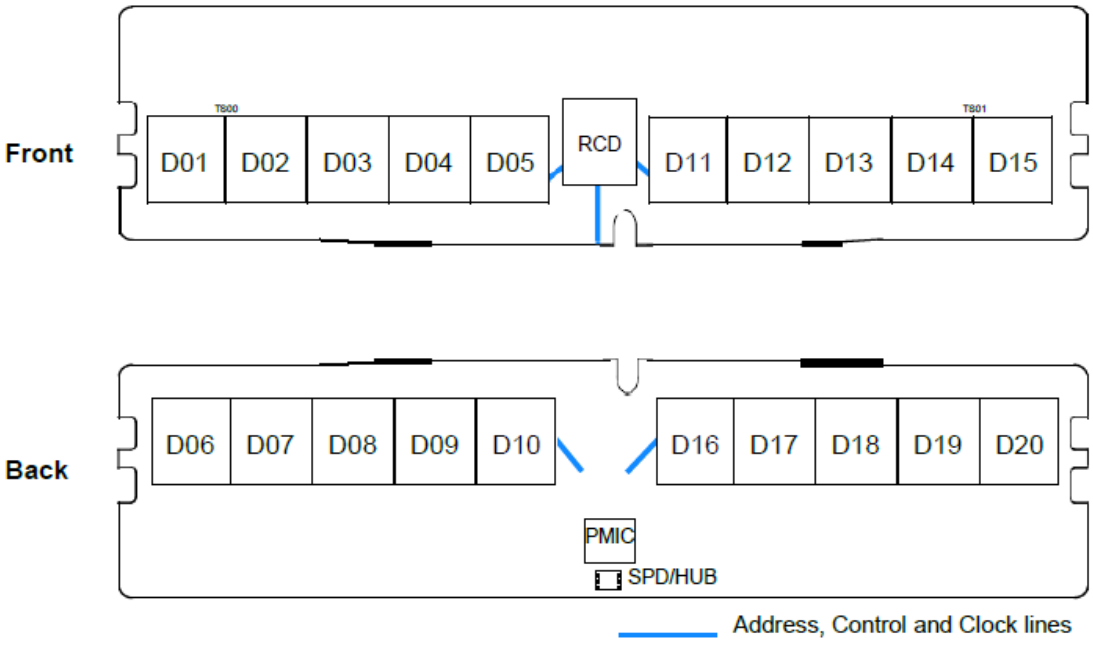
## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <div data-bbox="571 302 1635 849"></div> <p data-bbox="571 911 1860 1279">The memory slot connector on the host system's motherboard includes a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM via the set of data conduits, and address and control signals are delivered from the system board to the DIMM via the set of address and control conduits. For example, as Samsung's own datasheet (Samsung DDR5 RDIMM Datasheet) illustrates, the host computer and the DRAMs on the DIMMs have various interconnects for data (DQ) signals, control/address signals and power signals. Each interconnect provides a respective conduit to conduct the corresponding signals between the host and the DIMM. <i>See</i> Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 7–9.</p> |

U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |  <p data-bbox="961 690 1495 714"><b>Figure 17. Example of DDR5 Memory Interconnect</b></p> <p data-bbox="573 730 1822 844">Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at Figs. 10, 17 (showing DQ and DQS interconnects; there is a respective conduit for each DQ or DQS pin on the DIMM); <i>see also id.</i> at Figs. 13–14.</p> |

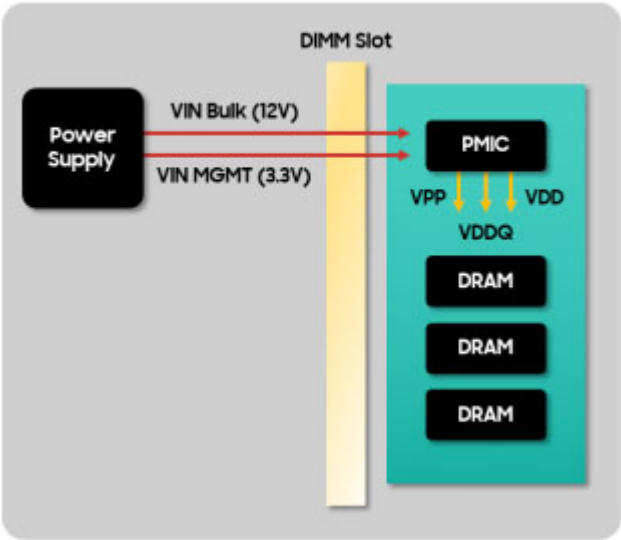
U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

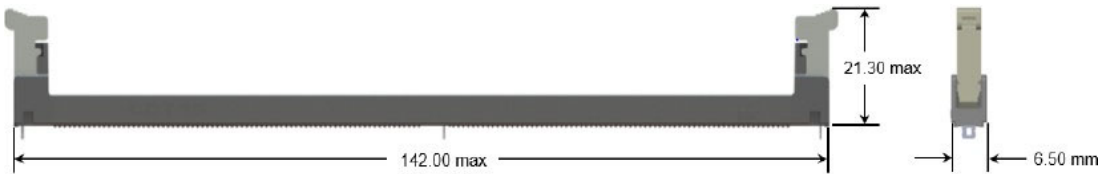
| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |  <p>The diagram illustrates the internal architecture of a Samsung DDR5 RDIMM. It is divided into two main sections: 'Front' and 'Back'.<br/> <b>Front View:</b> Shows two data buffers, TS00 (containing D01, D02, D03, D04, D05) and TS01 (containing D11, D12, D13, D14, D15). These buffers are connected to a central RCD (Refresh Control Driver) block. Blue lines indicate the connection of Address, Control, and Clock signals between the buffers and the RCD.<br/> <b>Back View:</b> Shows two data buffers, D06-D10 and D16-D20. Below these buffers are the PMIC (Power Management IC) and SPD/HUB (Serial Presence Detect/Hub) components. Blue lines connect the buffers to the PMIC and SPD/HUB, and also show the signal paths for Address, Control, and Clock lines.<br/> <b>Legend:</b> A blue line is defined as 'Address, Control and Clock lines'.</p> <p>Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 25 (each blue line includes multiple signal lines and the lines from the edge connection to the RCD are coupled to the respective control/address and clock conduits for transmitting respective address, control and clock signals); <i>see also id.</i> at 21, 29, 33.</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <div data-bbox="598 321 1596 727" data-label="Diagram"> <p>The diagram illustrates a power management integrated circuit (PMIC) and its associated power distribution network. The PMIC is shown as a central block with two main input pins: VIN_BULK and VIN_MGMT. VIN_BULK is connected to a network of capacitors and inductors that supply power to various components, including SPD_HUB, TS, D01-D20, and GND. VIN_MGMT is connected to a similar network that supplies power to D01-D20, SPD_HUB, TS, RCD, and GND. The diagram shows a complex interconnection of power planes and components, with various voltage levels (VDDSPD, VPP, VDD, VDDQ, VDDIO, VSS) and ground connections (GND) indicated.</p> </div> <p>Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 28 (showing VIN_BULK and VIN_MGMT being supplied to PMIC; these voltages would be supplied via the power conduits); <i>see also id.</i> at 24, 32, 36.</p> <p><i>See also</i> Ex. CC-104 (Optimized DDR5 DIMM Solutions) (showing VIN Bulk and VIN MGMT being supplied from a power supply via DIMM slot to PMIC).</p> |

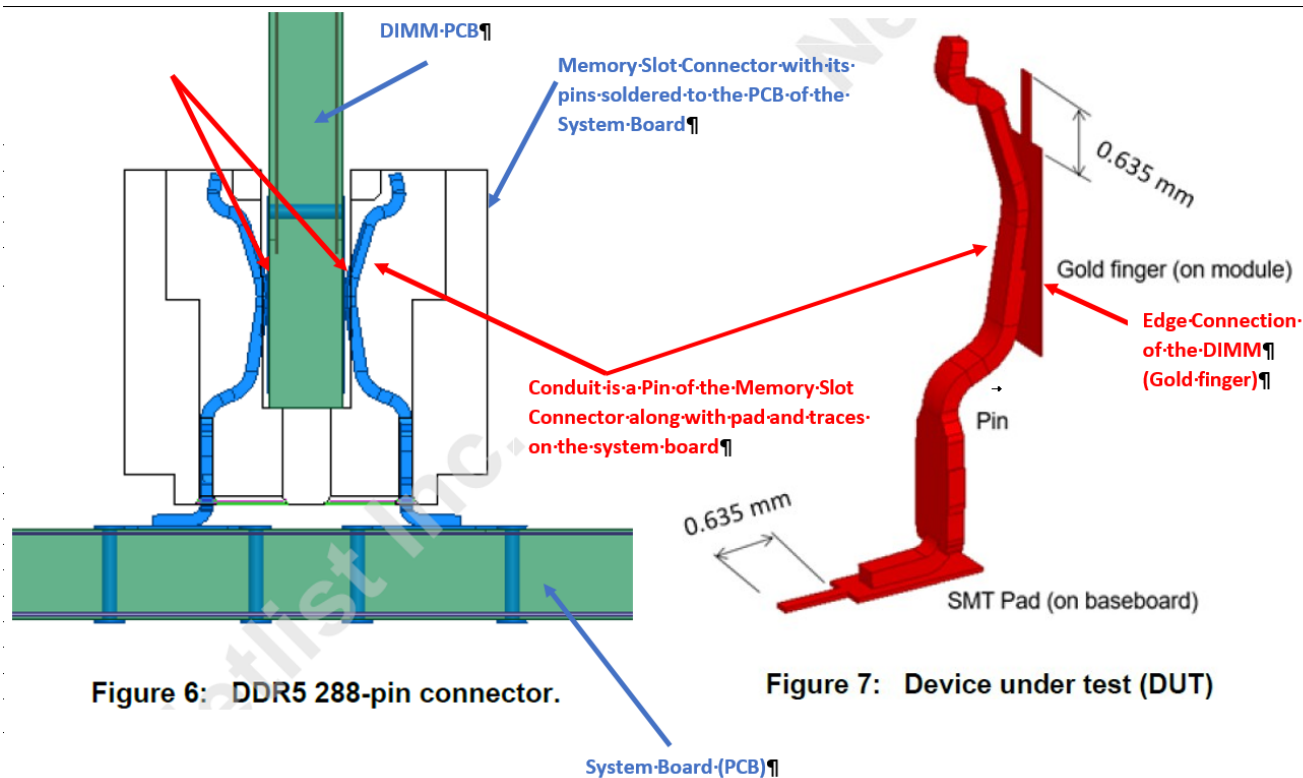
U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p data-bbox="1108 321 1318 362" style="text-align: center;"><b>DDR5 DIMM</b></p>  <p data-bbox="573 963 1841 1081">Additionally, Samsung's RDIMMs are "JEDEC standard compliant," (Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 5) and are therefore configured to fit into a JEDEC-compliant memory slot, as illustrated below.</p> |

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <div data-bbox="940 305 1524 941"></div> <div data-bbox="963 987 1497 1023"><p>Figure 1 DDR5 socket and module (SMT)</p></div> <div data-bbox="688 1068 1780 1240"></div> <div data-bbox="697 1308 1705 1344"><p>Figure 2 DDR5 U/R/LR DIMM Surface Mount (SMT) Connector Socket Outline</p></div> <div data-bbox="573 1377 1018 1412"><p>Ex. CC-73 (JEDEC PS-005B) at 6.</p></div> |

Claim 1

Evidence of Use



Ex. CC-73 (JEDEC PS-005B) at 13, Figs. 6–7 (annotated) (illustrating an edge connection on the DIMM being electrically connected with a corresponding pin on the memory slot connector).

The memory slot connector of the system board comprises conduits as illustrated above that correspond to Samsung DDR5 DIMM pin assignments, such as the one illustrated in Figure 1 in Samsung DDR5 RDIMM Datasheet. Each conduit of the memory slot connector corresponds to a pin number and associated signal name for power, data, and address and control conduits.

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

Claim 1

Evidence of Use

| <table><tr><th>Front Side Pin Label</th><th>Pin</th><th>Pin</th><th>Back side Pin Label</th></tr><tr><td>VIN_BULK</td><td>1</td><td>145</td><td>VIN_BULK</td></tr><tr><td>RFU</td><td>2</td><td>146</td><td>VIN_BULK</td></tr><tr><td>VIN_MGMT</td><td>3</td><td>147</td><td>PWR_GOOD/FAIL_n</td></tr><tr><td>HSCL</td><td>4</td><td>148</td><td>HSA</td></tr><tr><td>HSDA</td><td>5</td><td>149</td><td>RFU</td></tr><tr><td>VSS</td><td>6</td><td>150</td><td>RFU</td></tr><tr><td>DQ0_A</td><td>7</td><td>151</td><td>VSS</td></tr><tr><td>VSS</td><td>8</td><td>152</td><td>DQ2_A</td></tr><tr><td>DQ1_A</td><td>9</td><td>153</td><td>VSS</td></tr><tr><td>VSS</td><td>10</td><td>154</td><td>DQ3_A</td></tr><tr><td>DQS0_A_I</td><td>11</td><td>155</td><td>VSS</td></tr><tr><td>DQS0_A_c</td><td>12</td><td>156</td><td>DQS5_A_c, TDQS5_A_c</td></tr><tr><td>VSS</td><td>13</td><td>157</td><td>DQS5_A_I, TDQS5_A_I, DBI0_A_n</td></tr><tr><td>DQ4_A</td><td>14</td><td>158</td><td>VSS</td></tr><tr><td>VSS</td><td>15</td><td>159</td><td>DQ6_A</td></tr><tr><td>DQ5_A</td><td>16</td><td>160</td><td>VSS</td></tr><tr><td>VSS</td><td>17</td><td>161</td><td>DQ7_A</td></tr><tr><td>DQ8_A</td><td>18</td><td>162</td><td>VSS</td></tr><tr><td>VSS</td><td>19</td><td>163</td><td>DQ10_A</td></tr><tr><td>DQ9_A</td><td>20</td><td>164</td><td>VSS</td></tr><tr><td>VSS</td><td>21</td><td>165</td><td>DQ11_A</td></tr><tr><td>DQS1_A_I</td><td>22</td><td>166</td><td>VSS</td></tr><tr><td>DQS1_A_c</td><td>23</td><td>167</td><td>DQS6_A_c, TDQS6_A_c</td></tr><tr><td>VSS</td><td>24</td><td>168</td><td>DQS6_A_I, TDQS6_A_I, DBI1_A_n</td></tr><tr><td>DQ12_A</td><td>25</td><td>169</td><td>VSS</td></tr><tr><td>VSS</td><td>26</td><td>170</td><td>DQ14_A</td></tr><tr><td>DQ13_A</td><td>27</td><td>171</td><td>VSS</td></tr><tr><td>VSS</td><td>28</td><td>172</td><td>DQ15_A</td></tr><tr><td>DQ16_A</td><td>29</td><td>173</td><td>VSS</td></tr><tr><td>VSS</td><td>30</td><td>174</td><td>DQ18_A</td></tr><tr><td>DQ17_A</td><td>31</td><td>175</td><td>VSS</td></tr><tr><td>VSS</td><td>32</td><td>176</td><td>DQ19_A</td></tr><tr><td>DQS2_A_I</td><td>33</td><td>177</td><td>VSS</td></tr><tr><td>DQS2_A_c</td><td>34</td><td>178</td><td>DQS7_A_c, TDQS7_A_c</td></tr><tr><td>VSS</td><td>35</td><td>179</td><td>DQS7_A_I, TDQS7_A_I, DBI2_n</td></tr><tr><td>DQ20_A</td><td>36</td><td>180</td><td>VSS</td></tr><tr><td>VSS</td><td>37</td><td>181</td><td>DQ22_A</td></tr><tr><td>DQ21_A</td><td>38</td><td>182</td><td>VSS</td></tr></table> | Front Side Pin Label | Pin | Pin                           | Back side Pin Label | VIN_BULK | 1 | 145 | VIN_BULK | RFU | 2 | 146 | VIN_BULK | VIN_MGMT | 3 | 147 | PWR_GOOD/FAIL_n | HSCL | 4 | 148 | HSA | HSDA | 5 | 149 | RFU | VSS | 6 | 150 | RFU | DQ0_A | 7 | 151 | VSS | VSS | 8 | 152 | DQ2_A | DQ1_A | 9 | 153 | VSS | VSS | 10 | 154 | DQ3_A | DQS0_A_I | 11 | 155 | VSS | DQS0_A_c | 12 | 156 | DQS5_A_c, TDQS5_A_c | VSS | 13 | 157 | DQS5_A_I, TDQS5_A_I, DBI0_A_n | DQ4_A | 14 | 158 | VSS | VSS | 15 | 159 | DQ6_A | DQ5_A | 16 | 160 | VSS | VSS | 17 | 161 | DQ7_A | DQ8_A | 18 | 162 | VSS | VSS | 19 | 163 | DQ10_A | DQ9_A | 20 | 164 | VSS | VSS | 21 | 165 | DQ11_A | DQS1_A_I | 22 | 166 | VSS | DQS1_A_c | 23 | 167 | DQS6_A_c, TDQS6_A_c | VSS | 24 | 168 | DQS6_A_I, TDQS6_A_I, DBI1_A_n | DQ12_A | 25 | 169 | VSS | VSS | 26 | 170 | DQ14_A | DQ13_A | 27 | 171 | VSS | VSS | 28 | 172 | DQ15_A | DQ16_A | 29 | 173 | VSS | VSS | 30 | 174 | DQ18_A | DQ17_A | 31 | 175 | VSS | VSS | 32 | 176 | DQ19_A | DQS2_A_I | 33 | 177 | VSS | DQS2_A_c | 34 | 178 | DQS7_A_c, TDQS7_A_c | VSS | 35 | 179 | DQS7_A_I, TDQS7_A_I, DBI2_n | DQ20_A | 36 | 180 | VSS | VSS | 37 | 181 | DQ22_A | DQ21_A | 38 | 182 | VSS | <table><tr><th>Front Side Pin Label</th><th>Pin</th><th>Pin</th><th>Back side Pin Label</th></tr><tr><td>PAR_A</td><td>74</td><td>218</td><td>CK_c</td></tr><tr><td>VSS</td><td>75</td><td>219</td><td>VSS</td></tr><tr><td colspan="4">Key</td></tr><tr><td>CA0_B</td><td>76</td><td>220</td><td>RFU</td></tr><tr><td>VSS</td><td>77</td><td>221</td><td>CA1_B</td></tr><tr><td>CA2_B</td><td>78</td><td>222</td><td>VSS</td></tr><tr><td>VSS</td><td>79</td><td>223</td><td>CA3_B</td></tr><tr><td>CA4_B</td><td>80</td><td>224</td><td>VSS</td></tr><tr><td>VSS</td><td>81</td><td>225</td><td>CA5_B</td></tr><tr><td>CA6_B</td><td>82</td><td>226</td><td>VSS</td></tr><tr><td>VSS</td><td>83</td><td>227</td><td>PAR_B</td></tr><tr><td>CS0_B_n</td><td>84</td><td>228</td><td>VSS</td></tr><tr><td>VSS</td><td>85</td><td>229</td><td>CS1_B_n</td></tr><tr><td>LBD, RSP_A_n</td><td>86</td><td>230</td><td>VSS</td></tr><tr><td>LBS, RSP_B_n</td><td>87</td><td>231</td><td>RFU</td></tr><tr><td>VSS</td><td>88</td><td>232</td><td>RFU</td></tr><tr><td>CB4_B</td><td>89</td><td>233</td><td>VSS</td></tr><tr><td>VSS</td><td>90</td><td>234</td><td>CB6_B</td></tr><tr><td>CB5_B</td><td>91</td><td>235</td><td>VSS</td></tr><tr><td>VSS</td><td>92</td><td>236</td><td>CB7_B</td></tr><tr><td>DQS9_B_I, TDQS9_B_I, DBI4_B_n</td><td>93</td><td>237</td><td>VSS</td></tr><tr><td>DQS9_B_c, TDQS9_B_c</td><td>94</td><td>238</td><td>DQS4_B_c</td></tr><tr><td>VSS</td><td>95</td><td>239</td><td>DQS4_B_I</td></tr><tr><td>CB0_B</td><td>96</td><td>240</td><td>VSS</td></tr><tr><td>VSS</td><td>97</td><td>241</td><td>CB2_B</td></tr><tr><td>CB1_B</td><td>98</td><td>242</td><td>VSS</td></tr><tr><td>VSS</td><td>99</td><td>243</td><td>CB3_B</td></tr><tr><td>DQ0_B</td><td>100</td><td>244</td><td>VSS</td></tr><tr><td>VSS</td><td>101</td><td>245</td><td>DQ2_B</td></tr><tr><td>DQ1_B</td><td>102</td><td>246</td><td>VSS</td></tr><tr><td>VSS</td><td>103</td><td>247</td><td>DQ3_BDQ3_B</td></tr><tr><td>DQS0_B_I</td><td>104</td><td>248</td><td>VSS</td></tr><tr><td>DQS0_B_c</td><td>105</td><td>249</td><td>DQS5_B_c, TDQS5_B_c</td></tr><tr><td>VSS</td><td>106</td><td>250</td><td>DQS5_B_I, TDQS5_B_I, DBI0_B_n</td></tr><tr><td>DQ4_B</td><td>107</td><td>251</td><td>VSS</td></tr><tr><td>VSS</td><td>108</td><td>252</td><td>DQ6_B</td></tr><tr><td>DQ5_B</td><td>109</td><td>253</td><td>VSS</td></tr></table> | Front Side Pin Label | Pin | Pin | Back side Pin Label | PAR_A | 74 | 218 | CK_c | VSS | 75 | 219 | VSS | Key |  |  |  | CA0_B | 76 | 220 | RFU | VSS | 77 | 221 | CA1_B | CA2_B | 78 | 222 | VSS | VSS | 79 | 223 | CA3_B | CA4_B | 80 | 224 | VSS | VSS | 81 | 225 | CA5_B | CA6_B | 82 | 226 | VSS | VSS | 83 | 227 | PAR_B | CS0_B_n | 84 | 228 | VSS | VSS | 85 | 229 | CS1_B_n | LBD, RSP_A_n | 86 | 230 | VSS | LBS, RSP_B_n | 87 | 231 | RFU | VSS | 88 | 232 | RFU | CB4_B | 89 | 233 | VSS | VSS | 90 | 234 | CB6_B | CB5_B | 91 | 235 | VSS | VSS | 92 | 236 | CB7_B | DQS9_B_I, TDQS9_B_I, DBI4_B_n | 93 | 237 | VSS | DQS9_B_c, TDQS9_B_c | 94 | 238 | DQS4_B_c | VSS | 95 | 239 | DQS4_B_I | CB0_B | 96 | 240 | VSS | VSS | 97 | 241 | CB2_B | CB1_B | 98 | 242 | VSS | VSS | 99 | 243 | CB3_B | DQ0_B | 100 | 244 | VSS | VSS | 101 | 245 | DQ2_B | DQ1_B | 102 | 246 | VSS | VSS | 103 | 247 | DQ3_BDQ3_B | DQS0_B_I | 104 | 248 | VSS | DQS0_B_c | 105 | 249 | DQS5_B_c, TDQS5_B_c | VSS | 106 | 250 | DQS5_B_I, TDQS5_B_I, DBI0_B_n | DQ4_B | 107 | 251 | VSS | VSS | 108 | 252 | DQ6_B | DQ5_B | 109 | 253 | VSS | <table><tr><th>Front Side Pin Label</th><th>Pin</th><th>Pin</th><th>Back side Pin Label</th></tr><tr><td>VSS</td><td>39</td><td>183</td><td>DQ23_A</td></tr><tr><td>DQ24_A</td><td>40</td><td>184</td><td>VSS</td></tr><tr><td>VSS</td><td>41</td><td>185</td><td>DQ26_A</td></tr><tr><td>DQ25_A</td><td>42</td><td>186</td><td>VSS</td></tr><tr><td>VSS</td><td>43</td><td>187</td><td>DQ27_A</td></tr><tr><td>DQS3_A_I</td><td>44</td><td>188</td><td>VSS</td></tr><tr><td>DQS3_A_c</td><td>45</td><td>189</td><td>DQS8_A_c, TDQS8_A_c</td></tr><tr><td>VSS</td><td>46</td><td>190</td><td>DQS8_A_I, TDQS8_A_I, DBI3_A_n</td></tr><tr><td>DQ28_A</td><td>47</td><td>191</td><td>VSS</td></tr><tr><td>VSS</td><td>48</td><td>192</td><td>DQ30_A</td></tr><tr><td>DQ29_A</td><td>49</td><td>193</td><td>VSS</td></tr><tr><td>VSS</td><td>50</td><td>194</td><td>DQ31_A</td></tr><tr><td>CB0_A</td><td>51</td><td>195</td><td>VSS</td></tr><tr><td>VSS</td><td>52</td><td>196</td><td>CB2_A</td></tr><tr><td>CB1_A</td><td>53</td><td>197</td><td>VSS</td></tr><tr><td>VSS</td><td>54</td><td>198</td><td>CB3_A</td></tr><tr><td>DQS4_A_I</td><td>55</td><td>199</td><td>VSS</td></tr><tr><td>DQS4_A_c</td><td>56</td><td>200</td><td>DQS9_A_c, TDQS9_A_c</td></tr><tr><td>VSS</td><td>57</td><td>201</td><td>DQS9_A_I, TDQS9_A_I, DBI4_A_n</td></tr><tr><td>CB4_A</td><td>58</td><td>202</td><td>VSS</td></tr><tr><td>VSS</td><td>59</td><td>203</td><td>CB6_A</td></tr><tr><td>CB5_A</td><td>60</td><td>204</td><td>VSS</td></tr><tr><td>VSS</td><td>61</td><td>205</td><td>CB7_A</td></tr><tr><td>ALEFT_n</td><td>62</td><td>206</td><td>VSS</td></tr><tr><td>VSS</td><td>63</td><td>207</td><td>RESET_n</td></tr><tr><td>CS0_A_n</td><td>64</td><td>208</td><td>VSS</td></tr><tr><td>VSS</td><td>65</td><td>209</td><td>CS1_A_n</td></tr><tr><td>CA0_A</td><td>66</td><td>210</td><td>VSS</td></tr><tr><td>VSS</td><td>67</td><td>211</td><td>CA1_A</td></tr><tr><td>CA2</td><td>68</td><td>212</td><td>VSS</td></tr><tr><td>VSS</td><td>69</td><td>213</td><td>CA3_A</td></tr><tr><td>CA4_A</td><td>70</td><td>214</td><td>VSS</td></tr><tr><td>VSS</td><td>71</td><td>215</td><td>CA5_A</td></tr><tr><td>CA6_A</td><td>72</td><td>216</td><td>VSS</td></tr><tr><td>VSS</td><td>73</td><td>217</td><td>CK_I</td></tr></table> | Front Side Pin Label | Pin | Pin | Back side Pin Label | VSS | 39 | 183 | DQ23_A | DQ24_A | 40 | 184 | VSS | VSS | 41 | 185 | DQ26_A | DQ25_A | 42 | 186 | VSS | VSS | 43 | 187 | DQ27_A | DQS3_A_I | 44 | 188 | VSS | DQS3_A_c | 45 | 189 | DQS8_A_c, TDQS8_A_c | VSS | 46 | 190 | DQS8_A_I, TDQS8_A_I, DBI3_A_n | DQ28_A | 47 | 191 | VSS | VSS | 48 | 192 | DQ30_A | DQ29_A | 49 | 193 | VSS | VSS | 50 | 194 | DQ31_A | CB0_A | 51 | 195 | VSS | VSS | 52 | 196 | CB2_A | CB1_A | 53 | 197 | VSS | VSS | 54 | 198 | CB3_A | DQS4_A_I | 55 | 199 | VSS | DQS4_A_c | 56 | 200 | DQS9_A_c, TDQS9_A_c | VSS | 57 | 201 | DQS9_A_I, TDQS9_A_I, DBI4_A_n | CB4_A | 58 | 202 | VSS | VSS | 59 | 203 | CB6_A | CB5_A | 60 | 204 | VSS | VSS | 61 | 205 | CB7_A | ALEFT_n | 62 | 206 | VSS | VSS | 63 | 207 | RESET_n | CS0_A_n | 64 | 208 | VSS | VSS | 65 | 209 | CS1_A_n | CA0_A | 66 | 210 | VSS | VSS | 67 | 211 | CA1_A | CA2 | 68 | 212 | VSS | VSS | 69 | 213 | CA3_A | CA4_A | 70 | 214 | VSS | VSS | 71 | 215 | CA5_A | CA6_A | 72 | 216 | VSS | VSS | 73 | 217 | CK_I | <table><tr><th>Front Side Pin Label</th><th>Pin</th><th>Pin</th><th>Back side Pin Label</th></tr><tr><td>VSS</td><td>110</td><td>254</td><td>DQ7_B</td></tr><tr><td>DQ8_B</td><td>111</td><td>255</td><td>VSS</td></tr><tr><td>VSS</td><td>112</td><td>256</td><td>DQ10_B</td></tr><tr><td>DQ9_B</td><td>113</td><td>257</td><td>VSS</td></tr><tr><td>VSS</td><td>114</td><td>258</td><td>DQ11_B</td></tr><tr><td>DQS1_B_I</td><td>115</td><td>259</td><td>VSS</td></tr><tr><td>DQS1_B_c</td><td>116</td><td>260</td><td>DQS6_B_c, TDQS6_B_c</td></tr><tr><td>VSS</td><td>117</td><td>261</td><td>DQS6_B_I, TDQS6_B_I, DBI1_B_n</td></tr><tr><td>DQ12_B</td><td>118</td><td>262</td><td>VSS</td></tr><tr><td>VSS</td><td>119</td><td>263</td><td>DQ14_B</td></tr><tr><td>DQ13_B</td><td>120</td><td>264</td><td>VSS</td></tr><tr><td>VSS</td><td>121</td><td>265</td><td>DQ15_B</td></tr><tr><td>DQ16_B</td><td>122</td><td>266</td><td>VSS</td></tr><tr><td>VSS</td><td>123</td><td>267</td><td>DQ18_B</td></tr><tr><td>DQ17_B</td><td>124</td><td>268</td><td>VSS</td></tr><tr><td>VSS</td><td>125</td><td>269</td><td>DQ19_B</td></tr><tr><td>DQS2_B_I</td><td>126</td><td>270</td><td>VSS</td></tr><tr><td>DQS2_B_c</td><td>127</td><td>271</td><td>DQS7_B_c, TDQS7_B_c</td></tr><tr><td>VSS</td><td>128</td><td>272</td><td>DQS7_B_I, TDQS7_B_I, DBI2_B_n</td></tr><tr><td>DQ20_B</td><td>129</td><td>273</td><td>VSS</td></tr><tr><td>VSS</td><td>130</td><td>274</td><td>DQ22_B</td></tr><tr><td>DQ21_B</td><td>131</td><td>275</td><td>VSS</td></tr><tr><td>VSS</td><td>132</td><td>276</td><td>DQ23_B</td></tr><tr><td>DQ24_B</td><td>133</td><td>277</td><td>VSS</td></tr><tr><td>VSS</td><td>134</td><td>278</td><td>DQ26_B</td></tr><tr><td>DQ25_B</td><td>135</td><td>279</td><td>VSS</td></tr><tr><td>VSS</td><td>136</td><td>280</td><td>DQ27_B</td></tr><tr><td>DQS3_B_I</td><td>137</td><td>281</td><td>VSS</td></tr><tr><td>DQS3_B_c</td><td>138</td><td>282</td><td>DQS8_B_c, TDQS8_B_c</td></tr><tr><td>VSS</td><td>139</td><td>283</td><td>DQS8_B_I, TDQS8_B_I, DBI3_B_n</td></tr><tr><td>DQ28_B</td><td>140</td><td>284</td><td>VSS</td></tr><tr><td>VSS</td><td>141</td><td>285</td><td>DQ30_B</td></tr><tr><td>DQ29_B</td><td>142</td><td>286</td><td>VSS</td></tr><tr><td>VSS</td><td>143</td><td>287</td><td>DQ31_B</td></tr><tr><td>RFU</td><td>144</td><td>288</td><td>VSS</td></tr></table> | Front Side Pin Label | Pin | Pin | Back side Pin Label | VSS | 110 | 254 | DQ7_B | DQ8_B | 111 | 255 | VSS | VSS | 112 | 256 | DQ10_B | DQ9_B | 113 | 257 | VSS | VSS | 114 | 258 | DQ11_B | DQS1_B_I | 115 | 259 | VSS | DQS1_B_c | 116 | 260 | DQS6_B_c, TDQS6_B_c | VSS | 117 | 261 | DQS6_B_I, TDQS6_B_I, DBI1_B_n | DQ12_B | 118 | 262 | VSS | VSS | 119 | 263 | DQ14_B | DQ13_B | 120 | 264 | VSS | VSS | 121 | 265 | DQ15_B | DQ16_B | 122 | 266 | VSS | VSS | 123 | 267 | DQ18_B | DQ17_B | 124 | 268 | VSS | VSS | 125 | 269 | DQ19_B | DQS2_B_I | 126 | 270 | VSS | DQS2_B_c | 127 | 271 | DQS7_B_c, TDQS7_B_c | VSS | 128 | 272 | DQS7_B_I, TDQS7_B_I, DBI2_B_n | DQ20_B | 129 | 273 | VSS | VSS | 130 | 274 | DQ22_B | DQ21_B | 131 | 275 | VSS | VSS | 132 | 276 | DQ23_B | DQ24_B | 133 | 277 | VSS | VSS | 134 | 278 | DQ26_B | DQ25_B | 135 | 279 | VSS | VSS | 136 | 280 | DQ27_B | DQS3_B_I | 137 | 281 | VSS | DQS3_B_c | 138 | 282 | DQS8_B_c, TDQS8_B_c | VSS | 139 | 283 | DQS8_B_I, TDQS8_B_I, DBI3_B_n | DQ28_B | 140 | 284 | VSS | VSS | 141 | 285 | DQ30_B | DQ29_B | 142 | 286 | VSS | VSS | 143 | 287 | DQ31_B | RFU | 144 | 288 | VSS |
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| DQS1_A_c                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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| VSS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| VSS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| DQ13_A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| DQS9_B_I, TDQS9_B_I, DBI4_B_n                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          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| DQS9_B_c, TDQS9_B_c                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 94                   | 238 | DQS4_B_c                      |                     |          |   |     |          |     |   |     |          |          |   |     |                 |      |   |     |     |      |   |     |     |     |   |     |     |       |   |     |     |     |   |     |       |       |   |     |     |     |    |     |       |          |    |     |     |          |    |     |                     |     |    |     |                               |       |    |     |     |     |    |     |       |       |   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| DQS0_B_c                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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| VSS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| VSS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| DQS1_B_c                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 117                  | 261 | DQS6_B_I, TDQS6_B_I, DBI1_B_n |                     |          |   |     |          |     |   |     |          |          |   |     |                 |      |   |     |     |      |   |     |     |     |   |     |     |       |   |     |     |     |   |     |       |       |   |     |     |     |    |     |       |          |    |     |     |          |    |     |                     |     |    |     |                               |       |    |     |     |     |    |     |       |       |   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| VSS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 139                  | 283 | DQS8_B_I, TDQS8_B_I, DBI3_B_n |                     |          |   |     |          |     |   |     |          |          |   |     |                 |      |   |     |     |      |   |     |     |     |   |     |     |       |   |     |     |     |   |     |       |       |   |     |     |     |    |     |       |          |    |     |     |          |    |     |                     |     |    |     |                               |       |    |     |     |     |    |     |       |       |   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Figure 1. Wiring Assignments

Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 7–8.

The memory slot connector includes a set of power conduits, a set of data conduits, and a set of address and control conduits for transfer of corresponding signals between the system board and the DIMM:

Figure 1. Wiring Assignments

Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 7–8.

The memory slot connector includes a set of power conduits, a set of data conduits, and a set of address and control conduits for transfer of corresponding signals between the system board and the DIMM:

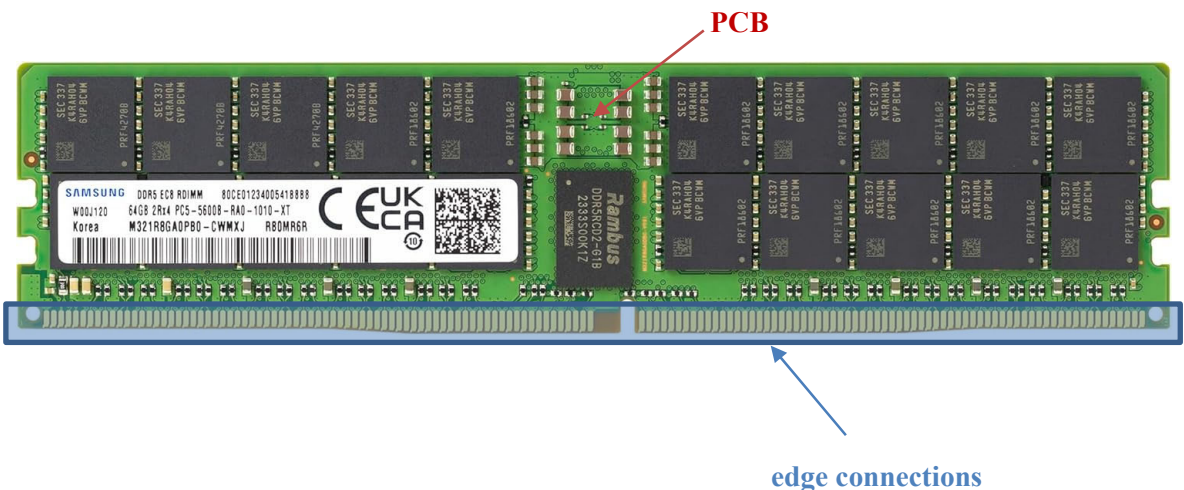
# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------|------------------------|-------------------------------|------------------------|-------------------|----------------|--------------------|------|------------------------------|------|------------------------------------|---------|---------------------------|---------|-------------------------|-------|--------------------------|------|---------------------------------------|------|--------------------------------------|-----|-----------------------------------------|------|-----------------------|
|                        | <p>connected to a set of address and control conduits →</p> <p>[Table 5] PIN Description</p> <table> <tr> <th>Pin Name</th><th>Description</th></tr> <tr> <td>CA[6:0]_A<br/>CA[6:0]_B</td><td>SDRAM Address and Command Bus</td></tr> <tr> <td>CS[1:0]_A<br/>CS[1:0]_B</td><td>SDRAM Chip Select</td></tr> <tr> <td>PAR_A<br/>PAR_B</td><td>SDRAM Parity input</td></tr> <tr> <td>CK_t</td><td>SDRAM Clocks (true/positive)</td></tr> <tr> <td>CK_c</td><td>SDRAM Clocks (complement/negative)</td></tr> <tr> <td>ALERT_n</td><td>SDRAM alert for CRC error</td></tr> <tr> <td>RESET_n</td><td>Set DRAM to known state</td></tr> <tr> <td>PCAMP</td><td>Control and Monitor Port</td></tr> <tr> <td>HSCL</td><td>I2C/I3C-Basic Host Sideband Bus Clock</td></tr> <tr> <td>HSDA</td><td>I2C/I3C-Basic Host Sideband Bus Data</td></tr> <tr> <td>HSA</td><td>I2C/I3C-Basic Host Sideband Bus Address</td></tr> <tr> <td>LBDQ</td><td>Loopback data output:</td></tr> </table> | Pin Name | Description | CA[6:0]_A<br>CA[6:0]_B | SDRAM Address and Command Bus | CS[1:0]_A<br>CS[1:0]_B | SDRAM Chip Select | PAR_A<br>PAR_B | SDRAM Parity input | CK_t | SDRAM Clocks (true/positive) | CK_c | SDRAM Clocks (complement/negative) | ALERT_n | SDRAM alert for CRC error | RESET_n | Set DRAM to known state | PCAMP | Control and Monitor Port | HSCL | I2C/I3C-Basic Host Sideband Bus Clock | HSDA | I2C/I3C-Basic Host Sideband Bus Data | HSA | I2C/I3C-Basic Host Sideband Bus Address | LBDQ | Loopback data output: |
| Pin Name               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| CA[6:0]_A<br>CA[6:0]_B | SDRAM Address and Command Bus                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| CS[1:0]_A<br>CS[1:0]_B | SDRAM Chip Select                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| PAR_A<br>PAR_B         | SDRAM Parity input                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| CK_t                   | SDRAM Clocks (true/positive)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| CK_c                   | SDRAM Clocks (complement/negative)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| ALERT_n                | SDRAM alert for CRC error                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| RESET_n                | Set DRAM to known state                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| PCAMP                  | Control and Monitor Port                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| HSCL                   | I2C/I3C-Basic Host Sideband Bus Clock                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| HSDA                   | I2C/I3C-Basic Host Sideband Bus Data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| HSA                    | I2C/I3C-Basic Host Sideband Bus Address                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| LBDQ                   | Loopback data output:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                        | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------|--------------------------|--------------------------------------|------------------------|-----------------------------------------|------------------------------|------------------------------------------------------------|------------------------------|------------------------------------------------------------|--------------------------------|-----------------------------------------------------------|--------------------------------|-----------------------------------------------------------|----------|---------------------------------------|----------|---------------------------------------|-----|------------------------------|-----|-------------------------|-------|-----------------------------|
|                                | <p>connected to a set of data conduits</p> <p>connected to a set of power conduits</p> <p><i>Id.</i> at 9 (annotated).</p> <table border="1"> <thead> <tr> <th>Pin Name</th><th>Description</th></tr> </thead> <tbody> <tr> <td>DQ[31:0]_A<br/>DQ[31:0]_B</td><td>DIMM memory data bus channel A and B</td></tr> <tr> <td>CB[7:0]_A<br/>CB[7:0]_B</td><td>DIMM ECC check bits(CB) channel A and B</td></tr> <tr> <td>DQS[9:0]_A_t<br/>DQS[9:0]_B_t</td><td>SDRAM data strobes<br/>(positive line of differential pair)</td></tr> <tr> <td>DQS[9:0]_A_c<br/>DQS[9:0]_B_c</td><td>SDRAM data strobes<br/>(negative line of differential pair)</td></tr> <tr> <td>TDQS[9:5]_A_t<br/>TDQS[9:5]_B_t</td><td>Not valid for x4 operation.<br/>Enabled via Mode Register.</td></tr> <tr> <td>TDQS[9:5]_A_c<br/>TDQS[9:5]_B_c</td><td>Not valid for x4 operation.<br/>Enabled via Mode Register.</td></tr> <tr> <td>VIN_BULK</td><td>DIMM Power Supply from system to PMIC</td></tr> <tr> <td>VIN_MGMT</td><td>DIMM Power Supply from system to PMIC</td></tr> <tr> <td>VSS</td><td>Power supply return (ground)</td></tr> <tr> <td>RFU</td><td>Reserved for future use</td></tr> <tr> <td>LBDQS</td><td>Loopback data strobe output</td></tr> </tbody> </table> | Pin Name | Description | DQ[31:0]_A<br>DQ[31:0]_B | DIMM memory data bus channel A and B | CB[7:0]_A<br>CB[7:0]_B | DIMM ECC check bits(CB) channel A and B | DQS[9:0]_A_t<br>DQS[9:0]_B_t | SDRAM data strobes<br>(positive line of differential pair) | DQS[9:0]_A_c<br>DQS[9:0]_B_c | SDRAM data strobes<br>(negative line of differential pair) | TDQS[9:5]_A_t<br>TDQS[9:5]_B_t | Not valid for x4 operation.<br>Enabled via Mode Register. | TDQS[9:5]_A_c<br>TDQS[9:5]_B_c | Not valid for x4 operation.<br>Enabled via Mode Register. | VIN_BULK | DIMM Power Supply from system to PMIC | VIN_MGMT | DIMM Power Supply from system to PMIC | VSS | Power supply return (ground) | RFU | Reserved for future use | LBDQS | Loopback data strobe output |
| Pin Name                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| DQ[31:0]_A<br>DQ[31:0]_B       | DIMM memory data bus channel A and B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| CB[7:0]_A<br>CB[7:0]_B         | DIMM ECC check bits(CB) channel A and B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| DQS[9:0]_A_t<br>DQS[9:0]_B_t   | SDRAM data strobes<br>(positive line of differential pair)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| DQS[9:0]_A_c<br>DQS[9:0]_B_c   | SDRAM data strobes<br>(negative line of differential pair)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| TDQS[9:5]_A_t<br>TDQS[9:5]_B_t | Not valid for x4 operation.<br>Enabled via Mode Register.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| TDQS[9:5]_A_c<br>TDQS[9:5]_B_c | Not valid for x4 operation.<br>Enabled via Mode Register.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| VIN_BULK                       | DIMM Power Supply from system to PMIC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| VIN_MGMT                       | DIMM Power Supply from system to PMIC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| VSS                            | Power supply return (ground)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| RFU                            | Reserved for future use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| LBDQS                          | Loopback data strobe output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |

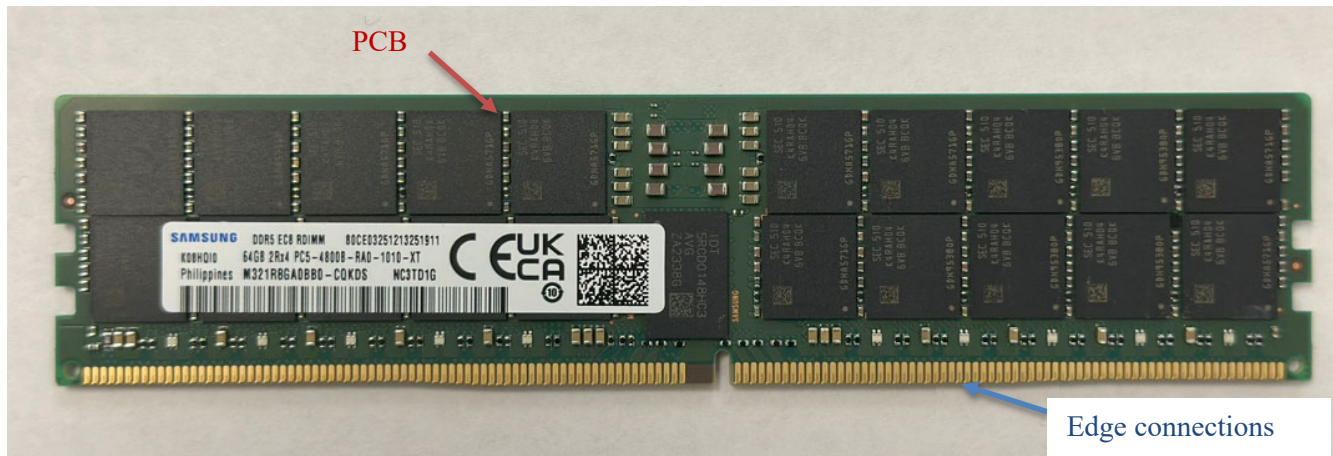
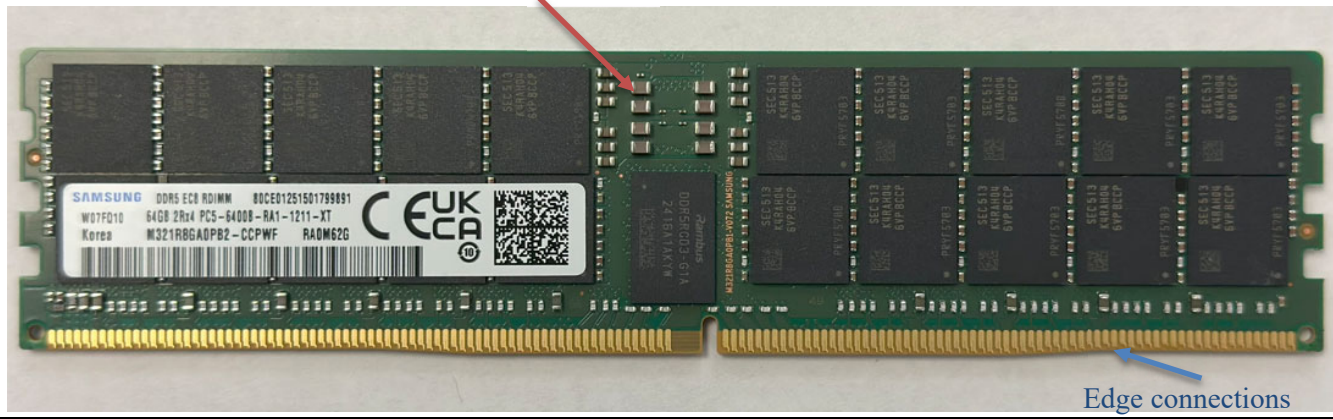
## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                                                                                                                                                     | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>[1.a] a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the system board, the PCB including:</p> | <p>The Accused Instrumentalities each comprise a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the system board. Analysis below uses Samsung's RDIMM format as an example, but MRDIMMs would operate in materially the same way in all relevant part. Ex. CC-13 (JEDEC Press Release) (JEDEC noting that other planned features for MRDIMMs include "Utilization of standard DDR5 DIMM components including DRAM, DIMM Form Factor &amp; Pinout, SPD, PMIC, and TS for ease of adoption"; "Efficient I/O scaling using RCD/DB logic process capability"; and "Leverage existing LRDIMM ecosystem for design and test infrastructure"); Ex. CC-145 (DDR5 MRDIMM: A Transformational Evolution For DDR5 DIMM) ("DDR5 MRDIMM is pin compatible with traditional DIMM, and it can be used in the same DIMM slots that are used for DDR5 RDIMMs").</p> <p>For example, a Samsung DDR5 based DIMM comprises a printed circuit board (PCB) that comprises edge connections, as shown below:</p> <div style="text-align: center;">  </div> <p>Ex. CC-117 (Samsung 5600MHz PC5-44800) (annotated).</p> |

U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p data-bbox="558 451 674 483"><i>See also:</i></p> <div data-bbox="558 459 1812 792"><p data-bbox="1037 459 1094 488">PCB</p><p data-bbox="1614 808 1822 837">Edge connections</p><p>This image shows a Samsung 64GB DDR5 PC5-38400 EC8 RDIMM 2Rx4 M321R8GA0BB0-CQKJ memory module. The module is a long, thin green printed circuit board (PCB) populated with numerous black integrated circuits (chips). A red arrow points to the top surface of the PCB, and a blue arrow points to the gold-plated edge connectors at the bottom. A white label on the left side of the module contains technical specifications and a barcode.</p></div> <p data-bbox="558 862 1772 894">Ex. CC-116 (Samsung 64GB DDR5 PC5-38400 EC8 RDIMM 2Rx4 M321R8GA0BB0-CQK).</p> <div data-bbox="558 1003 1875 1382"><p data-bbox="940 971 997 1000">PCB</p><p data-bbox="1507 1344 1715 1373">Edge connections</p><p>This image shows a Samsung 32GB DDR5 PC5-4800B-RD-1010-XT memory module. It is a green PCB populated with black chips. A red arrow points to the top surface of the PCB, and a blue arrow points to the gold-plated edge connectors at the bottom. A white label on the left side of the module contains technical specifications and a barcode.</p></div> |

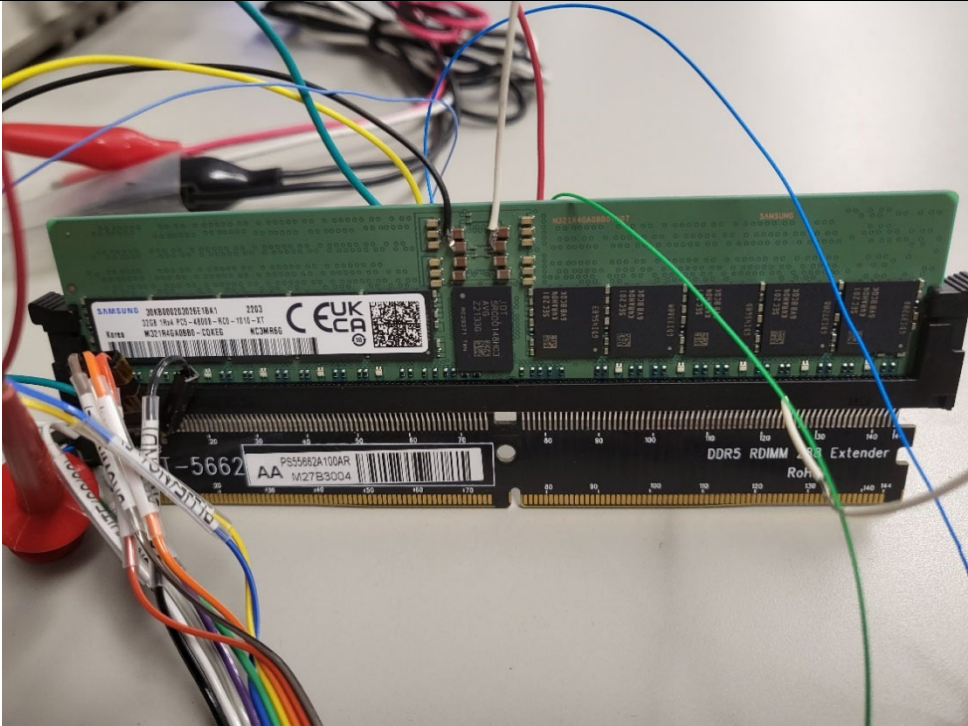
U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

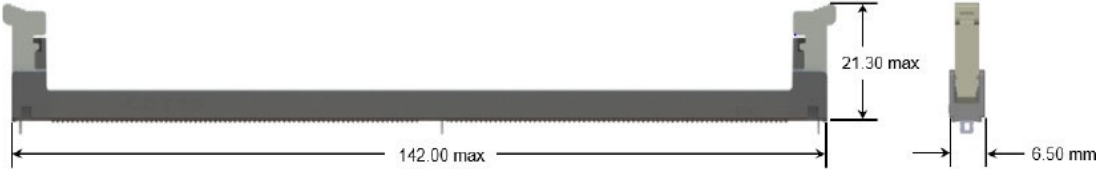
| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p data-bbox="554 305 1759 341">Ex. CC-121 (Samsung 32GB DDR5 PC5-4800 EC8 RDIMM 1Rx4 M321R4GA0BB0-CQK).</p> <div data-bbox="554 371 1879 823"><p data-bbox="898 396 961 423">PCB</p><p data-bbox="1612 781 1822 808">Edge connections</p></div> <p data-bbox="554 833 1751 909">Ex. CC-118 (Samsung 64GB DDR5 PC5-4800 EC8 RDIMM 2Rx4 M321R8GA0BB0-CQK) (annotated).</p> <div data-bbox="554 992 1879 1408"><p data-bbox="1033 959 1096 987">PCB</p><p data-bbox="1627 1373 1837 1401">Edge connections</p></div> |

## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

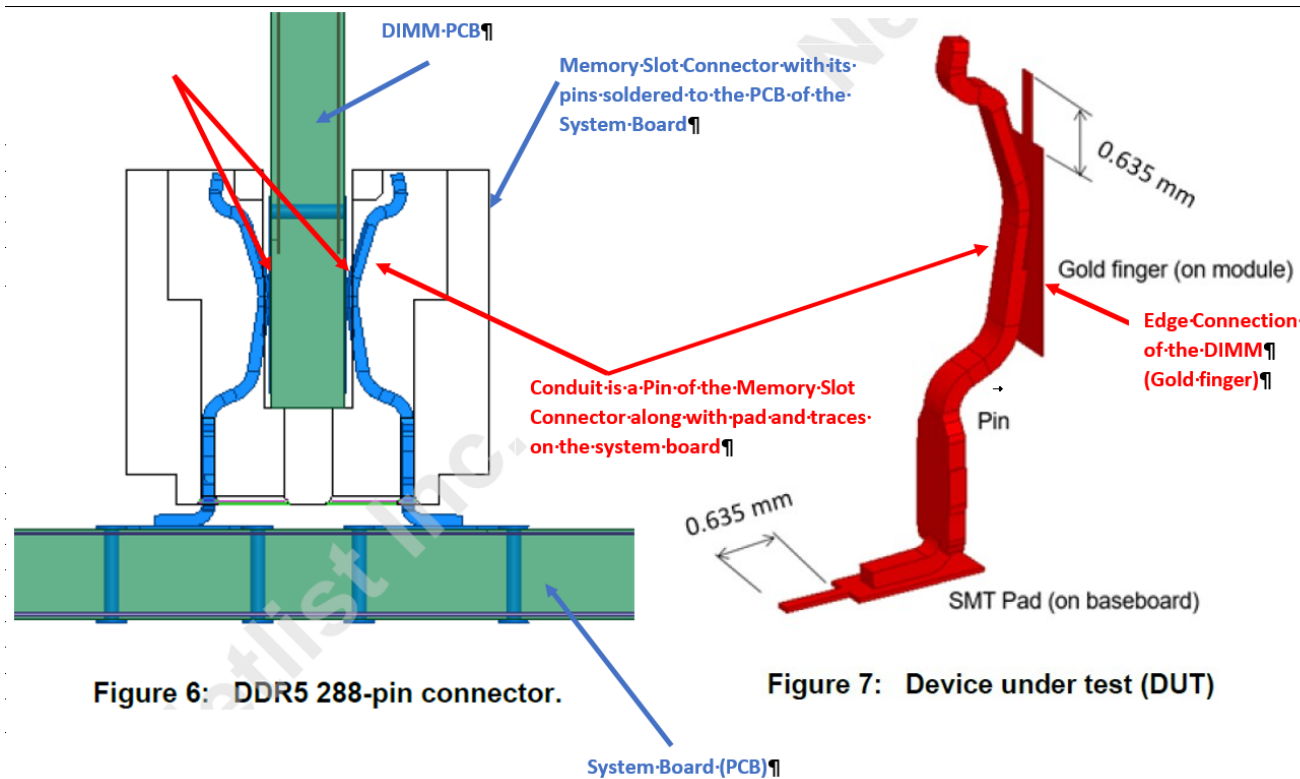
| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p data-bbox="556 305 1743 381">Ex. CC-119 (Samsung 64GB DDR5 PC5-6400 EC8 RDIMM 2Rx4 M321R8GA0PB2-CCP) (annotated).</p> <div data-bbox="556 422 1764 820"><p data-bbox="940 427 999 451">PCB</p><p data-bbox="1413 792 1623 816">Edge connections</p></div> <p data-bbox="556 857 1522 893">Ex. CC-111 (Samsung MRDIMM M327R8GA0EB0-CLVXB) (annotated).</p> <p data-bbox="556 933 1869 1039">The edge connections of the PCB of Samsung's accused DDR5 modules are configured to fit into the memory slot connector of the system board, for example, as illustrated below where a Samsung DDR5 DIMM was inserted into a corresponding memory slot.</p> |

## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <div data-bbox="739 303 1701 1024"></div> <p data-bbox="556 1062 1428 1097"><i>See also</i> Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 93–96.</p> <p data-bbox="556 1136 1791 1256">Additionally, as explained in 1-pre, Samsung’s RDIMMs are “JEDEC standard compliant,” (Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 5) and are therefore configured to fit into a JEDEC-compliant memory slot, as illustrated below.</p> |

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <div data-bbox="930 310 1509 943"></div> <div data-bbox="951 987 1482 1023"><p>Figure 1 DDR5 socket and module (SMT)</p></div> <div data-bbox="676 1073 1766 1240"></div> <div data-bbox="682 1308 1692 1343"><p>Figure 2 DDR5 U/R/LR DIMM Surface Mount (SMT) Connector Socket Outline</p></div> <div data-bbox="560 1378 1005 1412"><p>Ex. CC-73 (JEDEC PS-005B) at 6.</p></div> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p data-bbox="556 310 1848 375"><i>See also id.</i> at 13, Fig. 7 (annotated) (illustrating an edge connection on the DIMM being electrically connected with a corresponding pin on the memory slot connector):</p> <div data-bbox="556 407 1848 1182">  <p>The diagrams illustrate the electrical connection between a DIMM and a system board. Figure 6 shows a cross-section of a DDR5 288-pin connector with labels for the DIMM-PCB, Memory-Slot-Connector with pins soldered to the PCB of the System-Board, and the System-Board (PCB). Figure 7 shows a side view of the Device Under Test (DUT) with labels for the Gold finger (on module), Edge-Connection of the DIMM (Gold-finger), Pin, SMT Pad (on baseboard), and the 0.635 mm spacing. A red arrow points from the edge connection in Figure 7 to the corresponding pin in Figure 6, with a label: 'Conduit is a Pin of the Memory-Slot Connector along with pad and traces on the system-board'.</p> </div> <p data-bbox="556 1195 1354 1227">Ex. CC-73 (JEDEC PS-005B) at 13, Figs. 6 and 7 (annotated).</p> <p data-bbox="556 1268 1879 1409">Moreover, Netlist previously asserted claim 16 of U.S. Patent No. 11,016,918 against Samsung's DDR5 DIMMs, which recites, among other limitations, "a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module</p> |

**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>and the host system.” <i>See, e.g.</i>, Ex. CC-21 (EDTX Trial Tr. Day 2, 4 (2023)) at 333:9–12 (“Q. And can you tell us about claim 16 of the ’918 Patent? A. So here’s claim 16 of the ’918, and a claim analysis is just where I actually look at the claims and compare it to exactly how the products operate, the accused products.”). Netlist also asserted claim 1 of the ’918 patent, which recites, among other limitations, “a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system” and “at least one circuit coupled between a first portion of the plurality of edge connections and the plurality of SDRAM devices, the at least one circuit operable to (i) receive a first plurality of address and control signals via the first portion of the plurality of edge connections, and (ii) output a second plurality of address and control signals to the plurality of SDRAM devices, . . . .” <i>Id.</i> at 342:8–9 (“Next, I turned to claim 1 of the ’918. Now, as we’ll see, claim 1 only applies to the RDIMM products.”).</p> <p>Samsung did not dispute these limitations. <i>Id.</i> at 885:24–886:5 (“With regard to the ’918 claims, I have determined that—that these claims are not infringed by the Samsung DDR5 product, at least for the reason that there is a limitation in each of the independent claims directed to the requirement of a converter circuit structure, and that is not found on any of these Samsung DDR5 products. And what’s being pointed to is a linear drop-out regulator, but that’s not a converter circuit.”).</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                                                                                                                                                                                                                                                                                                                                                           | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>[1.a.1]</b> a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM;</p> | <p>The printed circuit board includes a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from power rails of the system board via the set of power conduits. Additionally, the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM.</p> <p>For example, Samsung illustrates that power is supplied to the DIMM via DIMM slots through the edge connections. <i>See</i> Ex. CC-104 (Optimized DDR5 DIMM Solutions). And this is the only power received by the DIMM.</p> <div data-bbox="930 760 1455 1286" data-label="Diagram"> <p>The diagram, titled "DDR5 DIMM", illustrates the power delivery architecture. On the left, a "Power Supply" block provides two input lines to a "DIMM Slot": "VIN Bulk (12V)" and "VIN MGMT (3.3V)". The "DIMM Slot" is represented by a vertical yellow bar. To the right of the slot is a teal-colored DIMM module. Inside this module, a "PMIC" (Power Management IC) is shown at the top, receiving power from the slot. Below the PMIC are three "DRAM" modules. Arrows indicate the power flow from the PMIC to the DRAM modules, labeled as "VPP", "VDD", and "VDDQ".</p> </div> <p>Ex. CC-104 (Optimized DDR5 DIMM Solutions).</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

## Claim 1

## Evidence of Use

As another example, as the Samsung DDR5 RDIMM Datasheet shows, the pin assignments for the edge connections of the PCB include a first set of edge connections (Vin\_Bulk, Vin\_Mgmt, and VSS) configured to be connected to power conduits.

| Front Side<br>Pin Label | Pin | Pin | Back side<br>Pin Label |
|-------------------------|-----|-----|------------------------|
| VIN_BULK                | 1   | 145 | VIN_BULK               |
| RDIM                    | 2   | 146 | VIN_BULK               |
| VIN_MGMT                | 3   | 147 | PWR_GOOD/FAIL_n        |
| HSC_L                   | 4   | 148 | HSA                    |
| HSD_A                   | 5   | 149 | RFU                    |
| VSS                     | 6   | 150 | RFU                    |
| DQ0_A                   | 7   | 151 | VSS                    |
| VSS                     | 8   | 152 | DQ2_A                  |
| DQ1_A                   | 9   | 153 | VSS                    |
| VSS                     | 10  | 154 | DQ3_A                  |
| DQS0_A_I                | 11  | 155 | VSS                    |
| DQS0_A_C                | 12  | 156 | DQS5_A_C, TDQS5_A_C    |
| VSS                     | 13  | 157 | DQS5_A_I, TDQS5_A_I    |
| DQ4_A                   | 14  | 158 | VSS                    |
| VSS                     | 15  | 159 | DQ6_A                  |
| DQ5_A                   | 16  | 160 | VSS                    |
| VSS                     | 17  | 161 | DQ7_A                  |
| DQ6_A                   | 18  | 162 | VSS                    |
| VSS                     | 19  | 163 | DQ10_A                 |
| DQ9_A                   | 20  | 164 | VSS                    |
| VSS                     | 21  | 165 | DQ11_A                 |
| DQS1_A_I                | 22  | 166 | VSS                    |
| DQS1_A_C                | 23  | 167 | DQS6_A_C, TDQS6_A_C    |
| VSS                     | 24  | 168 | DQS6_A_I, TDQS6_A_I    |
| DQ12_A                  | 25  | 169 | VSS                    |
| VSS                     | 26  | 170 | DQ14_A                 |
| DQ13_A                  | 27  | 171 | VSS                    |
| VSS                     | 28  | 172 | DQ15_A                 |
| DQ16_A                  | 29  | 173 | VSS                    |
| VSS                     | 30  | 174 | DQ18_A                 |
| DQ17_A                  | 31  | 175 | VSS                    |
| VSS                     | 32  | 176 | DQ19_A                 |
| DQS2_A_I                | 33  | 177 | VSS                    |
| DQS2_A_C                | 34  | 178 | DQS7_A_C, TDQS7_A_C    |
| VSS                     | 35  | 179 | DQS7_A_I, TDQS7_A_I    |
| DQ20_A                  | 36  | 180 | VSS                    |
| VSS                     | 37  | 181 | DQ22_A                 |
| DQ21_A                  | 38  | 182 | VSS                    |

| Front Side<br>Pin Label | Pin | Pin | Back side<br>Pin Label |
|-------------------------|-----|-----|------------------------|
| PAR_A                   | 74  | 218 | CK_C                   |
| VSS                     | 75  | 219 | VSS                    |
| Key                     |     |     |                        |
| CA0_B                   | 76  | 220 | RFU                    |
| VSS                     | 77  | 221 | CA1_B                  |
| CA2_B                   | 78  | 222 | VSS                    |
| VSS                     | 79  | 223 | CA3_B                  |
| CA4_B                   | 80  | 224 | VSS                    |
| VSS                     | 81  | 225 | CA5_B                  |
| CA6_B                   | 82  | 226 | VSS                    |
| VSS                     | 83  | 227 | PAR_B                  |
| CS0_B_n                 | 84  | 228 | VSS                    |
| VSS                     | 85  | 229 | CS1_B_n                |
| LBD, RSP_A_n            | 86  | 230 | VSS                    |
| LBS, RSP_B_n            | 87  | 231 | RFU                    |
| VSS                     | 88  | 232 | RFU                    |
| CB4_B                   | 89  | 233 | VSS                    |
| VSS                     | 90  | 234 | CB6_B                  |
| CB5_B                   | 91  | 235 | VSS                    |
| VSS                     | 92  | 236 | CB7_B                  |
| DQS9_B_I, TDQS9_B_I     | 93  | 237 | VSS                    |
| DQS9_B_C, TDQS9_B_C     | 94  | 238 | DQS4_B_C               |
| VSS                     | 95  | 239 | DQS4_B_I               |
| CB0_B                   | 96  | 240 | VSS                    |
| VSS                     | 97  | 241 | CB2_B                  |
| CB1_B                   | 98  | 242 | VSS                    |
| VSS                     | 99  | 243 | CB3_B                  |
| DQ0_B                   | 100 | 244 | VSS                    |
| VSS                     | 101 | 245 | DQ2_B                  |
| DQ1_B                   | 102 | 246 | VSS                    |
| VSS                     | 103 | 247 | DQ3_BDQ3_B             |
| DQS0_B_I                | 104 | 248 | VSS                    |
| DQS0_B_C                | 105 | 249 | DQS5_B_C, TDQS5_B_C    |
| VSS                     | 106 | 250 | DQS5_B_I, TDQS5_B_I    |
| DQ4_B                   | 107 | 251 | VSS                    |
| VSS                     | 108 | 252 | DQ6_B                  |
| DQ5_B                   | 109 | 253 | VSS                    |

| Front Side<br>Pin Label | Pin | Pin | Back side<br>Pin Label |
|-------------------------|-----|-----|------------------------|
| VSS                     | 39  | 183 | DQ23_A                 |
| DQ24_A                  | 40  | 184 | VSS                    |
| VSS                     | 41  | 185 | DQ26_A                 |
| DQ25_A                  | 42  | 186 | VSS                    |
| VSS                     | 43  | 187 | DQ27_A                 |
| DQS3_A_I                | 44  | 188 | VSS                    |
| DQS3_A_C                | 45  | 189 | DQS8_A_C, TDQS8_A_C    |
| VSS                     | 46  | 190 | DQS8_A_I, TDQS8_A_I    |
| DQ28_A                  | 47  | 191 | VSS                    |
| VSS                     | 48  | 192 | DQ30_A                 |
| DQ29_A                  | 49  | 193 | VSS                    |
| VSS                     | 50  | 194 | DQ31_A                 |
| CB0_A                   | 51  | 195 | VSS                    |
| VSS                     | 52  | 196 | CB2_A                  |
| CB1_A                   | 53  | 197 | VSS                    |
| VSS                     | 54  | 198 | CB3_A                  |
| DQS4_A_I                | 55  | 199 | VSS                    |
| DQS4_A_C                | 56  | 200 | DQS9_A_C, TDQS9_A_C    |
| VSS                     | 57  | 201 | DQS9_A_I, TDQS9_A_I    |
| CB4_A                   | 58  | 202 | VSS                    |
| VSS                     | 59  | 203 | CB6_A                  |
| CB5_A                   | 60  | 204 | VSS                    |
| VSS                     | 61  | 205 | CB7_A                  |
| ALEFT_n                 | 62  | 206 | VSS                    |
| VSS                     | 63  | 207 | RESET_n                |
| CS0_A_n                 | 64  | 208 | VSS                    |
| VSS                     | 65  | 209 | CS1_A_n                |
| CA0_A                   | 66  | 210 | VSS                    |
| VSS                     | 67  | 211 | CA1_A                  |
| CA2                     | 68  | 212 | VSS                    |
| VSS                     | 69  | 213 | CA3_A                  |
| CA4_A                   | 70  | 214 | VSS                    |
| VSS                     | 71  | 215 | CA5_A                  |
| CA6_A                   | 72  | 216 | VSS                    |
| VSS                     | 73  | 217 | CK_I                   |

| Front Side<br>Pin Label | Pin | Pin | Back side<br>Pin Label |
|-------------------------|-----|-----|------------------------|
| VSS                     | 110 | 254 | DQ7_B                  |
| DQ8_B                   | 111 | 255 | VSS                    |
| VSS                     | 112 | 256 | DQ10_B                 |
| DQ9_B                   | 113 | 257 | VSS                    |
| VSS                     | 114 | 258 | DQ11_B                 |
| DQS1_B_I                | 115 | 259 | VSS                    |
| DQS1_B_C                | 116 | 260 | DQS6_B_C, TDQS6_B_C    |
| VSS                     | 117 | 261 | DQS6_B_I, TDQS6_B_I    |
| DQ12_B                  | 118 | 262 | VSS                    |
| VSS                     | 119 | 263 | DQ14_B                 |
| DQ13_B                  | 120 | 264 | VSS                    |
| VSS                     | 121 | 265 | DQ15_B                 |
| DQ16_B                  | 122 | 266 | VSS                    |
| VSS                     | 123 | 267 | DQ18_B                 |
| DQ17_B                  | 124 | 268 | VSS                    |
| VSS                     | 125 | 269 | DQ19_B                 |
| DQS2_B_I                | 126 | 270 | VSS                    |
| DQS2_B_C                | 127 | 271 | DQS7_B_C, TDQS7_B_C    |
| VSS                     | 128 | 272 | DQS7_B_I, TDQS7_B_I    |
| DQ20_B                  | 129 | 273 | VSS                    |
| VSS                     | 130 | 274 | DQ22_B                 |
| DQ21_B                  | 131 | 275 | VSS                    |
| VSS                     | 132 | 276 | DQ23_B                 |
| DQ24_B                  | 133 | 277 | VSS                    |
| VSS                     | 134 | 278 | DQ26_B                 |
| DQ25_B                  | 135 | 279 | VSS                    |
| VSS                     | 136 | 280 | DQ27_B                 |
| DQS3_B_I                | 137 | 281 | VSS                    |
| DQS3_B_C                | 138 | 282 | DQS8_B_C, TDQS8_B_C    |
| VSS                     | 139 | 283 | DQS8_B_I, TDQS8_B_I    |
| DQ28_B                  | 140 | 284 | VSS                    |
| VSS                     | 141 | 285 | DQ30_B                 |
| DQ29_B                  | 142 | 286 | VSS                    |
| VSS                     | 143 | 287 | DQ31_B                 |
| RFU                     | 144 | 288 | VSS                    |

Figure 1. Wiring Assignments

Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 7–8, Fig. 1 (only a few of the power pins are annotated).

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                                                                                                     | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |                                                     |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------------------------------------|--------------------------|--------------------------------------|------------------------|-----------------------------------------|------------------------------|------------------------------------------------------------|------------------------------|------------------------------------------------------------|--------------------------------|-----------------------------------------------------------|--------------------------------|-----------------------------------------------------------|----------|---------------------------------------|----------|---------------------------------------|-----|------------------------------|-----|-------------------------|-------|-----------------------------|
| <p>a first set of edge connections configured to be electrically connected to the set of power conduits</p> | <table><thead><tr><th>Pin Name</th><th>Description</th></tr></thead><tbody><tr><td>DQ[31:0]_A<br/>DQ[31:0]_B</td><td>DIMM memory data bus channel A and B</td></tr><tr><td>CB[7:0]_A<br/>CB[7:0]_B</td><td>DIMM ECC check bits(CB) channel A and B</td></tr><tr><td>DQS[9:0]_A_t<br/>DQS[9:0]_B_t</td><td>SDRAM data strobes<br/>(positive line of differential pair)</td></tr><tr><td>DQS[9:0]_A_c<br/>DQS[9:0]_B_c</td><td>SDRAM data strobes<br/>(negative line of differential pair)</td></tr><tr><td>TDQS[9:5]_A_t<br/>TDQS[9:5]_B_t</td><td>Not valid for x4 operation.<br/>Enabled via Mode Register.</td></tr><tr><td>TDQS[9:5]_A_c<br/>TDQS[9:5]_B_c</td><td>Not valid for x4 operation.<br/>Enabled via Mode Register.</td></tr><tr><td>VIN_BULK</td><td>DIMM Power Supply from system to PMIC</td></tr><tr><td>VIN_MGMT</td><td>DIMM Power Supply from system to PMIC</td></tr><tr><td>VSS</td><td>Power supply return (ground)</td></tr><tr><td>RFU</td><td>Reserved for future use</td></tr><tr><td>LBDQS</td><td>Loopback data strobe output</td></tr></tbody></table> | Pin Name   | Description                                         | DQ[31:0]_A<br>DQ[31:0]_B | DIMM memory data bus channel A and B | CB[7:0]_A<br>CB[7:0]_B | DIMM ECC check bits(CB) channel A and B | DQS[9:0]_A_t<br>DQS[9:0]_B_t | SDRAM data strobes<br>(positive line of differential pair) | DQS[9:0]_A_c<br>DQS[9:0]_B_c | SDRAM data strobes<br>(negative line of differential pair) | TDQS[9:5]_A_t<br>TDQS[9:5]_B_t | Not valid for x4 operation.<br>Enabled via Mode Register. | TDQS[9:5]_A_c<br>TDQS[9:5]_B_c | Not valid for x4 operation.<br>Enabled via Mode Register. | VIN_BULK | DIMM Power Supply from system to PMIC | VIN_MGMT | DIMM Power Supply from system to PMIC | VSS | Power supply return (ground) | RFU | Reserved for future use | LBDQS | Loopback data strobe output |
| Pin Name                                                                                                    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |                                                     |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| DQ[31:0]_A<br>DQ[31:0]_B                                                                                    | DIMM memory data bus channel A and B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |                                                     |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| CB[7:0]_A<br>CB[7:0]_B                                                                                      | DIMM ECC check bits(CB) channel A and B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |                                                     |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| DQS[9:0]_A_t<br>DQS[9:0]_B_t                                                                                | SDRAM data strobes<br>(positive line of differential pair)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |                                                     |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| DQS[9:0]_A_c<br>DQS[9:0]_B_c                                                                                | SDRAM data strobes<br>(negative line of differential pair)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |                                                     |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| TDQS[9:5]_A_t<br>TDQS[9:5]_B_t                                                                              | Not valid for x4 operation.<br>Enabled via Mode Register.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                                                     |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| TDQS[9:5]_A_c<br>TDQS[9:5]_B_c                                                                              | Not valid for x4 operation.<br>Enabled via Mode Register.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                                                     |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| VIN_BULK                                                                                                    | DIMM Power Supply from system to PMIC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |                                                     |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| VIN_MGMT                                                                                                    | DIMM Power Supply from system to PMIC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |                                                     |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| VSS                                                                                                         | Power supply return (ground)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |                                                     |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| RFU                                                                                                         | Reserved for future use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |                                                     |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| LBDQS                                                                                                       | Loopback data strobe output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |                                                     |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
|                                                                                                             | <p>Id. at 9, Table 5 (annotated).</p> <table><thead><tr><th>Symbol</th><th>Type</th><th>I/O Levels</th><th>Function</th></tr></thead><tbody><tr><td>VIN_BULK</td><td>Supply</td><td></td><td>External power supply: 12V, 4.25V (min), 15V (max)</td></tr><tr><td>VIN_MGMT</td><td>Supply</td><td></td><td>External power supply: 3.3V, 3.0V (min), 3.6V (max)</td></tr><tr><td>VSS</td><td>Supply</td><td></td><td>Ground</td></tr></tbody></table> <p>Id. at 12.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Symbol     | Type                                                | I/O Levels               | Function                             | VIN_BULK               | Supply                                  |                              | External power supply: 12V, 4.25V (min), 15V (max)         | VIN_MGMT                     | Supply                                                     |                                | External power supply: 3.3V, 3.0V (min), 3.6V (max)       | VSS                            | Supply                                                    |          | Ground                                |          |                                       |     |                              |     |                         |       |                             |
| Symbol                                                                                                      | Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | I/O Levels | Function                                            |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| VIN_BULK                                                                                                    | Supply                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            | External power supply: 12V, 4.25V (min), 15V (max)  |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| VIN_MGMT                                                                                                    | Supply                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            | External power supply: 3.3V, 3.0V (min), 3.6V (max) |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| VSS                                                                                                         | Supply                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            | Ground                                              |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |

## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <div data-bbox="583 316 1354 633" data-label="Diagram"> </div> <p data-bbox="556 690 1879 803"><i>Id.</i> at 10 (showing inputs from VIN_BULK and VIN_MGMT from the board edge to PMIC and then output from PMIC to DIMM components); <i>see also id.</i> at 36, 32, 28, 24. As shown above, the only inputs to the PMIC are VIN_BULK and VIN_MGMT received from the host via the edge connections.</p> <p data-bbox="556 836 1879 1421">Moreover, Netlist previously asserted claim 16 of U.S. Patent No. 11,016,918 against Samsung’s DDR5 DIMMs, which recites, among other limitations, “a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system.” <i>See, e.g.,</i> Ex. CC-21 (EDTX Trial Tr. Day 2, 4 (2023)) at 333:9–12 (“Q. And can you tell us about claim 16 of the ’918 Patent? A. So here’s claim 16 of the ’918, and a claim analysis is just where I actually look at the claims and compare it to exactly how the products operate, the accused products.”). Netlist also asserted claim 1 of the ’918 patent, which recites, among other limitations, “a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system” and “at least one circuit coupled between a first portion of the plurality of edge connections and the plurality of SDRAM devices, the at least one circuit operable to (i) receive a first plurality of address and control signals via the first portion of the plurality of edge connections, and (ii) output a second plurality of address and control signals to the plurality of SD RAM devices, . . . .” <i>Id.</i> at 342:8–9 (“Next, I turned to claim 1 of the ’918. Now, as we’ll see, claim 1 only applies to the RDIMM products.”).</p> |

**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>Samsung did not dispute these limitations. <i>Id.</i> at 885:24–886:5 (“With regard to the ’918 claims, I have determined that—that these claims are not infringed by the Samsung DDR5 product, at least for the reason that there is a limitation in each of the independent claims directed to the requirement of a converter circuit structure, and that is not found on any of these Samsung DDR5 products. And what’s being pointed to is a linear drop-out regulator, but that’s not a converter circuit.”).</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                                                                                                                         | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------|--------------------------|--------------------------------------|------------------------|-----------------------------------------|------------------------------|------------------------------------------------------------|------------------------------|------------------------------------------------------------|--------------------------------|-----------------------------------------------------------|--------------------------------|-----------------------------------------------------------|----------|---------------------------------------|----------|---------------------------------------|-----|------------------------------|-----|-------------------------|-------|-----------------------------|
| <p><b>[1.a.2]</b> a second set of edge connections configured to be electrically connected to the set of data conduits, and</p> | <p>The Accused Instrumentalities each comprise a printed circuit board (PCB) having a second set of edge connections, such as DQ[31:0]_A, DQ[31:0]_B, CB[7:0]_A, and CB[7:0]_B configured to be electrically connected to the set of data conduits.</p> <p>For example, the Samsung DDR5 RDIMM Datasheet shows pin assignment for edge connections, which include pins for data transmission, such as DQ[31:0]_A, DQ[31:0]_B, CB[7:0]_A, and CB[7:0]_B configured to be connected to data conduits:</p> <div style="display: flex; align-items: center; margin-top: 20px;"> <div style="flex: 1;"> <p>a second set of edge connections configured to be electrically connected to the set of data conduits</p>  </div> <table border="1"> <thead> <tr> <th>Pin Name</th><th>Description</th></tr> </thead> <tbody> <tr> <td>DQ[31:0]_A<br/>DQ[31:0]_B</td><td>DIMM memory data bus channel A and B</td></tr> <tr> <td>CB[7:0]_A<br/>CB[7:0]_B</td><td>DIMM ECC check bits(CB) channel A and B</td></tr> <tr> <td>DQS[9:0]_A_t<br/>DQS[9:0]_B_t</td><td>SDRAM data strobes<br/>(positive line of differential pair)</td></tr> <tr> <td>DQS[9:0]_A_c<br/>DQS[9:0]_B_c</td><td>SDRAM data strobes<br/>(negative line of differential pair)</td></tr> <tr> <td>TDQS[9:5]_A_t<br/>TDQS[9:5]_B_t</td><td>Not valid for x4 operation.<br/>Enabled via Mode Register.</td></tr> <tr> <td>TDQS[9:5]_A_c<br/>TDQS[9:5]_B_c</td><td>Not valid for x4 operation.<br/>Enabled via Mode Register.</td></tr> <tr> <td>VIN_BULK</td><td>DIMM Power Supply from system to PMIC</td></tr> <tr> <td>VIN_MGMT</td><td>DIMM Power Supply from system to PMIC</td></tr> <tr> <td>VSS</td><td>Power supply return (ground)</td></tr> <tr> <td>RFU</td><td>Reserved for future use</td></tr> <tr> <td>LBDQS</td><td>Loopback data strobe output</td></tr> </tbody> </table> </div> <p>Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 9, Table 5 (annotated).</p> | Pin Name | Description | DQ[31:0]_A<br>DQ[31:0]_B | DIMM memory data bus channel A and B | CB[7:0]_A<br>CB[7:0]_B | DIMM ECC check bits(CB) channel A and B | DQS[9:0]_A_t<br>DQS[9:0]_B_t | SDRAM data strobes<br>(positive line of differential pair) | DQS[9:0]_A_c<br>DQS[9:0]_B_c | SDRAM data strobes<br>(negative line of differential pair) | TDQS[9:5]_A_t<br>TDQS[9:5]_B_t | Not valid for x4 operation.<br>Enabled via Mode Register. | TDQS[9:5]_A_c<br>TDQS[9:5]_B_c | Not valid for x4 operation.<br>Enabled via Mode Register. | VIN_BULK | DIMM Power Supply from system to PMIC | VIN_MGMT | DIMM Power Supply from system to PMIC | VSS | Power supply return (ground) | RFU | Reserved for future use | LBDQS | Loopback data strobe output |
| Pin Name                                                                                                                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| DQ[31:0]_A<br>DQ[31:0]_B                                                                                                        | DIMM memory data bus channel A and B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| CB[7:0]_A<br>CB[7:0]_B                                                                                                          | DIMM ECC check bits(CB) channel A and B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| DQS[9:0]_A_t<br>DQS[9:0]_B_t                                                                                                    | SDRAM data strobes<br>(positive line of differential pair)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| DQS[9:0]_A_c<br>DQS[9:0]_B_c                                                                                                    | SDRAM data strobes<br>(negative line of differential pair)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| TDQS[9:5]_A_t<br>TDQS[9:5]_B_t                                                                                                  | Not valid for x4 operation.<br>Enabled via Mode Register.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| TDQS[9:5]_A_c<br>TDQS[9:5]_B_c                                                                                                  | Not valid for x4 operation.<br>Enabled via Mode Register.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| VIN_BULK                                                                                                                        | DIMM Power Supply from system to PMIC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| VIN_MGMT                                                                                                                        | DIMM Power Supply from system to PMIC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| VSS                                                                                                                             | Power supply return (ground)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| RFU                                                                                                                             | Reserved for future use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| LBDQS                                                                                                                           | Loopback data strobe output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |

## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

Claim 1

Evidence of Use

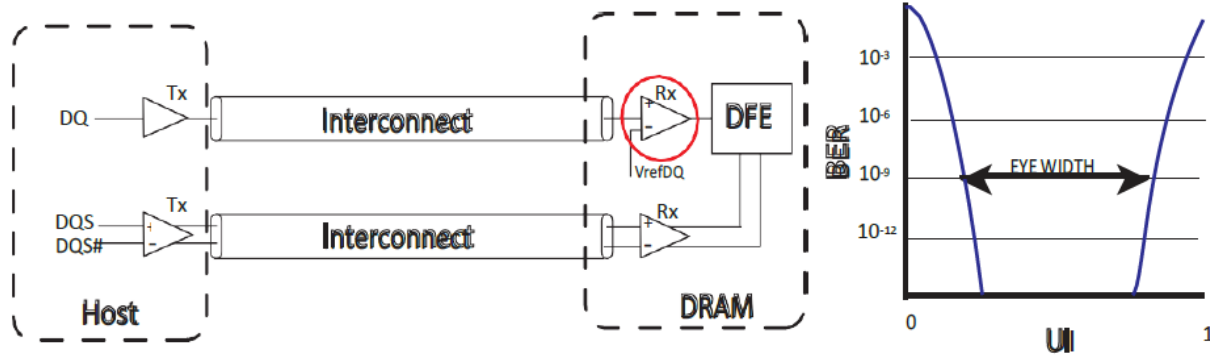
|                          |                  |     |                                                                                                                                                                                                                                                                                                                   |
|--------------------------|------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DQ[31:0]_A<br>DQ[31:0]_B | Input/<br>Output | VDD | Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used. |
| CB[7:0]_A<br>CB[7:0]_B   | Input/<br>Output | VDD | ECC Check Bits Input/ Output: Bi-directional data bus.                                                                                                                                                                                                                                                            |

Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 11.

| <table><tr><th>Front Side<br/>Pin Label</th><th>Pin</th><th>Pin</th><th>Back side<br/>Pin Label</th></tr><tr><td>VIN_BULK</td><td>1</td><td>145</td><td>VIN_BULK</td></tr><tr><td>RFU</td><td>2</td><td>146</td><td>VIN_BULK</td></tr><tr><td>VIN_MGMT</td><td>3</td><td>147</td><td>PWR_GOOD/FAIL_n</td></tr><tr><td>HSC_L</td><td>4</td><td>148</td><td>HSA</td></tr><tr><td>HSD_A</td><td>5</td><td>149</td><td>RFU</td></tr><tr><td>VSS</td><td>6</td><td>150</td><td>RFU</td></tr><tr><td>DQ0_A</td><td>7</td><td>151</td><td>VSS</td></tr><tr><td>VSS</td><td>8</td><td>152</td><td>DQ2_A</td></tr><tr><td>DQ1_A</td><td>9</td><td>153</td><td>VSS</td></tr><tr><td>VSS</td><td>10</td><td>154</td><td>DQ3_A</td></tr><tr><td>DQS0_A_I</td><td>11</td><td>155</td><td>VSS</td></tr><tr><td>DQS0_A_C</td><td>12</td><td>156</td><td>DQS5_A_c, TDQS5_A_c</td></tr><tr><td>VSS</td><td>13</td><td>157</td><td>DQS5_A_I, TDQS5_A_I<br/>DBI0_A_n</td></tr><tr><td>DQ4_A</td><td>14</td><td>158</td><td>VSS</td></tr><tr><td>VSS</td><td>15</td><td>159</td><td>DQ6_A</td></tr><tr><td>DQ5_A</td><td>16</td><td>160</td><td>VSS</td></tr><tr><td>VSS</td><td>17</td><td>161</td><td>DQ7_A</td></tr><tr><td>DQ8_A</td><td>18</td><td>162</td><td>VSS</td></tr><tr><td>VSS</td><td>19</td><td>163</td><td>DQ10_A</td></tr><tr><td>DQ9_A</td><td>20</td><td>164</td><td>VSS</td></tr><tr><td>VSS</td><td>21</td><td>165</td><td>DQ11_A</td></tr><tr><td>DQS1_A_I</td><td>22</td><td>166</td><td>VSS</td></tr><tr><td>DQS1_A_C</td><td>23</td><td>167</td><td>DQS6_A_c, TDQS6_A_c</td></tr><tr><td>VSS</td><td>24</td><td>168</td><td>DQS6_A_I, TDQS6_A_I<br/>DBI1_A_n</td></tr><tr><td>DQ12_A</td><td>25</td><td>169</td><td>VSS</td></tr><tr><td>VSS</td><td>26</td><td>170</td><td>DQ14_A</td></tr><tr><td>DQ13_A</td><td>27</td><td>171</td><td>VSS</td></tr><tr><td>VSS</td><td>28</td><td>172</td><td>DQ15_A</td></tr><tr><td>DQ16_A</td><td>29</td><td>173</td><td>VSS</td></tr><tr><td>VSS</td><td>30</td><td>174</td><td>DQ18_A</td></tr><tr><td>DQ17_A</td><td>31</td><td>175</td><td>VSS</td></tr><tr><td>VSS</td><td>32</td><td>176</td><td>DQ19_A</td></tr><tr><td>DQS2_A_I</td><td>33</td><td>177</td><td>VSS</td></tr><tr><td>DQS2_A_C</td><td>34</td><td>178</td><td>DQS7_A_c, TDQS7_A_c</td></tr><tr><td>VSS</td><td>35</td><td>179</td><td>DQS7_A_I, TDQS7_A_I<br/>DBI2_n</td></tr><tr><td>DQ20_A</td><td>36</td><td>180</td><td>VSS</td></tr><tr><td>VSS</td><td>37</td><td>181</td><td>DQ22_A</td></tr><tr><td>DQ21_A</td><td>38</td><td>182</td><td>VSS</td></tr></table> | Front Side<br>Pin Label | Pin | Pin                             | Back side<br>Pin Label | VIN_BULK | 1 | 145 | VIN_BULK | RFU | 2 | 146 | VIN_BULK | VIN_MGMT | 3 | 147 | PWR_GOOD/FAIL_n | HSC_L | 4 | 148 | HSA | HSD_A | 5 | 149 | RFU | VSS | 6 | 150 | RFU | DQ0_A | 7 | 151 | VSS | VSS | 8 | 152 | DQ2_A | DQ1_A | 9 | 153 | VSS | VSS | 10 | 154 | DQ3_A | DQS0_A_I | 11 | 155 | VSS | DQS0_A_C | 12 | 156 | DQS5_A_c, TDQS5_A_c | VSS | 13 | 157 | DQS5_A_I, TDQS5_A_I<br>DBI0_A_n | DQ4_A | 14 | 158 | VSS | VSS | 15 | 159 | DQ6_A | DQ5_A | 16 | 160 | VSS | VSS | 17 | 161 | DQ7_A | DQ8_A | 18 | 162 | VSS | VSS | 19 | 163 | DQ10_A | DQ9_A | 20 | 164 | VSS | VSS | 21 | 165 | DQ11_A | DQS1_A_I | 22 | 166 | VSS | DQS1_A_C | 23 | 167 | DQS6_A_c, TDQS6_A_c | VSS | 24 | 168 | DQS6_A_I, TDQS6_A_I<br>DBI1_A_n | DQ12_A | 25 | 169 | VSS | VSS | 26 | 170 | DQ14_A | DQ13_A | 27 | 171 | VSS | VSS | 28 | 172 | DQ15_A | DQ16_A | 29 | 173 | VSS | VSS | 30 | 174 | DQ18_A | DQ17_A | 31 | 175 | VSS | VSS | 32 | 176 | DQ19_A | DQS2_A_I | 33 | 177 | VSS | DQS2_A_C | 34 | 178 | DQS7_A_c, TDQS7_A_c | VSS | 35 | 179 | DQS7_A_I, TDQS7_A_I<br>DBI2_n | DQ20_A | 36 | 180 | VSS | VSS | 37 | 181 | DQ22_A | DQ21_A | 38 | 182 | VSS | <table><tr><th>Front Side<br/>Pin Label</th><th>Pin</th><th>Pin</th><th>Back side<br/>Pin Label</th></tr><tr><td>PAR_A</td><td>74</td><td>218</td><td>CK_C</td></tr><tr><td>VSS</td><td>75</td><td>219</td><td>VSS</td></tr><tr><td colspan="4">Key</td></tr><tr><td>CA0_B</td><td>76</td><td>220</td><td>RFU</td></tr><tr><td>VSS</td><td>77</td><td>221</td><td>CA1_B</td></tr><tr><td>CA2_B</td><td>78</td><td>222</td><td>VSS</td></tr><tr><td>VSS</td><td>79</td><td>223</td><td>CA3_B</td></tr><tr><td>CA4_B</td><td>80</td><td>224</td><td>VSS</td></tr><tr><td>VSS</td><td>81</td><td>225</td><td>CA5_B</td></tr><tr><td>CA6_B</td><td>82</td><td>226</td><td>VSS</td></tr><tr><td>VSS</td><td>83</td><td>227</td><td>PAR_B</td></tr><tr><td>CS0_B_n</td><td>84</td><td>228</td><td>VSS</td></tr><tr><td>VSS</td><td>85</td><td>229</td><td>CS1_B_n</td></tr><tr><td>LBD, RSP_A_n</td><td>86</td><td>230</td><td>VSS</td></tr><tr><td>LBS, RSP_B_n</td><td>87</td><td>231</td><td>RFU</td></tr><tr><td>VSS</td><td>88</td><td>232</td><td>RFU</td></tr><tr><td>CB4_B</td><td>89</td><td>233</td><td>VSS</td></tr><tr><td>VSS</td><td>90</td><td>234</td><td>CB6_B</td></tr><tr><td>CB5_B</td><td>91</td><td>235</td><td>VSS</td></tr><tr><td>VSS</td><td>92</td><td>236</td><td>CB7_B</td></tr><tr><td>DQS9_B_I, TDQS9_B_I<br/>DBI4_B_n</td><td>93</td><td>237</td><td>VSS</td></tr><tr><td>DQS9_B_C, TDQS9_B_C</td><td>94</td><td>238</td><td>DQS4_B_c</td></tr><tr><td>VSS</td><td>95</td><td>239</td><td>DQS4_B_I</td></tr><tr><td>CB0_B</td><td>96</td><td>240</td><td>VSS</td></tr><tr><td>VSS</td><td>97</td><td>241</td><td>CB2_B</td></tr><tr><td>CB1_B</td><td>98</td><td>242</td><td>VSS</td></tr><tr><td>VSS</td><td>99</td><td>243</td><td>CB3_B</td></tr><tr><td>DQ0_B</td><td>100</td><td>244</td><td>VSS</td></tr><tr><td>VSS</td><td>101</td><td>245</td><td>DQ2_B</td></tr><tr><td>DQ1_B</td><td>102</td><td>246</td><td>VSS</td></tr><tr><td>VSS</td><td>103</td><td>247</td><td>DQ3_BDQ3_B</td></tr><tr><td>DQS0_B_I</td><td>104</td><td>248</td><td>VSS</td></tr><tr><td>DQS0_B_C</td><td>105</td><td>249</td><td>DQS5_B_c, TDQS5_B_c</td></tr><tr><td>VSS</td><td>106</td><td>250</td><td>DQS5_B_I, TDQS5_B_I<br/>DBI0_B_n</td></tr><tr><td>DQ4_B</td><td>107</td><td>251</td><td>VSS</td></tr><tr><td>VSS</td><td>108</td><td>252</td><td>DQ6_B</td></tr><tr><td>DQ5_B</td><td>109</td><td>253</td><td>VSS</td></tr></table> | Front Side<br>Pin Label | Pin | Pin | Back side<br>Pin Label | PAR_A | 74 | 218 | CK_C | VSS | 75 | 219 | VSS | Key |  |  |  | CA0_B | 76 | 220 | RFU | VSS | 77 | 221 | CA1_B | CA2_B | 78 | 222 | VSS | VSS | 79 | 223 | CA3_B | CA4_B | 80 | 224 | VSS | VSS | 81 | 225 | CA5_B | CA6_B | 82 | 226 | VSS | VSS | 83 | 227 | PAR_B | CS0_B_n | 84 | 228 | VSS | VSS | 85 | 229 | CS1_B_n | LBD, RSP_A_n | 86 | 230 | VSS | LBS, RSP_B_n | 87 | 231 | RFU | VSS | 88 | 232 | RFU | CB4_B | 89 | 233 | VSS | VSS | 90 | 234 | CB6_B | CB5_B | 91 | 235 | VSS | VSS | 92 | 236 | CB7_B | DQS9_B_I, TDQS9_B_I<br>DBI4_B_n | 93 | 237 | VSS | DQS9_B_C, TDQS9_B_C | 94 | 238 | DQS4_B_c | VSS | 95 | 239 | DQS4_B_I | CB0_B | 96 | 240 | VSS | VSS | 97 | 241 | CB2_B | CB1_B | 98 | 242 | VSS | VSS | 99 | 243 | CB3_B | DQ0_B | 100 | 244 | VSS | VSS | 101 | 245 | DQ2_B | DQ1_B | 102 | 246 | VSS | VSS | 103 | 247 | DQ3_BDQ3_B | DQS0_B_I | 104 | 248 | VSS | DQS0_B_C | 105 | 249 | DQS5_B_c, TDQS5_B_c | VSS | 106 | 250 | DQS5_B_I, TDQS5_B_I<br>DBI0_B_n | DQ4_B | 107 | 251 | VSS | VSS | 108 | 252 | DQ6_B | DQ5_B | 109 | 253 | VSS | <table><tr><th>Front Side<br/>Pin Label</th><th>Pin</th><th>Pin</th><th>Back side<br/>Pin Label</th></tr><tr><td>VSS</td><td>39</td><td>183</td><td>DQ23_A</td></tr><tr><td>DQ24_A</td><td>40</td><td>184</td><td>VSS</td></tr><tr><td>VSS</td><td>41</td><td>185</td><td>DQ26_A</td></tr><tr><td>DQ25_A</td><td>42</td><td>186</td><td>VSS</td></tr><tr><td>VSS</td><td>43</td><td>187</td><td>DQ27_A</td></tr><tr><td>DQS3_A_I</td><td>44</td><td>188</td><td>VSS</td></tr><tr><td>DQS3_A_C</td><td>45</td><td>189</td><td>DQS8_A_c, TDQS8_A_c</td></tr><tr><td>VSS</td><td>46</td><td>190</td><td>DQS8_A_I, TDQS8_A_I<br/>DBI3_A_n</td></tr><tr><td>DQ28_A</td><td>47</td><td>191</td><td>VSS</td></tr><tr><td>VSS</td><td>48</td><td>192</td><td>DQ30_A</td></tr><tr><td>DQ29_A</td><td>49</td><td>193</td><td>VSS</td></tr><tr><td>VSS</td><td>50</td><td>194</td><td>DQ31_A</td></tr><tr><td>CB0_A</td><td>51</td><td>195</td><td>VSS</td></tr><tr><td>VSS</td><td>52</td><td>196</td><td>CB2_A</td></tr><tr><td>CB1_A</td><td>53</td><td>197</td><td>VSS</td></tr><tr><td>VSS</td><td>54</td><td>198</td><td>CB3_A</td></tr><tr><td>DQS4_A_I</td><td>55</td><td>199</td><td>VSS</td></tr><tr><td>DQS4_A_C</td><td>56</td><td>200</td><td>DQS9_A_c, TDQS9_A_c</td></tr><tr><td>VSS</td><td>57</td><td>201</td><td>DQS9_A_I, TDQS9_A_I<br/>DBI4_A_n</td></tr><tr><td>CB4_A</td><td>58</td><td>202</td><td>VSS</td></tr><tr><td>VSS</td><td>59</td><td>203</td><td>CB6_A</td></tr><tr><td>CB5_A</td><td>60</td><td>204</td><td>VSS</td></tr><tr><td>VSS</td><td>61</td><td>205</td><td>CB7_A</td></tr><tr><td>ALEFT_n</td><td>62</td><td>206</td><td>VSS</td></tr><tr><td>VSS</td><td>63</td><td>207</td><td>RESET_n</td></tr><tr><td>CS0_A_n</td><td>64</td><td>208</td><td>VSS</td></tr><tr><td>VSS</td><td>65</td><td>209</td><td>CS1_A_n</td></tr><tr><td>CA0_A</td><td>66</td><td>210</td><td>VSS</td></tr><tr><td>VSS</td><td>67</td><td>211</td><td>CA1_A</td></tr><tr><td>CA2</td><td>68</td><td>212</td><td>VSS</td></tr><tr><td>VSS</td><td>69</td><td>213</td><td>CA3_A</td></tr><tr><td>CA4_A</td><td>70</td><td>214</td><td>VSS</td></tr><tr><td>VSS</td><td>71</td><td>215</td><td>CA5_A</td></tr><tr><td>CA6_A</td><td>72</td><td>216</td><td>VSS</td></tr><tr><td>VSS</td><td>73</td><td>217</td><td>CK_I</td></tr></table> | Front Side<br>Pin Label | Pin | Pin | Back side<br>Pin Label | VSS | 39 | 183 | DQ23_A | DQ24_A | 40 | 184 | VSS | VSS | 41 | 185 | DQ26_A | DQ25_A | 42 | 186 | VSS | VSS | 43 | 187 | DQ27_A | DQS3_A_I | 44 | 188 | VSS | DQS3_A_C | 45 | 189 | DQS8_A_c, TDQS8_A_c | VSS | 46 | 190 | DQS8_A_I, TDQS8_A_I<br>DBI3_A_n | DQ28_A | 47 | 191 | VSS | VSS | 48 | 192 | DQ30_A | DQ29_A | 49 | 193 | VSS | VSS | 50 | 194 | DQ31_A | CB0_A | 51 | 195 | VSS | VSS | 52 | 196 | CB2_A | CB1_A | 53 | 197 | VSS | VSS | 54 | 198 | CB3_A | DQS4_A_I | 55 | 199 | VSS | DQS4_A_C | 56 | 200 | DQS9_A_c, TDQS9_A_c | VSS | 57 | 201 | DQS9_A_I, TDQS9_A_I<br>DBI4_A_n | CB4_A | 58 | 202 | VSS | VSS | 59 | 203 | CB6_A | CB5_A | 60 | 204 | VSS | VSS | 61 | 205 | CB7_A | ALEFT_n | 62 | 206 | VSS | VSS | 63 | 207 | RESET_n | CS0_A_n | 64 | 208 | VSS | VSS | 65 | 209 | CS1_A_n | CA0_A | 66 | 210 | VSS | VSS | 67 | 211 | CA1_A | CA2 | 68 | 212 | VSS | VSS | 69 | 213 | CA3_A | CA4_A | 70 | 214 | VSS | VSS | 71 | 215 | CA5_A | CA6_A | 72 | 216 | VSS | VSS | 73 | 217 | CK_I | <table><tr><th>Front Side<br/>Pin Label</th><th>Pin</th><th>Pin</th><th>Back side<br/>Pin Label</th></tr><tr><td>VSS</td><td>110</td><td>254</td><td>DQ7_B</td></tr><tr><td>DQ8_B</td><td>111</td><td>255</td><td>VSS</td></tr><tr><td>VSS</td><td>112</td><td>256</td><td>DQ10_B</td></tr><tr><td>DQ9_B</td><td>113</td><td>257</td><td>VSS</td></tr><tr><td>VSS</td><td>114</td><td>258</td><td>DQ11_B</td></tr><tr><td>DQS1_B_I</td><td>115</td><td>259</td><td>VSS</td></tr><tr><td>DQS1_B_C</td><td>116</td><td>260</td><td>DQS6_B_c, TDQS6_B_c</td></tr><tr><td>VSS</td><td>117</td><td>261</td><td>DQS6_B_I, TDQS6_B_I<br/>DBI1_B_n</td></tr><tr><td>DQ12_B</td><td>118</td><td>262</td><td>VSS</td></tr><tr><td>VSS</td><td>119</td><td>263</td><td>DQ14_B</td></tr><tr><td>DQ13_B</td><td>120</td><td>264</td><td>VSS</td></tr><tr><td>VSS</td><td>121</td><td>265</td><td>DQ15_B</td></tr><tr><td>DQ16_B</td><td>122</td><td>266</td><td>VSS</td></tr><tr><td>VSS</td><td>123</td><td>267</td><td>DQ18_B</td></tr><tr><td>DQ17_B</td><td>124</td><td>268</td><td>VSS</td></tr><tr><td>VSS</td><td>125</td><td>269</td><td>DQ19_B</td></tr><tr><td>DQS2_B_I</td><td>126</td><td>270</td><td>VSS</td></tr><tr><td>DQS2_B_C</td><td>127</td><td>271</td><td>DQS7_B_c, TDQS7_B_c</td></tr><tr><td>VSS</td><td>128</td><td>272</td><td>DQS7_B_I, TDQS7_B_I<br/>DBI2_B_n</td></tr><tr><td>DQ20_B</td><td>129</td><td>273</td><td>VSS</td></tr><tr><td>VSS</td><td>130</td><td>274</td><td>DQ22_B</td></tr><tr><td>DQ21_B</td><td>131</td><td>275</td><td>VSS</td></tr><tr><td>VSS</td><td>132</td><td>276</td><td>DQ23_B</td></tr><tr><td>DQ24_B</td><td>133</td><td>277</td><td>VSS</td></tr><tr><td>VSS</td><td>134</td><td>278</td><td>DQ26_B</td></tr><tr><td>DQ25_B</td><td>135</td><td>279</td><td>VSS</td></tr><tr><td>VSS</td><td>136</td><td>280</td><td>DQ27_B</td></tr><tr><td>DQS3_B_I</td><td>137</td><td>281</td><td>VSS</td></tr><tr><td>DQS3_B_C</td><td>138</td><td>282</td><td>DQS8_B_c, TDQS8_B_c</td></tr><tr><td>VSS</td><td>139</td><td>283</td><td>DQS8_B_I, TDQS8_B_I<br/>DBI3_B_n</td></tr><tr><td>DQ28_B</td><td>140</td><td>284</td><td>VSS</td></tr><tr><td>VSS</td><td>141</td><td>285</td><td>DQ30_B</td></tr><tr><td>DQ29_B</td><td>142</td><td>286</td><td>VSS</td></tr><tr><td>VSS</td><td>143</td><td>287</td><td>DQ31_B</td></tr><tr><td>RFU</td><td>144</td><td>288</td><td>VSS</td></tr></table> | Front Side<br>Pin Label | Pin | Pin | Back side<br>Pin Label | VSS | 110 | 254 | DQ7_B | DQ8_B | 111 | 255 | VSS | VSS | 112 | 256 | DQ10_B | DQ9_B | 113 | 257 | VSS | VSS | 114 | 258 | DQ11_B | DQS1_B_I | 115 | 259 | VSS | DQS1_B_C | 116 | 260 | DQS6_B_c, TDQS6_B_c | VSS | 117 | 261 | DQS6_B_I, TDQS6_B_I<br>DBI1_B_n | DQ12_B | 118 | 262 | VSS | VSS | 119 | 263 | DQ14_B | DQ13_B | 120 | 264 | VSS | VSS | 121 | 265 | DQ15_B | DQ16_B | 122 | 266 | VSS | VSS | 123 | 267 | DQ18_B | DQ17_B | 124 | 268 | VSS | VSS | 125 | 269 | DQ19_B | DQS2_B_I | 126 | 270 | VSS | DQS2_B_C | 127 | 271 | DQS7_B_c, TDQS7_B_c | VSS | 128 | 272 | DQS7_B_I, TDQS7_B_I<br>DBI2_B_n | DQ20_B | 129 | 273 | VSS | VSS | 130 | 274 | DQ22_B | DQ21_B | 131 | 275 | VSS | VSS | 132 | 276 | DQ23_B | DQ24_B | 133 | 277 | VSS | VSS | 134 | 278 | DQ26_B | DQ25_B | 135 | 279 | VSS | VSS | 136 | 280 | DQ27_B | DQS3_B_I | 137 | 281 | VSS | DQS3_B_C | 138 | 282 | DQS8_B_c, TDQS8_B_c | VSS | 139 | 283 | DQS8_B_I, TDQS8_B_I<br>DBI3_B_n | DQ28_B | 140 | 284 | VSS | VSS | 141 | 285 | DQ30_B | DQ29_B | 142 | 286 | VSS | VSS | 143 | 287 | DQ31_B | RFU | 144 | 288 | VSS |
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| Front Side<br>Pin Label                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Pin                     | Pin | Back side<br>Pin Label          |                        |          |   |     |          |     |   |     |          |          |   |     |                 |       |   |     |     |       |   |     |     |     |   |     |     |       |   |     |     |     |   |     |       |       |   |     |     |     |    |     |       |          |    |     |     |          |    |     |                     |     |    |     |                                 |       |    |     |     |     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| DQS9_B_C, TDQS9_B_C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 117                     | 261 | DQS6_B_I, TDQS6_B_I<br>DBI1_B_n |                        |          |   |     |          |     |   |     |          |          |   |     |                 |       |   |     |     |       |   |     |     |     |   |     |     |       |   |     |     |     |   |     |       |       |   |     |     |     |    |     |       |          |    |     |     |          |    |     |                     |     |    |     |                                 |       |    |     |     |     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| VSS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| VSS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| DQ29_B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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Figure 1. Wiring Assignments

Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 7–8, Fig. 1.

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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|         | <p>These data edge connections are configured to be electrically connected to the set of data conduits:</p>  <p style="text-align: center;"><b>Figure 17. Example of DDR5 Memory Interconnect</b></p> <p>Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at Figs. 10, 17; <i>see also id.</i> at Figs. 13–14.</p> <p>Moreover, Netlist previously asserted claim 16 of U.S. Patent No. 11,016,918 against Samsung’s DDR5 DIMMs, which recites, among other limitations, “a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, <b>the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system.</b>” <i>See, e.g.,</i> Ex. CC-21 (EDTX Trial Tr. Day 2, 4 (2023)) at 333:9–12 (“Q. And can you tell us about claim 16 of the ’918 Patent? A. So here’s claim 16 of the ’918, and a claim analysis is just where I actually look at the claims and compare it to exactly how the products operate, the accused products.”). Netlist also asserted claim 1 of the ’918 patent, which recites, among other limitations, “a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system” and “at least one circuit coupled between a first portion of the plurality of edge connections and the plurality of SDRAM</p> |

**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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|         | <p>devices, the at least one circuit operable to (i) receive a first plurality of address and control signals via the first portion of the plurality of edge connections, and (ii) output a second plurality of address and control signals to the plurality of SD RAM devices, . . . .” <i>Id.</i> at 342:8–9 (“Next, I turned to claim 1 of the ’918. Now, as we’ll see, claim 1 only applies to the RDIMM products.”).</p> <p>Samsung did not dispute these limitations. <i>Id.</i> at 885:24–886:5 (“With regard to the ’918 claims, I have determined that—that these claims are not infringed by the Samsung DDR5 product, at least for the reason that there is a limitation in each of the independent claims directed to the requirement of a converter circuit structure, and that is not found on any of these Samsung DDR5 products. And what’s being pointed to is a linear drop-out regulator, but that’s not a converter circuit.”).</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                                                                                                                                   | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
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| <p><b>[1.a.3]</b> a third set of edge connections configured to be electrically connected to the set of address and control conduits,</p> | <p>The Accused Instrumentalities each comprise a printed circuit board (PCB) having a third set of edge connections such as CA[6:0]_A, CA[6:0]_B, CS[1:0]_A_n, CS[1:0]_B_n configured to be electrically connected to the set of address and control conduits in the memory slot.</p> <p>For example, the Samsung DDR5 RDIMM Datasheet shows pin assignment for edge connections, which include pins for data transmission, such as CA[6:0]_A, CA[6:0]_B, CS[1:0]_A_n, CS[1:0]_B_n, configured to be connected to data conduits:</p> <div style="display: flex; align-items: center;"> <div style="margin-right: 20px;"> <p>a third set of edge connections configured to be electrically connected to the set of address and control conduits</p>  </div> <table border="1"> <caption>[Table 5] PIN Description</caption> <thead> <tr> <th>Pin Name</th><th>Description</th></tr> </thead> <tbody> <tr> <td>CA[6:0]_A<br/>CA[6:0]_B</td><td>SDRAM Address and Command Bus</td></tr> <tr> <td>CS[1:0]_A<br/>CS[1:0]_B</td><td>SDRAM Chip Select</td></tr> <tr> <td>PAR_A<br/>PAR_B</td><td>SDRAM Parity input</td></tr> <tr> <td>CK_t</td><td>SDRAM Clocks (true/positive)</td></tr> <tr> <td>CK_c</td><td>SDRAM Clocks (complement/negative)</td></tr> <tr> <td>ALERT_n</td><td>SDRAM alert for CRC error</td></tr> <tr> <td>RESET_n</td><td>Set DRAM to known state</td></tr> <tr> <td>PCAMP</td><td>Control and Monitor Port</td></tr> <tr> <td>HSCL</td><td>I2C/I3C-Basic Host Sideband Bus Clock</td></tr> <tr> <td>HSDA</td><td>I2C/I3C-Basic Host Sideband Bus Data</td></tr> <tr> <td>HSA</td><td>I2C/I3C-Basic Host Sideband Bus Address</td></tr> <tr> <td>LBDQ</td><td>Loopback data output:</td></tr> </tbody> </table> </div> <p>Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 9, Table 5 (annotated).</p> | Pin Name | Description | CA[6:0]_A<br>CA[6:0]_B | SDRAM Address and Command Bus | CS[1:0]_A<br>CS[1:0]_B | SDRAM Chip Select | PAR_A<br>PAR_B | SDRAM Parity input | CK_t | SDRAM Clocks (true/positive) | CK_c | SDRAM Clocks (complement/negative) | ALERT_n | SDRAM alert for CRC error | RESET_n | Set DRAM to known state | PCAMP | Control and Monitor Port | HSCL | I2C/I3C-Basic Host Sideband Bus Clock | HSDA | I2C/I3C-Basic Host Sideband Bus Data | HSA | I2C/I3C-Basic Host Sideband Bus Address | LBDQ | Loopback data output: |
| Pin Name                                                                                                                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| CA[6:0]_A<br>CA[6:0]_B                                                                                                                    | SDRAM Address and Command Bus                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| CS[1:0]_A<br>CS[1:0]_B                                                                                                                    | SDRAM Chip Select                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| PAR_A<br>PAR_B                                                                                                                            | SDRAM Parity input                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| CK_t                                                                                                                                      | SDRAM Clocks (true/positive)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| CK_c                                                                                                                                      | SDRAM Clocks (complement/negative)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| ALERT_n                                                                                                                                   | SDRAM alert for CRC error                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| RESET_n                                                                                                                                   | Set DRAM to known state                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| PCAMP                                                                                                                                     | Control and Monitor Port                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| HSCL                                                                                                                                      | I2C/I3C-Basic Host Sideband Bus Clock                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| HSDA                                                                                                                                      | I2C/I3C-Basic Host Sideband Bus Data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| HSA                                                                                                                                       | I2C/I3C-Basic Host Sideband Bus Address                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |
| LBDQ                                                                                                                                      | Loopback data output:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |             |                        |                               |                        |                   |                |                    |      |                              |      |                                    |         |                           |         |                         |       |                          |      |                                       |      |                                      |     |                                         |      |                       |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

Claim 1

Evidence of Use

| Front Side<br>Pin Label | Pin | Pin | Back side<br>Pin Label |
|-------------------------|-----|-----|------------------------|
| VIN_BULK                | 1   | 145 | VIN_BULK               |
| RFU                     | 2   | 146 | VIN_BULK               |
| VIN_MGMT                | 3   | 147 | PWR_GOOD/FAIL_n        |
| HSQL                    | 4   | 148 | HSA                    |
| HSDA                    | 5   | 149 | RFU                    |
| VSS                     | 6   | 150 | RFU                    |
| DQ0_A                   | 7   | 151 | VSS                    |
| VSS                     | 8   | 152 | DQ2_A                  |
| DQ1_A                   | 9   | 153 | VSS                    |
| VSS                     | 10  | 154 | DQ3_A                  |
| DQS0_A_I                | 11  | 155 | VSS                    |
| DQS0_A_c                | 12  | 156 | DQS5_A_c, TDQS5_A_c    |
| VSS                     | 13  | 157 | DQS5_A_I, TDQS5_A_I    |
| DQ4_A                   | 14  | 158 | VSS                    |
| VSS                     | 15  | 159 | DQ6_A                  |
| DQ5_A                   | 16  | 160 | VSS                    |
| VSS                     | 17  | 161 | DQ7_A                  |
| DQ8_A                   | 18  | 162 | VSS                    |
| VSS                     | 19  | 163 | DQ10_A                 |
| DQ9_A                   | 20  | 164 | VSS                    |
| VSS                     | 21  | 165 | DQ11_A                 |
| DQS1_A_I                | 22  | 166 | VSS                    |
| DQS1_A_c                | 23  | 167 | DQS6_A_c, TDQS6_A_c    |
| VSS                     | 24  | 168 | DQS6_A_I, TDQS6_A_I    |
| DQ12_A                  | 25  | 169 | VSS                    |
| VSS                     | 26  | 170 | DQ14_A                 |
| DQ13_A                  | 27  | 171 | VSS                    |
| VSS                     | 28  | 172 | DQ15_A                 |
| DQ16_A                  | 29  | 173 | VSS                    |
| VSS                     | 30  | 174 | DQ18_A                 |
| DQ17_A                  | 31  | 175 | VSS                    |
| VSS                     | 32  | 176 | DQ19_A                 |
| DQS2_A_I                | 33  | 177 | VSS                    |
| DQS2_A_c                | 34  | 178 | DQS7_A_c, TDQS7_A_c    |
| VSS                     | 35  | 179 | DQS7_A_I, TDQS7_A_I    |
| DQ20_A                  | 36  | 180 | VSS                    |
| VSS                     | 37  | 181 | DQ22_A                 |
| DQ21_A                  | 38  | 182 | VSS                    |

| Front Side<br>Pin Label | Pin | Pin | Back side<br>Pin Label |
|-------------------------|-----|-----|------------------------|
| PAR_A                   | 74  | 218 | CK_c                   |
| VSS                     | 75  | 219 | VSS                    |
| Key                     |     |     |                        |
| CA0_B                   | 76  | 220 | RFU                    |
| VSS                     | 77  | 221 | CA1_B                  |
| CA2_B                   | 78  | 222 | VSS                    |
| VSS                     | 79  | 223 | CA3_B                  |
| CA4_B                   | 80  | 224 | VSS                    |
| VSS                     | 81  | 225 | CA5_B                  |
| CA6_B                   | 82  | 226 | VSS                    |
| VSS                     | 83  | 227 | PAR_B                  |
| CS0_B_n                 | 84  | 228 | VSS                    |
| VSS                     | 85  | 229 | CS1_B_n                |
| LBD, RSP_A_n            | 86  | 230 | VSS                    |
| LBS, RSP_B_n            | 87  | 231 | RFU                    |
| VSS                     | 88  | 232 | RFU                    |
| CB4_B                   | 89  | 233 | VSS                    |
| VSS                     | 90  | 234 | CB6_B                  |
| CB5_B                   | 91  | 235 | VSS                    |
| VSS                     | 92  | 236 | CB7_B                  |
| DQS9_B_I, TDQS9_B_I     | 93  | 237 | VSS                    |
| DQS9_B_c, TDQS9_B_c     | 94  | 238 | DQS4_B_c               |
| VSS                     | 95  | 239 | DQS4_B_I               |
| CB0_B                   | 96  | 240 | VSS                    |
| VSS                     | 97  | 241 | CB2_B                  |
| CB1_B                   | 98  | 242 | VSS                    |
| VSS                     | 99  | 243 | CB3_B                  |
| DQ0_B                   | 100 | 244 | VSS                    |
| VSS                     | 101 | 245 | DQ2_B                  |
| DQ1_B                   | 102 | 246 | VSS                    |
| VSS                     | 103 | 247 | DQ3_BDQ3_B             |
| DQS0_B_I                | 104 | 248 | VSS                    |
| DQS0_B_c                | 105 | 249 | DQS5_B_c, TDQS5_B_c    |
| VSS                     | 106 | 250 | DQS5_B_I, TDQS5_B_I    |
| DQ4_B                   | 107 | 251 | VSS                    |
| VSS                     | 108 | 252 | DQ6_B                  |
| DQ5_B                   | 109 | 253 | VSS                    |

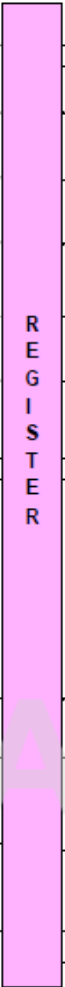
| Front Side<br>Pin Label | Pin | Pin | Back side<br>Pin Label |
|-------------------------|-----|-----|------------------------|
| VSS                     | 39  | 183 | DQ23_A                 |
| DQ24_A                  | 40  | 184 | VSS                    |
| VSS                     | 41  | 185 | DQ26_A                 |
| DQ25_A                  | 42  | 186 | VSS                    |
| VSS                     | 43  | 187 | DQ27_A                 |
| DQS3_A_I                | 44  | 188 | VSS                    |
| DQS3_A_c                | 45  | 189 | DQS8_A_c, TDQS8_A_c    |
| VSS                     | 46  | 190 | DQS8_A_I, TDQS8_A_I    |
| DQ28_A                  | 47  | 191 | VSS                    |
| VSS                     | 48  | 192 | DQ30_A                 |
| DQ29_A                  | 49  | 193 | VSS                    |
| VSS                     | 50  | 194 | DQ31_A                 |
| CB0_A                   | 51  | 195 | VSS                    |
| VSS                     | 52  | 196 | CB2_A                  |
| CB1_A                   | 53  | 197 | VSS                    |
| VSS                     | 54  | 198 | CB3_A                  |
| DQS4_A_I                | 55  | 199 | VSS                    |
| DQS4_A_c                | 56  | 200 | DQS9_A_c, TDQS9_A_c    |
| VSS                     | 57  | 201 | DQS9_A_I, TDQS9_A_I    |
| CB4_A                   | 58  | 202 | VSS                    |
| VSS                     | 59  | 203 | CB6_A                  |
| CB5_A                   | 60  | 204 | VSS                    |
| VSS                     | 61  | 205 | CB7_A                  |
| ALEFT_n                 | 62  | 206 | VSS                    |
| VSS                     | 63  | 207 | RESET_n                |
| CS0_A_n                 | 64  | 208 | VSS                    |
| VSS                     | 65  | 209 | CS1_A_n                |
| CA0_A                   | 66  | 210 | VSS                    |
| VSS                     | 67  | 211 | CA1_A                  |
| CA2                     | 68  | 212 | VSS                    |
| VSS                     | 69  | 213 | CA3_A                  |
| CA4_A                   | 70  | 214 | VSS                    |
| VSS                     | 71  | 215 | CA5_A                  |
| CA6_A                   | 72  | 216 | VSS                    |
| VSS                     | 73  | 217 | CK_I                   |

| Front Side<br>Pin Label | Pin | Pin | Back side<br>Pin Label |
|-------------------------|-----|-----|------------------------|
| VSS                     | 110 | 254 | DQ7_B                  |
| DQ8_B                   | 111 | 255 | VSS                    |
| VSS                     | 112 | 256 | DQ10_B                 |
| DQ9_B                   | 113 | 257 | VSS                    |
| VSS                     | 114 | 258 | DQ11_B                 |
| DQS1_B_I                | 115 | 259 | VSS                    |
| DQS1_B_c                | 116 | 260 | DQS6_B_c, TDQS6_B_c    |
| VSS                     | 117 | 261 | DQS6_B_I, TDQS6_B_I    |
| DQ12_B                  | 118 | 262 | VSS                    |
| VSS                     | 119 | 263 | DQ14_B                 |
| DQ13_B                  | 120 | 264 | VSS                    |
| VSS                     | 121 | 265 | DQ15_B                 |
| DQ16_B                  | 122 | 266 | VSS                    |
| VSS                     | 123 | 267 | DQ18_B                 |
| DQ17_B                  | 124 | 268 | VSS                    |
| VSS                     | 125 | 269 | DQ19_B                 |
| DQS2_B_I                | 126 | 270 | VSS                    |
| DQS2_B_c                | 127 | 271 | DQS7_B_c, TDQS7_B_c    |
| VSS                     | 128 | 272 | DQS7_B_I, TDQS7_B_I    |
| DQ20_B                  | 129 | 273 | VSS                    |
| VSS                     | 130 | 274 | DQ22_B                 |
| DQ21_B                  | 131 | 275 | VSS                    |
| VSS                     | 132 | 276 | DQ23_B                 |
| DQ24_B                  | 133 | 277 | VSS                    |
| VSS                     | 134 | 278 | DQ26_B                 |
| DQ25_B                  | 135 | 279 | VSS                    |
| VSS                     | 136 | 280 | DQ27_B                 |
| DQS3_B_I                | 137 | 281 | VSS                    |
| DQS3_B_c                | 138 | 282 | DQS8_B_c, TDQS8_B_c    |
| VSS                     | 139 | 283 | DQS8_B_I, TDQS8_B_I    |
| DQ28_B                  | 140 | 284 | VSS                    |
| VSS                     | 141 | 285 | DQ30_B                 |
| DQ29_B                  | 142 | 286 | VSS                    |
| VSS                     | 143 | 287 | DQ31_B                 |
| RFU                     | 144 | 288 | VSS                    |

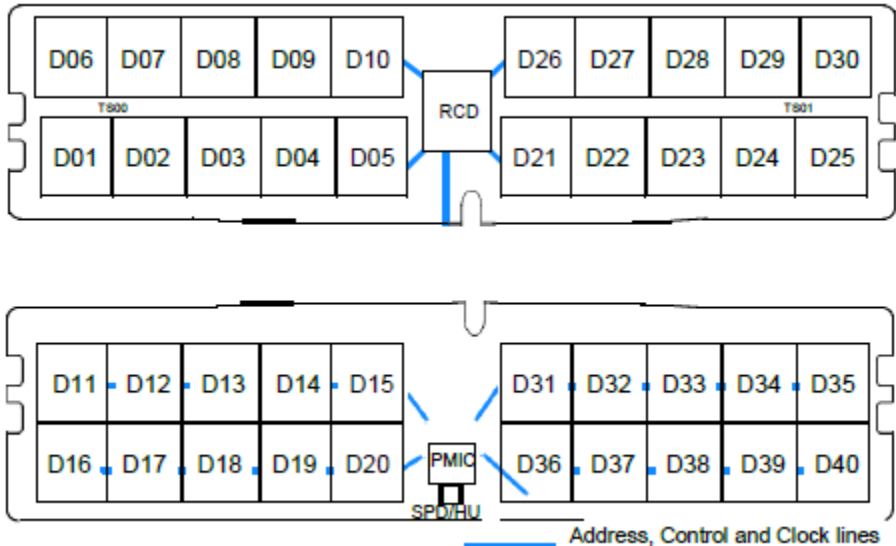
Figure 1. Wiring Assignments

Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 7–8, Fig. 1.

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |  <p> DCA[6:0]_A ———→ QACA [13:0]_A → CA [13:0] : SDRAMs D [5:1, 15:11]<br/> QBCA [13:0]_A → CA [13:0] : SDRAMs D [10:6, 20:16]<br/> <br/> DCS0_A_n ———→ QACS0_A → CS_n : SDRAMs D [5:1]<br/> QBCS0_A → CS_n : SDRAMs D [10:6]<br/> <br/> DCS1_A_n ———→ QACS1_A → CS_n : SDRAMs D [15:11]<br/> QBCS1_A → CS_n : SDRAMs D [20:16]<br/> <br/> DCA[6:0]_B ———→ QACA [13:0]_B → CA [13:0] : SDRAMs D [25:21, 35:31]<br/> QBCA [13:0]_B → CA [13:0] : SDRAMs D [30:26, 40:36]<br/> <br/> DCS0_B_n ———→ QACS0_B → CS_n : SDRAMs D [25:21]<br/> QBCS0_B → CS_n : SDRAMs D [30:26]<br/> <br/> DCS1_B_n ———→ QACS1_B → CS_n : SDRAMs D [35:31]<br/> QBCS1_B → CS_n : SDRAMs D [40:36]<br/> <br/> DCK_t / DCK_c ———→ QACK_A_t → CK_t : SDRAMs D [5:1] / QACK_A_c → CK_c : SDRAMs D [5:1]<br/> QBCK_A_t → CK_t : SDRAMs D [10:6] / QBCK_A_c → CK_c : SDRAMs D [10:6]<br/> QCK_A_t → CK_t : SDRAMs D [15:11] / QCK_A_c → CK_c : SDRAMs D [15:11]<br/> QDCK_A_t → CK_t : SDRAMs D [20:16] / QDCK_A_c → CK_c : SDRAMs D [20:16]<br/> <br/> QACK_B_t → CK_t : SDRAMs D [25:21] / QACK_B_c → CK_c : SDRAMs D [25:21]<br/> QBCK_B_t → CK_t : SDRAMs D [30:26] / QBCK_B_c → CK_c : SDRAMs D [30:26]<br/> QCK_B_t → CK_t : SDRAMs D [35:31] / QCK_B_c → CK_c : SDRAMs D [35:31]<br/> QDCK_B_t → CK_t : SDRAMs D [40:36] / QDCK_B_c → CK_c : SDRAMs D [40:36]<br/> <br/> QLBD ———→ DLBD_A → LBDQ : SDRAMs D [20:1]<br/> DLBD_B → LBDQ : SDRAMs D [40:21]<br/> QLBS ———→ DLBS_A → LBDQS : SDRAMs D [20:1]<br/> DLBS_B → LBDQS : SDRAMs D [40:21]<br/> <br/> DRST_n ———→ QRT_A_n → RESET_n : SDRAMs D [20:1]<br/> QRT_B_n → RESET_n : SDRAMs D [40:21]<br/> <br/> ALERT_n ———→ DERROR_IN_A_n → ALERT_n : SDRAMs D [20:1]<br/> DERROR_IN_B_n → ALERT_n : SDRAMs D [40:21] </p> <p>Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 33 (2Rx4); <i>see also id.</i> at 29 (1Rx4), 25 (2Rx8), 21 (1Rx8).</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |  <p>The diagram illustrates the internal structure and connections of a Samsung DDR5 RDIMM. It shows two main sections: a top section connected to an RCD (Row Address Decoder) and a bottom section connected to a PMIC (Power Management IC). The top section contains two rows of data cells: D06-D10 and D26-D30 in the top row, and D01-D05 and D21-D25 in the bottom row. The bottom section contains two rows of data cells: D11-D15 and D31-D35 in the top row, and D16-D20 and D36-D40 in the bottom row. Blue lines represent Address, Control and Clock lines connecting the data cells to the RCD and PMIC. A legend at the bottom indicates that blue lines represent Address, Control and Clock lines.</p> <p>Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 33 (showing control and address lines from edge connections to RCD for a 2Rx4 RDIMM); <i>see also id.</i> at 29 (1Rx4), 25 (2Rx8), 21 (1Rx8). These edge connections are configured to be electrically connected to the corresponding connectors of the control/address conduits in the memory slot.</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

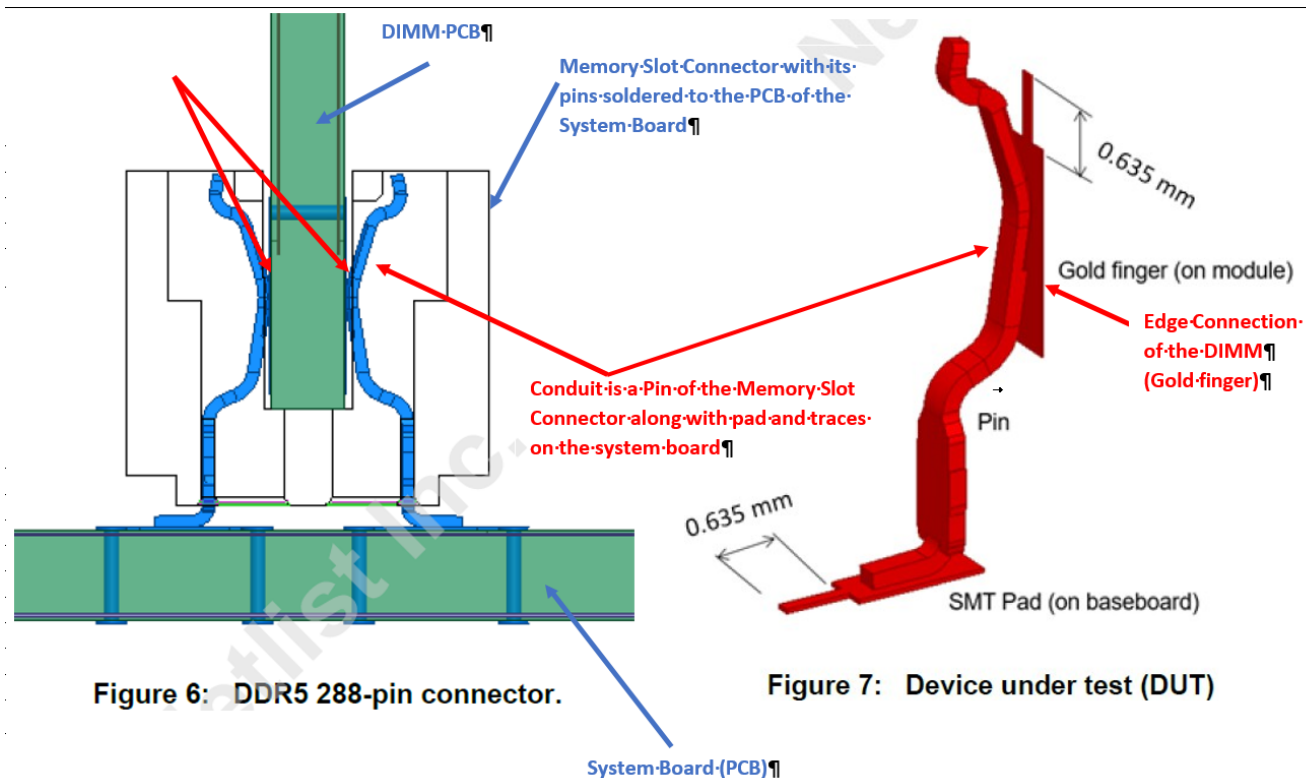


Figure 6: DDR5 288-pin connector.

Figure 7: Device under test (DUT)

Ex. CC-73 (JEDEC PS-005B) at 13, Figs. 6 and 7 (annotated).

Moreover, Netlist previously asserted claim 16 of U.S. Patent No. 11,016,918 against Samsung's DDR5 DIMMs, which recites, among other limitations, "a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, **the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system.**" *See, e.g.,* Ex. CC-21 (EDTX Trial Tr. Day 2, 4 (2023)) at 333:9–12 ("Q. And can you tell us about claim 16 of the '918 Patent? A. So here's claim 16 of the '918, and a claim analysis is just where I actually look at the claims and compare it to exactly how the products operate,

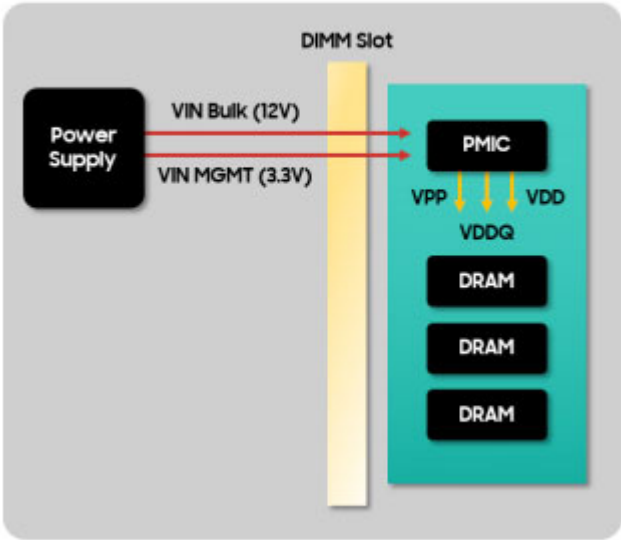
## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

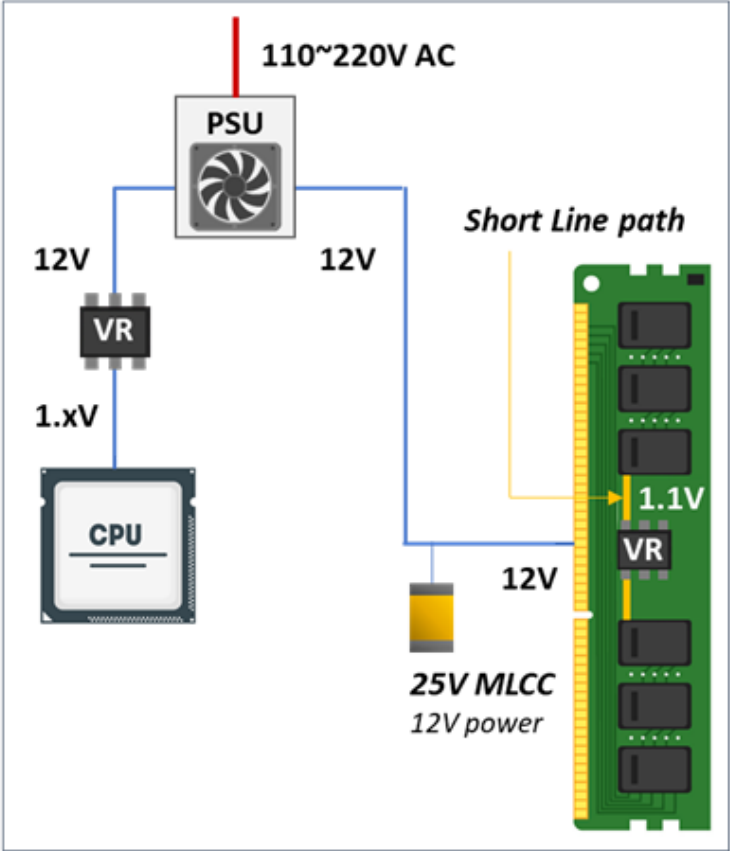
| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>the accused products.”). Netlist also asserted claim 1 of the ’918 patent, which recites, among other limitations, “a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system” and “at least one circuit coupled between a first portion of the plurality of edge connections and the plurality of SDRAM devices, the at least one circuit operable to (i) receive a first plurality of address and control signals via the first portion of the plurality of edge connections, and (ii) output a second plurality of address and control signals to the plurality of SD RAM devices, . . .” <i>Id.</i> at 342:8–9 (“Next, I turned to claim 1 of the ’918. Now, as we’ll see, claim 1 only applies to the RDIMM products.”).</p> <p>Samsung did not dispute these limitations. <i>Id.</i> at 885:24–886:5 (“With regard to the ’918 claims, I have determined that—that these claims are not infringed by the Samsung DDR5 product, at least for the reason that there is a limitation in each of the independent claims directed to the requirement of a converter circuit structure, and that is not found on any of these Samsung DDR5 products. And what’s being pointed to is a linear drop-out regulator, but that’s not a converter circuit.”).</p> |

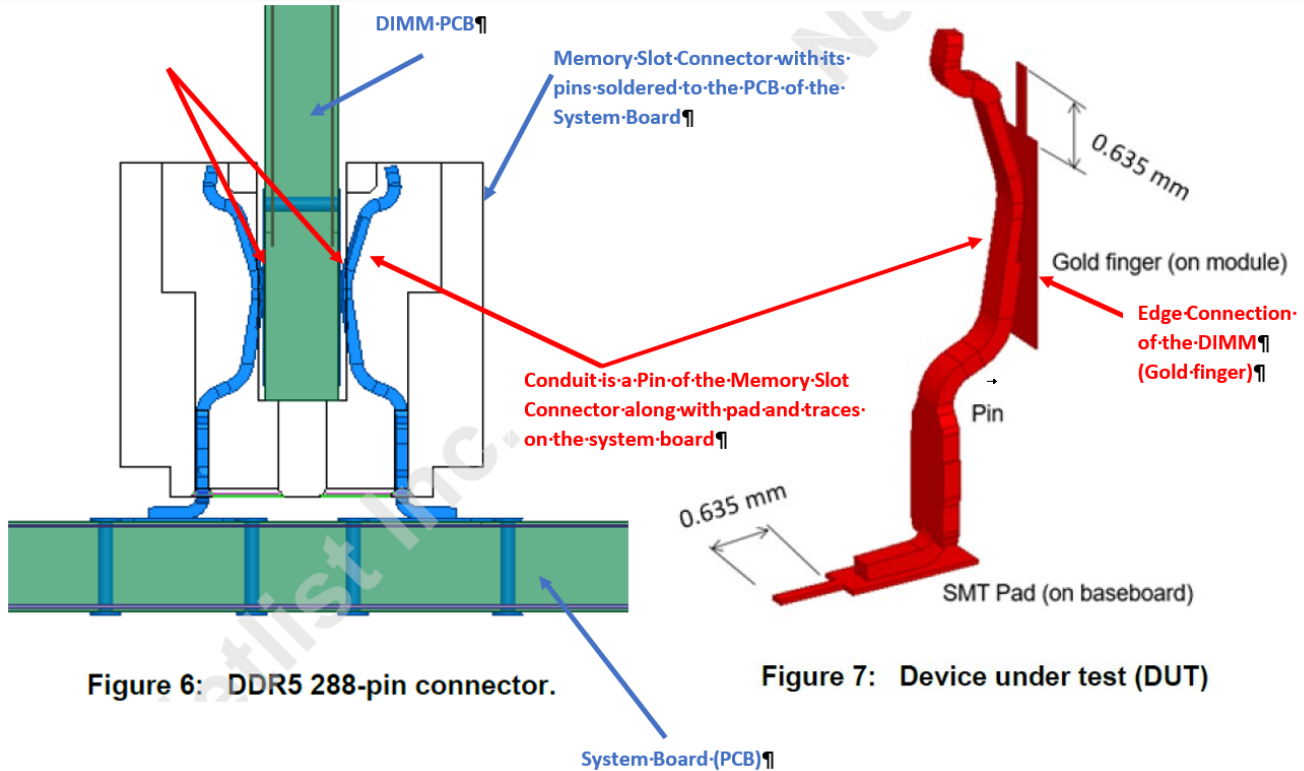
# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                                                                                                                                                                                                                                                                                                                       | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>[1.a.4]</b> a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM,</p> | <p>The Accused Instrumentalities each comprise a printed circuit board (PCB) that has a set of voltage supply lines electrically connected to the first set of edge connections. For example, as illustrated in Samsung's DDR5 RDIMM datasheet, there is a set of voltage supply lines electrically connecting the VIN_BULK and VIN_MGMT edge connections on the RDIMM with the corresponding power supply input pins on the PMIC (<i>see below</i>). As illustrated, the set of voltage supply lines is configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM (<i>e.g.</i>, VIN_BULK and VIN_MGMT pins).</p> <div data-bbox="821 779 1589 1097" data-label="Diagram"> <p>The diagram shows a PMIC block with two input pins on the left: VIN_BULK and VIN_MGMT. Inside the PMIC, these inputs connect to a network of internal nodes and capacitors. On the right side of the PMIC, there are six output pins: VDDSPD, VPP, VDD, VDDQ, VDDIO, and VSS. These pins are connected to external components: VDDSPD to SPD_HUB, TS; VPP to D1 - D20; VDD to D1 - D20; VDDQ to D1 - D20 SPD_HUB, TS, RCD; VDDIO to TS, RCD; and VSS to GND. Each output pin has a capacitor connected to ground.</p> </div> <p>Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 10 (showing inputs from VIN_BULK and VIN_MGMT from the board edge to PMIC and then output from PMIC to DIMM components); <i>see also id.</i> at 36, 32, 28, 24.</p> <p>As another example, Samsung illustrates that power is supplied to the DIMM from a power supply off the DIMM via DIMM slots through the edge connections. <i>See</i> Ex. CC-104 (Optimized DDR5 DIMM Solutions).</p> |

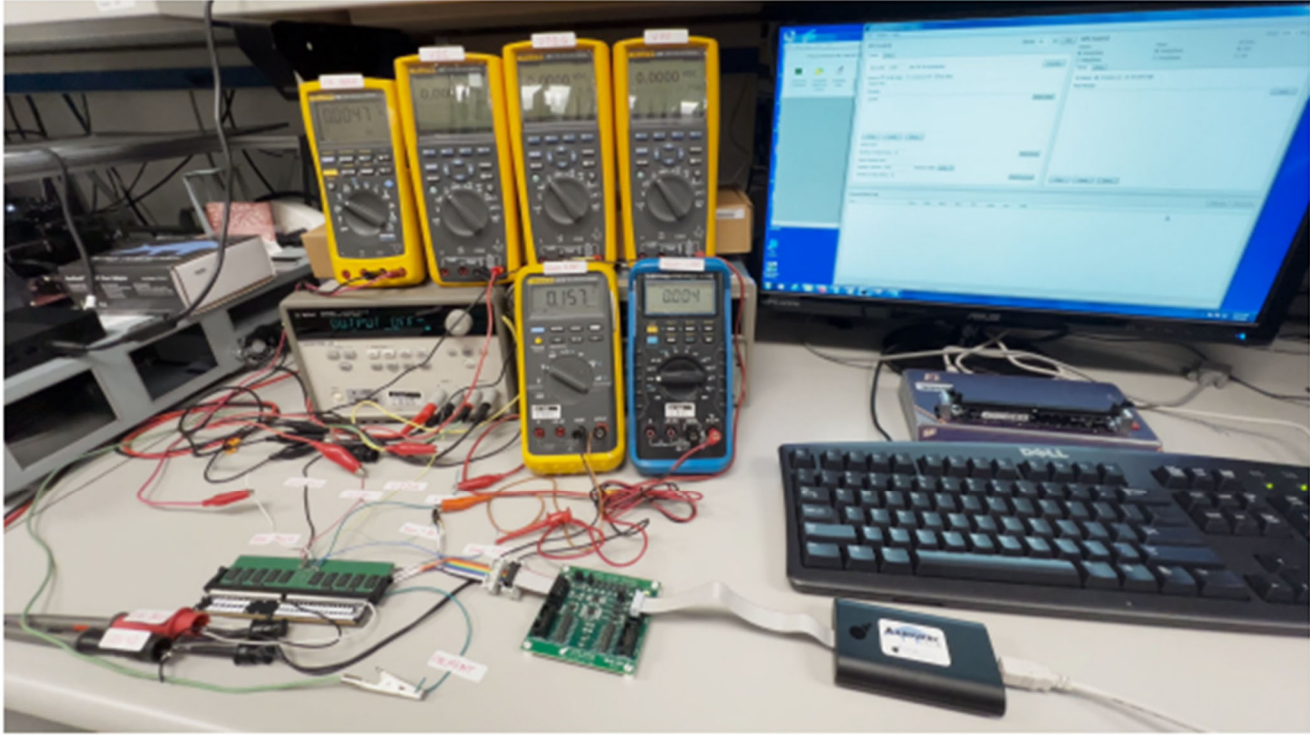
**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p align="center"><b>DDR5 DIMM</b></p>  <p>The diagram illustrates the power delivery architecture for a DDR5 DIMM. A Power Supply block on the left is connected to a DIMM Slot. Two red arrows represent the power rails: VIN Bulk (12V) and VIN MGMT (3.3V). These rails pass through the DIMM Slot to a PMIC (Power Management Integrated Circuit) block on the right. The PMIC then provides three output signals: VPP, VDD, and VDDQ, which are connected to three DRAM blocks stacked vertically.</p> <p>The external power supply is delivered from the power rails of the system board by way of the memory slot connector of the system board, as Samsung illustrates at Ex. CC-120 (Samsung, “Memory VR changes in the system by DDR5 adoption”) (the off-DIMM 12V line is an example of power rails of the system board, which would then be fed to the DIMM slot and from the DIMM slot to the power pins of the DIMM’s edge connections).</p> |

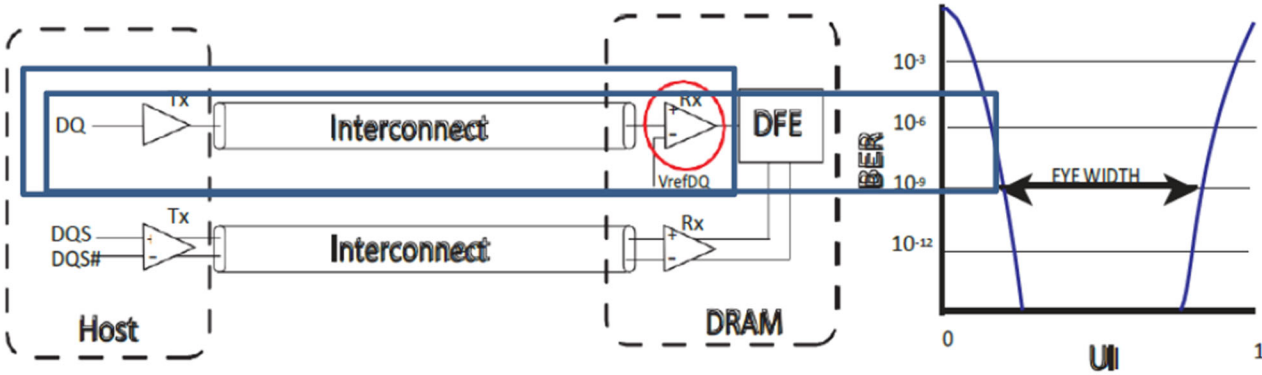
| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p data-bbox="1003 331 1381 370">[ DDR5 Based-System ]</p>  <p>The diagram illustrates the power distribution in a DDR5-based system. It shows a PSU connected to 110~220V AC. The PSU outputs 12V to a VR (Voltage Regulator) which then provides 1.xV to the CPU. Another 12V line from the PSU goes to a 25V MLCC (Multilayer Ceramic Capacitor) labeled '12V power'. A 'Short Line path' is shown connecting this 12V power source to the memory VR, which provides 1.1V to the memory modules.</p> <p data-bbox="558 1268 1608 1300">Ex. CC-120 (Samsung, “Memory VR changes in the system by DDR5 adoption”).</p> |

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |  <p>The evidence consists of two technical diagrams. Figure 6, titled 'DDR5 288-pin connector', is a cross-sectional view of a connector assembly. It shows a green 'DIMM-PCB' at the top, a white 'Memory-Slot-Connector-with-its-pins-soldered-to-the-PCB-of-the-System-Board' in the middle, and a green 'System-Board-(PCB)' at the bottom. Blue traces connect the DIMM-PCB to the system board. A red arrow points to a specific connection point. Figure 7, titled 'Device under test (DUT)', is a 3D perspective view of a red component. It shows a 'Gold finger (on module)' at the top, an 'Edge-Connection-of-the-DIMM (Gold-finger)' in the middle, and an 'SMT Pad (on baseboard)' at the bottom. A 'Pin' is indicated between the gold finger and the SMT pad. Two dimensions of '0.635 mm' are shown. Red arrows from Figure 6 point to the corresponding parts in Figure 7, illustrating the electrical connection between the connector and the DUT.</p> <p><b>Figure 6: DDR5 288-pin connector.</b></p> <p><b>Figure 7: Device under test (DUT)</b></p> <p>Ex. CC-73 (JEDEC PS-005B) at 13, Figs. 6 and 7 (annotated) (showing an electrical connection between a connector in a memory slot and a corresponding edge connection of the DIMM).</p> |

U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                        |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p data-bbox="556 349 1675 386"><i>See also</i> testing setup below (voltmeters were connected to voltage inputs and outputs).</p>  |

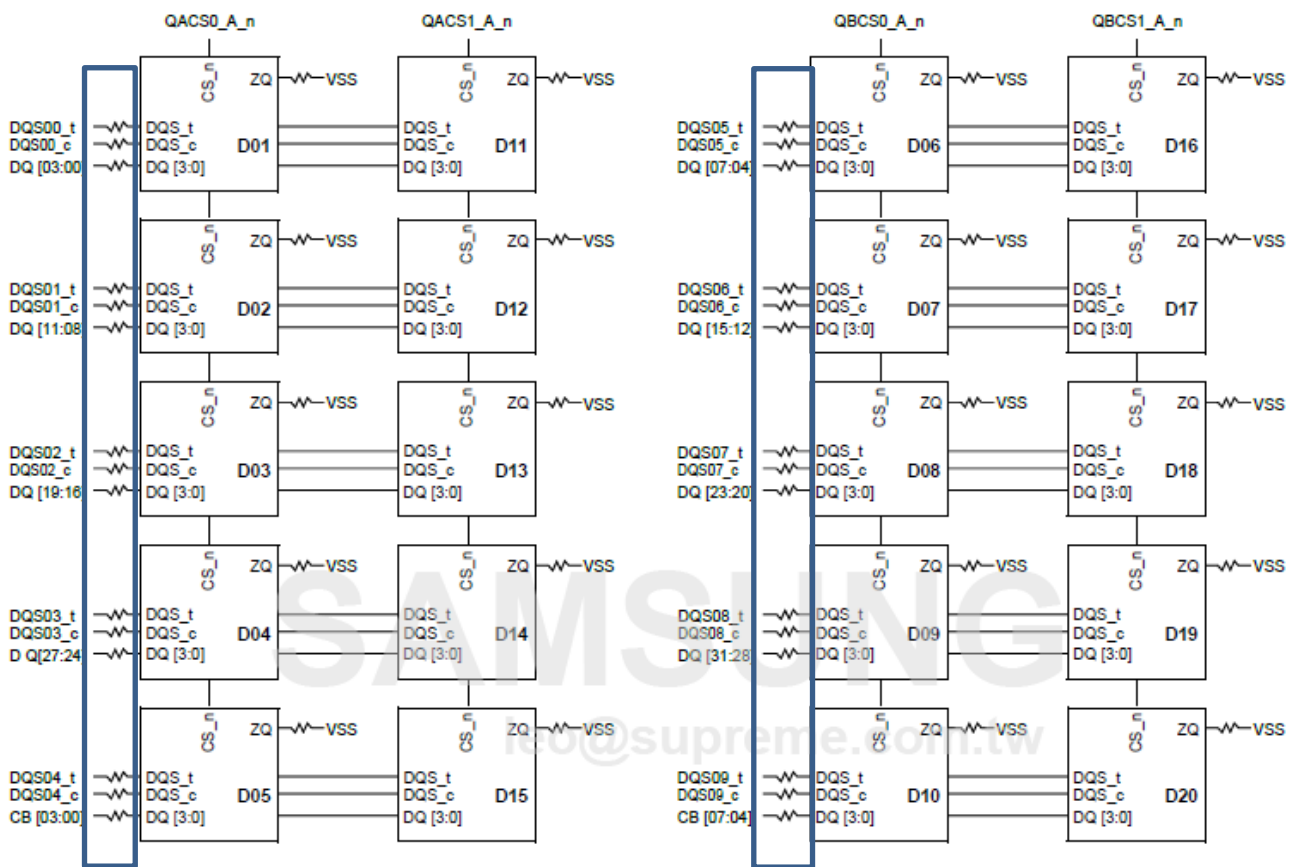
# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                                                                                                     | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>[1.a.5]</b> a set of data lines electrically connected to the second set of edge connections, and</p> | <p>The Accused Instrumentalities each comprise a printed circuit board (PCB) having a set of data lines electrically connected to the second set of edge connections (<i>e.g.</i>, connections with pin assignment DQ[31:0]_A, DQ[31:0]_B, CB[7:0]_A, and CB[7:0]_B).</p> <p>For example, as illustrated by Samsung's DDR5 RDIMM Datasheet, there are data interconnects coupling the host's DQ data lines with the DRAM's data lines. The portions of the data lines that are between the DIMM's edge connections for data signals and the corresponding DQ pins of the DRAM packages on the DIMM would be an example of a set of data lines electrically connected to the second set of edge connections for data:</p>  <p><b>Figure 17. Example of DDR5 Memory Interconnect</b></p> <p>Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at Figs. 10, 17; <i>see also id.</i> at Figs. 13–14.</p> <p><i>See also id.</i> at 34–35 (annotated for a 2Rx4 RDIMM) (showing data lines electrically connected to the second set of edge connections (<i>e.g.</i>, lines connecting DQ pins to DRAMs)):</p> |

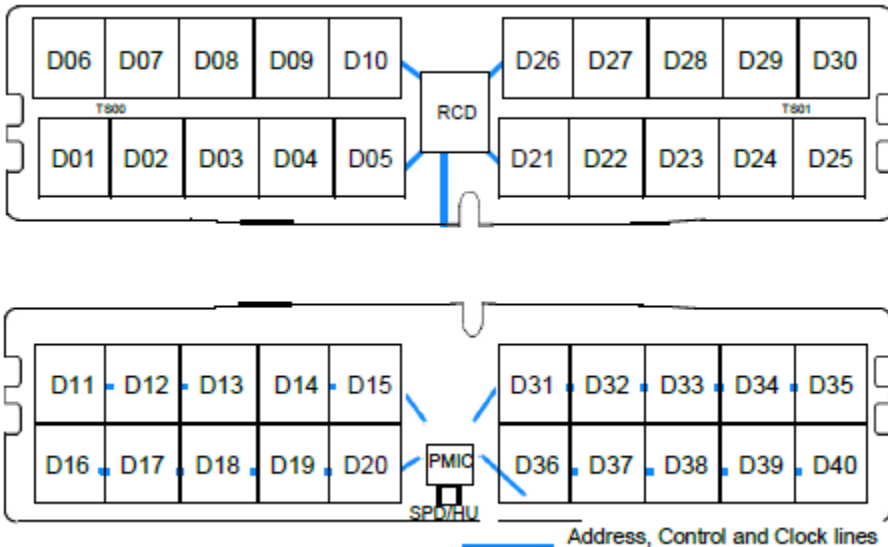
# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                        |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p><b>Channel A</b></p> <p>Examples of data lines connected to 2<sup>nd</sup> set of edge connections; for simplicity, only a subset is annotated.</p> |

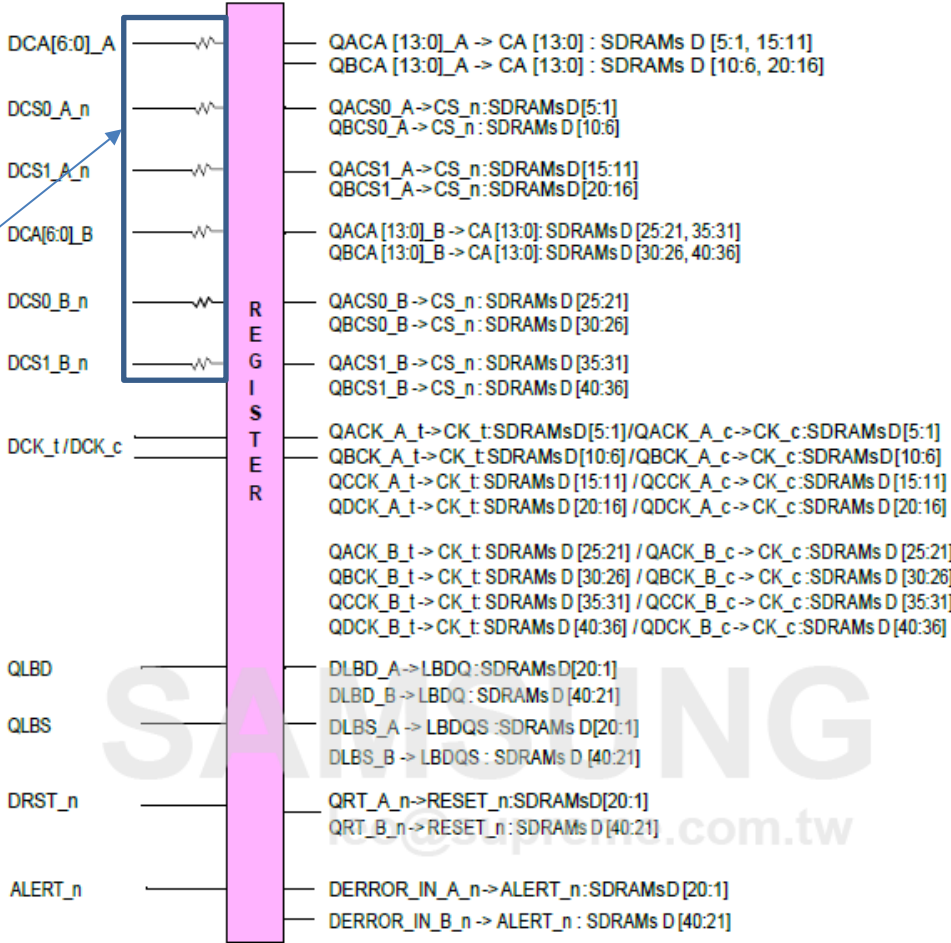
U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p><b>Channel B</b></p>  <p>The diagram illustrates the electrical connections for Channel B. It shows two columns of memory devices. The left column contains devices D01 through D05, and the right column contains devices D11 through D15. A second set of devices, D06 through D10 and D16 through D20, is shown to the right. Each device is connected to a common signal line (QCS0_A_n or QCS1_A_n), a ZQ line, and a VSS line. Data signals (DQS_t, DQS_c, DQ) are shown for each device. Two vertical blue boxes highlight the DQS_t and DQS_c signals for devices D01-D05 and D06-D10.</p> <p>See also <i>id.</i> at 30–31 (1Rx4 RDIMM), 26–27 (2Rx8 RDIMM), 22–23 (1Rx8 RDIMM).</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                                                                                                               | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>[1.a.6]</b> a set of address and control lines electrically connected to the third set of edge connections;</p> | <p>The Accused Instrumentalities each comprise a printed circuit board (PCB) and a set of address and control lines electrically connected to the third set of edge connections (<i>e.g.</i>, connections with pin assignment CA[6:0]_A, CA[6:0]_B, CS[1:0]_A_n, CS[1:0]_B_n).</p> <p>For example, as illustrated by Samsung's DDR5 RDIMM Datasheet, there are lines going from the edge connections to the RCD inputs and then from the RCD to be output to the corresponding DRAMs:</p>  <p>The diagram illustrates the internal structure of a DDR5 RDIMM. It shows two main sections of DRAMs, each organized into two rows of five cells. The top section contains DRAMs D06-D10 and D26-D30 in the top row, and D01-D05 and D21-D25 in the bottom row. The bottom section contains DRAMs D11-D15 and D31-D35 in the top row, and D16-D20 and D36-D40 in the bottom row. A central RCD (Row Address Decoder) is connected to the edge connections (indicated by blue lines) and outputs signals to the DRAMs. A PMIC (Power Management IC) is also shown, connected to the edge connections and the DRAMs. A legend at the bottom indicates that blue lines represent 'Address, Control and Clock lines'.</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                                                                                                                   | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div data-bbox="252 638 480 829"> <p>C/A lines<br/>connected to 3<sup>rd</sup><br/>set of edge<br/>connections</p> </div> |  <p>The diagram illustrates a memory module connector with various signal lines. A central vertical pink bar is labeled 'REGISTER'. To the left of this bar, several lines are labeled: DCA[6:0]_A, DCS0_A_n, DCS1_A_n, DCA[6:0]_B, DCS0_B_n, DCS1_B_n, DCK_t/DCK_c, QLBD, QLBS, DRST_n, and ALERT_n. To the right of the bar, corresponding connections to SDRAMs are listed. A blue box highlights the DCA and DCS lines, with an arrow pointing from the text box on the left. A large 'SAMSUNG' watermark is visible in the background.</p> <p>QACA [13:0]_A → CA [13:0] : SDRAMs D [5:1, 15:11]<br/> QBCA [13:0]_A → CA [13:0] : SDRAMs D [10:6, 20:16]<br/> QACS0_A → CS_n : SDRAMs D [5:1]<br/> QBCS0_A → CS_n : SDRAMs D [10:6]<br/> QACS1_A → CS_n : SDRAMs D [15:11]<br/> QBCS1_A → CS_n : SDRAMs D [20:16]<br/> QACA [13:0]_B → CA [13:0] : SDRAMs D [25:21, 35:31]<br/> QBCA [13:0]_B → CA [13:0] : SDRAMs D [30:26, 40:36]<br/> QACS0_B → CS_n : SDRAMs D [25:21]<br/> QBCS0_B → CS_n : SDRAMs D [30:26]<br/> QACS1_B → CS_n : SDRAMs D [35:31]<br/> QBCS1_B → CS_n : SDRAMs D [40:36]<br/> QACK_A_t → CK_t : SDRAMs D [5:1] / QACK_A_c → CK_c : SDRAMs D [5:1]<br/> QBCK_A_t → CK_t : SDRAMs D [10:6] / QBCK_A_c → CK_c : SDRAMs D [10:6]<br/> QCCK_A_t → CK_t : SDRAMs D [15:11] / QCCK_A_c → CK_c : SDRAMs D [15:11]<br/> QDCK_A_t → CK_t : SDRAMs D [20:16] / QDCK_A_c → CK_c : SDRAMs D [20:16]<br/> QACK_B_t → CK_t : SDRAMs D [25:21] / QACK_B_c → CK_c : SDRAMs D [25:21]<br/> QBCK_B_t → CK_t : SDRAMs D [30:26] / QBCK_B_c → CK_c : SDRAMs D [30:26]<br/> QCCK_B_t → CK_t : SDRAMs D [35:31] / QCCK_B_c → CK_c : SDRAMs D [35:31]<br/> QDCK_B_t → CK_t : SDRAMs D [40:36] / QDCK_B_c → CK_c : SDRAMs D [40:36]<br/> DLBD_A → LBDQ : SDRAMs D [20:1]<br/> DLBD_B → LBDQ : SDRAMs D [40:21]<br/> DLBS_A → LBDQS : SDRAMs D [20:1]<br/> DLBS_B → LBDQS : SDRAMs D [40:21]<br/> QRT_A_n → RESET_n : SDRAMs D [20:1]<br/> QRT_B_n → RESET_n : SDRAMs D [40:21]<br/> DERROR_IN_A_n → ALERT_n : SDRAMs D [20:1]<br/> DERROR_IN_B_n → ALERT_n : SDRAMs D [40:21]</p> <p>Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 33 (annotated for a 2Rx4 RDIMM) (showing control and address lines from edge connections to RCD); <i>see also id.</i> at 29 (1Rx4 RDIMM), 25 (2Rx8 RDIMM), 21 (1Rx8 RDIMM). These edge connections are configured to be electrically connected to the corresponding connectors of the control/address conduits in the memory slot.</p> |

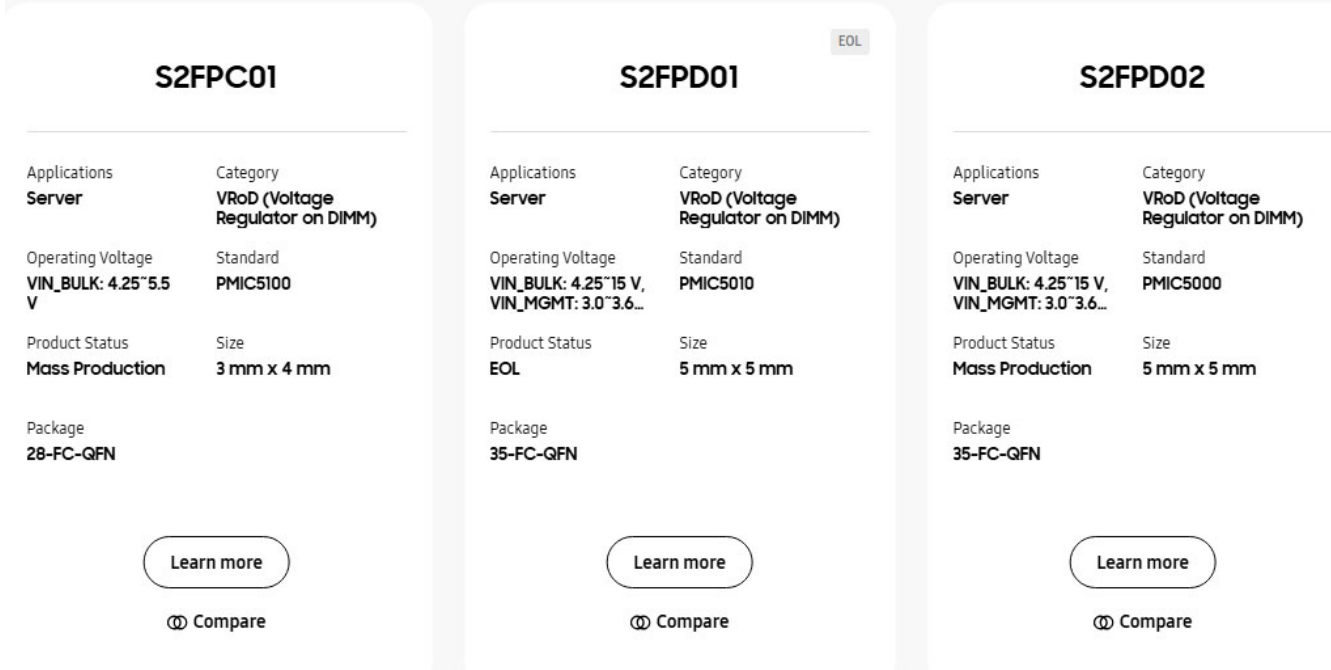
## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                                                                                      | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>[1.b]</b> a controller including a voltage monitor circuit and nonvolatile memory,</p> | <p>The Accused Instrumentalities include a controller (such as the non-voltage-converter/regulator portions of the power management IC (“PMIC”)) that includes a voltage monitor circuit and non-volatile memory.</p> <p>Analysis below uses Samsung’s RDIMM format as an example, but MRDIMMs would operate in materially the same way in all relevant part. Ex. CC-13 (JEDEC Press Release) (JEDEC noting that other planned features for MRDIMMs include “Utilization of standard DDR5 DIMM components including DRAM, DIMM Form Factor &amp; Pinout, SPD, PMIC, and TS for ease of adoption”).</p> <p>For example, a Samsung web site illustrates that power is supplied from an external power supply to a PMIC on the DIMM. See Ex. CC-104 (Optimized DDR5 DIMM Solutions).</p> <div data-bbox="898 763 1491 1347" data-label="Diagram"> <p style="text-align: center; color: blue; font-weight: bold;">DDR5 DIMM</p> <p>The diagram illustrates the power architecture of a DDR5 DIMM. On the left, a black box labeled 'Power Supply' is connected to a yellow vertical bar representing the 'DIMM Slot'. Two red arrows originate from the Power Supply: one labeled 'VIN Bulk (12V)' and another labeled 'VIN MGMT (3.3V)'. These arrows point to a black box labeled 'PMIC' (Power Management IC) located on the DIMM. From the PMIC, three yellow arrows point downwards to three stacked black boxes labeled 'DRAM'. The arrows from the PMIC are labeled 'VPP', 'VDD', and 'VDDQ' respectively.</p> </div> <p>Ex. CC-104 (Optimized DDR5 DIMM Solutions).</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>As another example, the Samsung DDR5 RDIMM Datasheet shows a PMIC that receives power from VIN_BULK and VIN_MGMT power supply pins and outputs power to other components on the DIMM:</p> <div data-bbox="819 534 1589 852" data-label="Diagram"> </div> <p>Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 10 (showing inputs from VIN_BULK and VIN_MGMT from the board edge to PMIC and then output from PMIC to DIMM components), 24 (same), 28 (same), 32 (same), 36 (same); <i>see also id.</i> at 21 (showing a PMIC coupled to the PCB), 25 (same), 29 (same), 33 (same, reproduced below).</p> <div data-bbox="758 1107 1638 1359" data-label="Diagram"> </div> <p><i>Id.</i> at 33.</p> |

## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>According to Samsung's web site, server DIMMs "use a 4-channel buck converter and a dual low-dropout regulator (2LDO)." Ex. CC-104 (Optimized DDR5 DIMM Solutions).</p> <p>Samsung uses PMICs made by several suppliers, including those by Samsung, Renesas (IDT), and MPS (based on examination of Samsung DDR5 DIMMs), among others.</p> <p>Each PMIC includes a voltage monitoring circuit. For example, the PMICs advertised on Samsung's web site are all specified to operate with certain ranges of voltages of VIN_BULK and VIN_MGMT. <i>See</i> Ex. CC-146 (Samsung Memory Power IC).</p>  <p>The screenshot displays three product cards for PMICs. The first card, S2FPC01, is for a 3 mm x 4 mm package. The second card, S2FPD01, is marked as EOL and for a 5 mm x 5 mm package. The third card, S2FPD02, is also for a 5 mm x 5 mm package. All three cards specify an operating voltage range of VIN_BULK: 4.25~5.5 V and VIN_MGMT: 3.0~3.6 V. Each card includes a 'Learn more' button and a 'Compare' button.</p> |

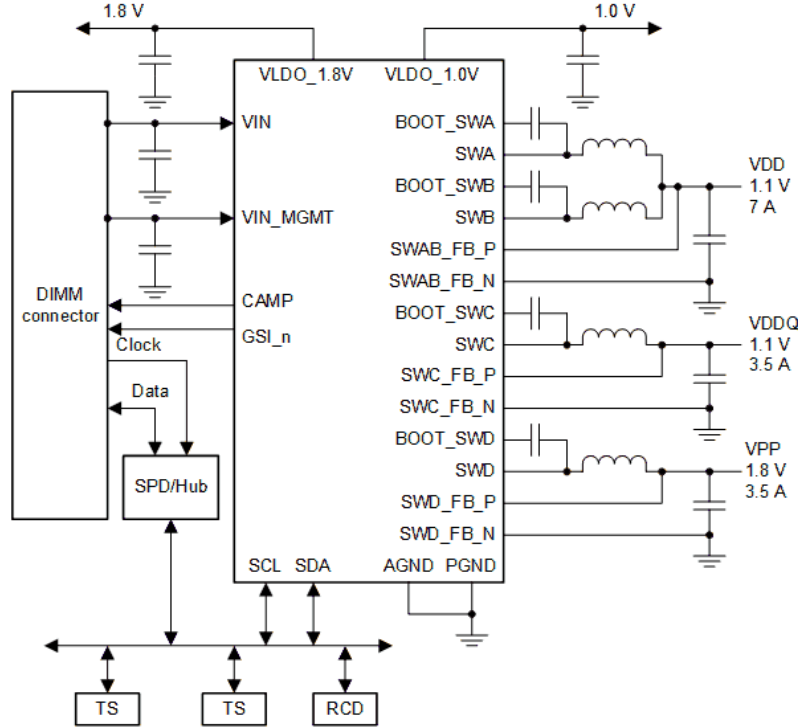
## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <div><div>S2FPD31</div><div><div><div><div>Applications</div><div>Server</div></div><div><div>Category</div><div>VRoD (Voltage Regulator on DIMM)</div></div></div><div><div><div>Operating Voltage</div><div>VIN_BULK: 4.25~5.5 V</div></div><div><div>Standard</div><div>PMIC5120</div></div></div><div><div><div>Product Status</div><div>Under Development</div></div><div><div>Size</div><div>3 mm x 4 mm</div></div></div><div><div><div>Package</div><div>28-FC-QFN</div></div></div></div><div><div>Learn more</div><div>⌕ Compare</div></div></div> | <div><div>S2FPD73</div><div><div><div><div>Applications</div><div>Server</div></div><div><div>Category</div><div>VRoD (Voltage Regulator on DIMM)</div></div></div><div><div><div>Operating Voltage</div><div>VIN_BULK: 4.25~15 V, VIN_MGMT: 3.0~3.6...</div></div><div><div>Standard</div><div>PMIC5020</div></div></div><div><div><div>Product Status</div><div>Under Development</div></div><div><div>Size</div><div>5 mm x 5 mm</div></div></div><div><div><div>Package</div><div>35-FC-QFN</div></div></div></div><div><div>Learn more</div><div>⌕ Compare</div></div></div> |
|         | <p>Ex. CC-146 (Samsung Memory Power IC).</p> <p>This indicates that there is a voltage monitoring circuit so that the PMIC can react when the input voltage is outside the specified operating range.</p> <p>As another example, Renesas supplies P8900 and P8910 for server DIMMs such as RDIMMs. Renesas notes that P8900 includes “VIN Bulk input supply protection feature: Input overvoltage.” Ex. CC-75</p>                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

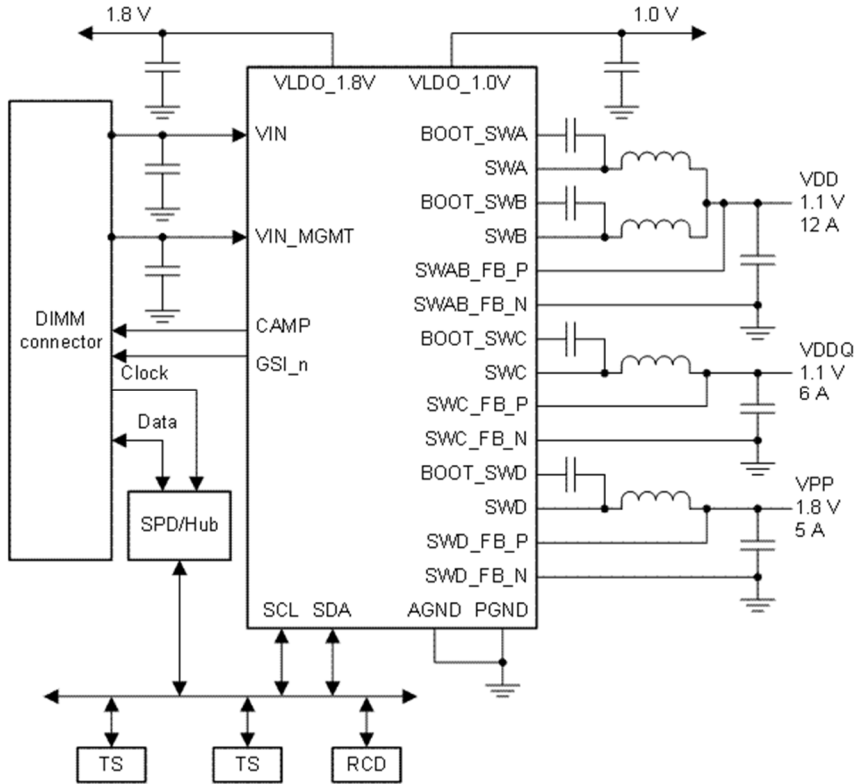
# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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|         | <p>(Renesas P8900 Short-Form Datasheet). Renesas' P8910 includes "VIN_Bulk input supply protection features: Input overvoltage and input undervoltage." Ex. CC-74 (Renesas P8910 Short-Form Datasheet). These again suggest the presence of an input voltage monitor circuit in the Renesas PMICs.</p> <p>Similarly, DDR5 PMICs supplied by MPS are advertised as including "over-voltage protection" (OVP) and "under-voltage protection" (UVP). <i>See</i> Ex. CC-27 (MPQ8896); Ex. CC-28 (MPQ8895); Ex. CC-123 (MPQ8895F); Ex. CC-29 (MPQ8894).</p> <p><i>See also</i> Ex. CC-124 (TPS53832A Product Overview) &amp; Ex. CC-125 (TPS53832A Data Sheet) ("The converter provides a full set of telemetry, including input voltage, output voltage, and output current. [O]vervoltage, undervoltage, overcurrent limit and over-temperature protections are provided as well.").</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |  <p>The diagram illustrates the internal architecture and external connections of the TPS53830A. Key components include:</p> <ul style="list-style-type: none"> <li><b>Power Input/Output:</b> VIN (1.8V), VIN_MGMT, and VDD (1.1V, 7A).</li> <li><b>Control and Status:</b> SCL, SDA, AGND, PGND, and various feedback pins (SWAB_FB_P, SWAB_FB_N, SWC_FB_P, SWC_FB_N, SWD_FB_P, SWD_FB_N).</li> <li><b>Internal Blocks:</b> VLDO_1.8V, VLDO_1.0V, BOOT_SWA, SWA, BOOT_SWB, SWB, BOOT_SWC, SWC, BOOT_SWD, SWD, and the SPD/Hub.</li> <li><b>External Connections:</b> DIMM connector, SPD/Hub, and various power rails (VDDQ, VPP).</li> </ul> <p>Ex. CC-126 (TPS53830A Product Overview) &amp; Ex. CC-127 (TPS53830A Data Sheet) (“The converter provides a full set of telemetry, including input voltage, output voltage, and output current. [O]vervoltage, undervoltage, overcurrent limit and over-temperature protections are provided as well.”).</p> |

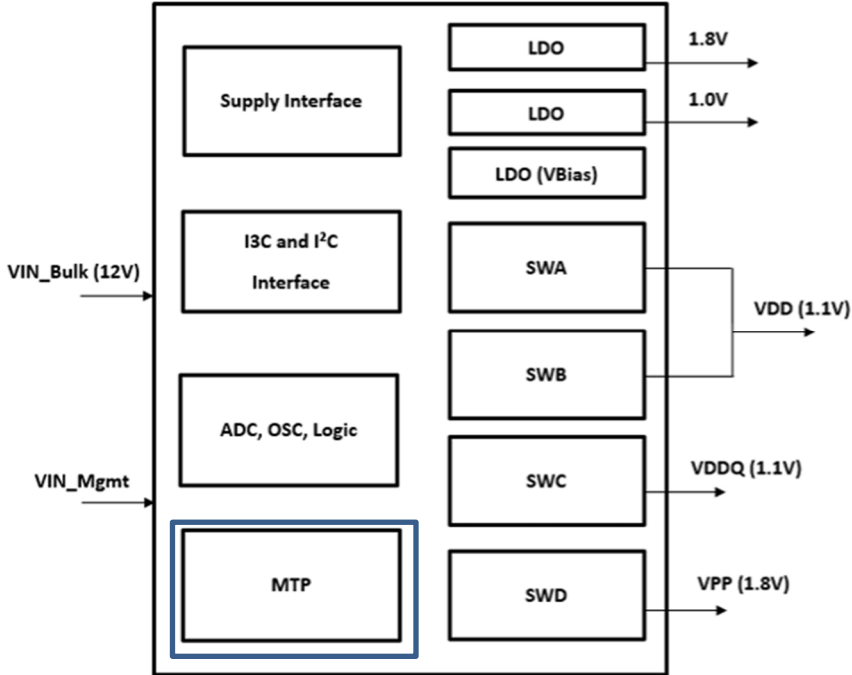
# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |  <p>The diagram shows a PMIC (TPS53830) with two main voltage regulators: VLDO_1.8V and VLDO_1.0V. The VLDO_1.8V regulator is connected to a 1.8V input (VIN) and provides output to a DIMM connector (VIN_MGMT) and a SPD/Hub. The VLDO_1.0V regulator is connected to a 1.0V input (VIN_MGMT) and provides output to a DIMM connector (CAMP) and a SPD/Hub. The DIMM connector is connected to a DIMM module, which provides Clock and Data signals to the SPD/Hub. The SPD/Hub is connected to the DIMM connector and provides SCL, SDA, and AGND signals to the PMIC. The PMIC also has PGND and AGND pins connected to ground. The PMIC is connected to three power rails: VDD (1.1V, 12A), VDDQ (1.1V, 6A), and VPP (1.8V, 5A). Each power rail has a boot switch (BOOT_SWA, BOOT_SWB, BOOT_SWC, BOOT_SWD) and feedback pins (SWAB_FB_P, SWAB_FB_N, SWC_FB_P, SWC_FB_N, SWD_FB_P, SWD_FB_N). The PMIC is also connected to two temperature sensors (TS) and a resistor divider (RCD).</p> <p><i>See also</i> Ex. CC-128 (TPS53830 Product Overview); Ex. CC-129 (TPS53830 Data Sheet); Ex. CC-130 (TPS53832 Product Overview); Ex. CC-26 (TPS53832 Data Sheet).</p> <p>The PMICs on Samsung's DDR5 DIMMs also each include non-volatile memory, such as in the form of MTP (multi-time programmable non-volatile memory) or e-flash. <i>E.g.</i>, Ex. CC-131 (S2FPD01 Brochure) at 2 (annotated):</p> |

U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                       | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |         |                               |             |          |             |                  |                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------|-------------------------------|-------------|----------|-------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <table><tr><th></th><th>S2FPD01</th></tr><tr><td>VIN_BULK (VINA/INB/VINC/VIND)</td><td>4.25V ~ 15V</td></tr><tr><td>VIN_MGMT</td><td>3.0V ~ 3.6V</td></tr><tr><td>Power Management</td><td>Input UVLO/OVLO, Thermal Shut Down function<br/>Read/Write I2C/I3C for Register Control<br/>Interface up to 1MHz for I2C and 12.5MHz for I3C<br/>Error Log Count &amp; Storage<br/><u>Built-in Nonvolatile Memory registers</u><br/>Dynamic Voltage Scaling controlled by I2C/I3C<br/>Programmable Soft Start &amp; Soft Stop Time<br/>Current Limit, Over/Under voltage protection<br/>Power Meter</td></tr></table> <p>Ex. CC-74 (Renesas P8910 Short-Form Datasheet) (annotated):</p> <ul style="list-style-type: none"><li>• Multi Time Programmable Non-Volatile Memory</li></ul> <p>Ex. CC-75 (Renesas P8900 Short-Form Datasheet) at 1 (annotated):</p> |  | S2FPD01 | VIN_BULK (VINA/INB/VINC/VIND) | 4.25V ~ 15V | VIN_MGMT | 3.0V ~ 3.6V | Power Management | Input UVLO/OVLO, Thermal Shut Down function<br>Read/Write I2C/I3C for Register Control<br>Interface up to 1MHz for I2C and 12.5MHz for I3C<br>Error Log Count & Storage<br><u>Built-in Nonvolatile Memory registers</u><br>Dynamic Voltage Scaling controlled by I2C/I3C<br>Programmable Soft Start & Soft Stop Time<br>Current Limit, Over/Under voltage protection<br>Power Meter |
|                               | S2FPD01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |         |                               |             |          |             |                  |                                                                                                                                                                                                                                                                                                                                                                                     |
| VIN_BULK (VINA/INB/VINC/VIND) | 4.25V ~ 15V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |         |                               |             |          |             |                  |                                                                                                                                                                                                                                                                                                                                                                                     |
| VIN_MGMT                      | 3.0V ~ 3.6V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |         |                               |             |          |             |                  |                                                                                                                                                                                                                                                                                                                                                                                     |
| Power Management              | Input UVLO/OVLO, Thermal Shut Down function<br>Read/Write I2C/I3C for Register Control<br>Interface up to 1MHz for I2C and 12.5MHz for I3C<br>Error Log Count & Storage<br><u>Built-in Nonvolatile Memory registers</u><br>Dynamic Voltage Scaling controlled by I2C/I3C<br>Programmable Soft Start & Soft Stop Time<br>Current Limit, Over/Under voltage protection<br>Power Meter                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |         |                               |             |          |             |                  |                                                                                                                                                                                                                                                                                                                                                                                     |

U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p style="text-align: center;"><b>Block Diagram</b></p>  <p style="text-align: center;">▪ Multi Time Programmable Non-Volatile Memory</p> <p>Ex. CC-27 (MPQ8896) (annotated):</p> <p style="background-color: #e6f2ff; padding: 5px;">✓ Configurable Slave Address via the Multiple-Time Programmable (MTP) Memory</p> <p><i>See also</i> Ex. CC-28 (MPQ8895) (“Configurable Slave Address via the Multiple-Time Programmable</p> |

## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>(MTP) Memory”); Ex. CC-123 (MPQ8895F) (“Configurable Slave Address via the Multiple-Time Programmable (MTP) Memory”); Ex. CC-29 (MPQ8894) (“Configurable Slave Address via the Multiple-Time Programmable (MTP) Memory”).</p> <p>Additionally, Samsung’s DIMMs allegedly use PMICs that comply with corresponding JEDEC standards. Ex. CC-104 (Optimized DDR5 DIMM Solutions):</p> <p>As PMICs have evolved, the number and type of power channels has increased accordingly, with server products generally requiring more channels and higher power capacity than client products. This is why server products use a 4-channel buck converter and a dual low-dropout regulator (2LDO)<sup>1</sup> — and client products use a 3-channel buck converter and 2LDO.<sup>2</sup> Initial plans were to have only one product each for servers and clients, but as DRAM modules increased in capacity and speed, it became clear that necessary power would have to be supplied in different ways, leading to the diversification of products.</p> <p><sup>1</sup>) Server Models: PMIC5000 5010 5020 5030.</p> <p><sup>2</sup>) Client Models: PMIC 5100 5120 5200.</p> <p><i>See also</i> PMIC supplier web links above (JEDEC compliant); Ex. CC-131 (S2FPD01 Brochure) at 1 (“The S2FPD01 fully supports JEDEC standard, as well as PMIC50x0 (Power Management IC Specification Rev1.8, JESD301-1.)”); Ex. CC-132 (S2FPD01 Memory Power IC) (PMIC5010); Ex. CC-133 (S2FPD02 Memory Power IC) (Standard PMIC5000); Ex. CC-134 (S2FPD73 Memory Power IC) (Standard PMIC5020).</p> <p><i>See also</i> Ex. CC-135 (TPS53840 Product Overview) &amp; Ex. CC-136 (TPS53840 Data Sheet) at 1 (“Compliant to JEDEC PMIC5020 for DDR5 MRDIMM and RDIMM”).</p> <p>According to the JEDEC Power Management IC specifications, the PMICs are each to include an input</p> |

## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>voltage monitoring circuit and non-volatile memory. For example, the JESD301-1 standard for PMIC5000/PMIC5010 contains the following relevant description.</p> <p><b>2.2 Common Features summary</b></p> <ul style="list-style-type: none"> <li>• VIN_Bulk input supply protection feature: Input over voltage</li> <li>• Multi Time Programmable Non-Volatile Memory</li> </ul> <p>Ex. CC-85 (JESD301-1) (PMIC5000/PMIC5010 standard) at 2; Ex. CC-81 (JESD301-1A) at 2.</p> <p><b>8.15 Input Over Voltage Protection</b></p> <p>An input over voltage protection mechanism is implemented to limit the voltages to the PMIC. The PMIC actively monitors the input voltage VIN_Bulk and VIN_Mgmt rail.</p> <p>Ex. CC-85 (JESD301-1) at 40 (annotated); Ex. CC-81 (JESD301-1A) at 38; <i>see also</i> Ex. CC-83 (JESD301-4A) (PMIC5020 specification) at 2, 38; Ex. CC-137 (JESD301-5) (PMIC5030 specification) at 2, 50.</p> <p>Moreover, Netlist previously asserted claims 16 and 18–19 of U.S. Patent No. 11,016,918 against Samsung’s DDR5 DIMMs, which recites, among other limitations, “<b>a voltage monitor circuit</b> configured to monitor an input voltage received via a first portion of the plurality of edge connections, the voltage monitor circuit configured to produce a signal in response to the input voltage having a voltage amplitude that is greater than a first threshold voltage,” “<b>a controller coupled to the voltage monitor circuit</b> and configured to receive the signal, wherein the controller executes a write operation in response to the signal,” and “wherein the write operation includes writing data information into <b>non-volatile memory</b>.” ’918 Patent, claims 16, 18–19 (emphasis added). <i>See, e.g.</i>, Ex. CC-21 (EDTX Trial Tr. Day 2, 4 (2023)) at 333:9–12 (“Q. And can you tell us about claim 16 of the ’918 Patent? A. So here’s claim 16 of the ’918, and a claim analysis is just where I actually look at the claims and compare it to exactly how the products operate, the accused products.”), 381:24–25 (“Q: And what claim did you analyze next? A: Next, I turned to claim 18 . . . .”), 341:3 (“Q: Let’s turn to ’918, claim 19.”). Netlist previously asserted claim 16 of U.S. Patent No.</p> |

**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>11,232,054 against Samsung’s DDR5 DIMMs, which requires among other things, “a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system” and “a controller coupled to the PCB, the controller including a voltage monitor circuit coupled to an input voltage received from the host system via the interface.” <i>Id.</i> at 345:12–13 (“Q: which claim did you analyze next? A: Next, I turned to claim 16 of the ’054 Patent.”).</p> <p>Samsung did not dispute these limitations. <i>Id.</i> at 885:24–886:9 (“With regard to the ’918 claims, I have determined that—that these claims are not infringed by the Samsung DDR5 product, at least for the reason that there is a limitation in each of the independent claims directed to the requirement of a converter circuit structure, and that is not found on any of these Samsung DDR5 products. And what’s being pointed to is a linear drop-out regulator, but that’s not a converter circuit. As relates to the ’054 claims, the DDR5 product has a normal operating state. The other state is it’s turned off. So it does not have a second operable state as required by the ’054 claims.”).</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                                                                                                                         | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>[1.b.1]</b> the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines,</p> | <p>The Accused Instrumentalities each include a controller (portions of the PMICs that are not voltage regulators) that is coupled to the PCB and to an input voltage supply line (<i>e.g.</i>, to the voltage supply line connected to VIN_BULK and/or to the voltage supply line connected to the VIN_MGMT) of the set of voltage supply lines.</p> <p>For example, a Samsung web site illustrates a PMIC mounted on a DIMM and coupled to the VIN_BULK and VIN_MGMT power supply lines. <i>See</i> Ex. CC-104 (Optimized DDR5 DIMM Solutions). The controller, being part of the PMIC, is thus also mounted on and coupled to the DIMM.</p> <div data-bbox="917 844 1470 1391" data-label="Diagram"> <p>The diagram, titled "DDR5 DIMM", illustrates the power architecture of a DIMM. On the left, a "Power Supply" block is connected to a "DIMM Slot" (represented by a yellow vertical bar). Two red arrows indicate the input voltages: "VIN Bulk (12V)" and "VIN MGMT (3.3V)". These lines connect to a "PMIC" (Power Management IC) block located on the DIMM. From the PMIC, three yellow arrows labeled "VPP", "VDDQ", and "VDD" point to three stacked "DRAM" blocks, showing the distribution of power to the memory modules.</p> </div> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>Ex. CC-104 (Optimized DDR5 DIMM Solutions).</p> <p>As another example, the Samsung DDR5 RDIMM Datasheet shows a PMIC that receives power from VIN_BULK and VIN_MGMT power supply pins and outputs power to other components on the DIMM:</p> <div data-bbox="819 535 1596 860"> </div> <p>Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 10, 24, 28, 32, 36 (showing inputs from VIN_BULK and VIN_MGMT from the board edge to PMIC and then output from PMIC to DIMM components); <i>see also id.</i> at 21 (showing a PMIC coupled to the PCB), 25 (same), 29 (same), 33 (same):</p> <div data-bbox="756 1104 1659 1364"> </div> <p><i>Id.</i> at 33.</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                        | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------|--------------------------|--------------------------------------|------------------------|-----------------------------------------|------------------------------|------------------------------------------------------------|------------------------------|------------------------------------------------------------|--------------------------------|-----------------------------------------------------------|--------------------------------|-----------------------------------------------------------|----------|---------------------------------------|----------|---------------------------------------|-----|------------------------------|-----|-------------------------|-------|-----------------------------|
|                                | <table border="1"> <thead> <tr> <th>Pin Name</th><th>Description</th></tr> </thead> <tbody> <tr> <td>DQ[31:0]_A<br/>DQ[31:0]_B</td><td>DIMM memory data bus channel A and B</td></tr> <tr> <td>CB[7:0]_A<br/>CB[7:0]_B</td><td>DIMM ECC check bits(CB) channel A and B</td></tr> <tr> <td>DQS[9:0]_A_t<br/>DQS[9:0]_B_t</td><td>SDRAM data strobes<br/>(positive line of differential pair)</td></tr> <tr> <td>DQS[9:0]_A_c<br/>DQS[9:0]_B_c</td><td>SDRAM data strobes<br/>(negative line of differential pair)</td></tr> <tr> <td>TDQS[9:5]_A_t<br/>TDQS[9:5]_B_t</td><td>Not valid for x4 operation.<br/>Enabled via Mode Register.</td></tr> <tr> <td>TDQS[9:5]_A_c<br/>TDQS[9:5]_B_c</td><td>Not valid for x4 operation.<br/>Enabled via Mode Register.</td></tr> <tr> <td>VIN_BULK</td><td>DIMM Power Supply from system to PMIC</td></tr> <tr> <td>VIN_MGMT</td><td>DIMM Power Supply from system to PMIC</td></tr> <tr> <td>VSS</td><td>Power supply return (ground)</td></tr> <tr> <td>RFU</td><td>Reserved for future use</td></tr> <tr> <td>LBDQS</td><td>Loopback data strobe output</td></tr> </tbody> </table> <p>Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 9, Table 5 (annotated).</p> <p>Furthermore, as noted above, the controller includes a voltage monitor circuit that monitors the input overvoltage. Thus, the controller is coupled to the VIN_BULK and VIN_MGMT voltage supply lines, because otherwise, the voltage monitor circuit would not be able to monitor these voltage supply lines.</p> <p>Moreover, Netlist previously asserted claims 16 and 5 of U.S. Patent No. 11,016,918 against Samsung's DDR5 DIMMs, which recites, among other limitations, <b>"a voltage monitor circuit configured to monitor an input voltage received via a first portion of the plurality of edge connections"</b> and <b>"a voltage monitor circuit configured to monitor a power input voltage received via a second portion of the plurality of edge connections, . . ."</b> '918 Patent, claims 16 and 5 (emphasis added). <i>See, e.g.</i>, Ex. CC-21 (EDTX Trial Tr. Day 2, 4 (2023)) at 333:9–12 ("Q. And can you tell us about claim 16 of the '918 Patent? A. So here's</p> | Pin Name | Description | DQ[31:0]_A<br>DQ[31:0]_B | DIMM memory data bus channel A and B | CB[7:0]_A<br>CB[7:0]_B | DIMM ECC check bits(CB) channel A and B | DQS[9:0]_A_t<br>DQS[9:0]_B_t | SDRAM data strobes<br>(positive line of differential pair) | DQS[9:0]_A_c<br>DQS[9:0]_B_c | SDRAM data strobes<br>(negative line of differential pair) | TDQS[9:5]_A_t<br>TDQS[9:5]_B_t | Not valid for x4 operation.<br>Enabled via Mode Register. | TDQS[9:5]_A_c<br>TDQS[9:5]_B_c | Not valid for x4 operation.<br>Enabled via Mode Register. | VIN_BULK | DIMM Power Supply from system to PMIC | VIN_MGMT | DIMM Power Supply from system to PMIC | VSS | Power supply return (ground) | RFU | Reserved for future use | LBDQS | Loopback data strobe output |
| Pin Name                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| DQ[31:0]_A<br>DQ[31:0]_B       | DIMM memory data bus channel A and B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| CB[7:0]_A<br>CB[7:0]_B         | DIMM ECC check bits(CB) channel A and B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| DQS[9:0]_A_t<br>DQS[9:0]_B_t   | SDRAM data strobes<br>(positive line of differential pair)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| DQS[9:0]_A_c<br>DQS[9:0]_B_c   | SDRAM data strobes<br>(negative line of differential pair)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| TDQS[9:5]_A_t<br>TDQS[9:5]_B_t | Not valid for x4 operation.<br>Enabled via Mode Register.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| TDQS[9:5]_A_c<br>TDQS[9:5]_B_c | Not valid for x4 operation.<br>Enabled via Mode Register.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| VIN_BULK                       | DIMM Power Supply from system to PMIC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| VIN_MGMT                       | DIMM Power Supply from system to PMIC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| VSS                            | Power supply return (ground)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| RFU                            | Reserved for future use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |
| LBDQS                          | Loopback data strobe output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |             |                          |                                      |                        |                                         |                              |                                                            |                              |                                                            |                                |                                                           |                                |                                                           |          |                                       |          |                                       |     |                              |     |                         |       |                             |

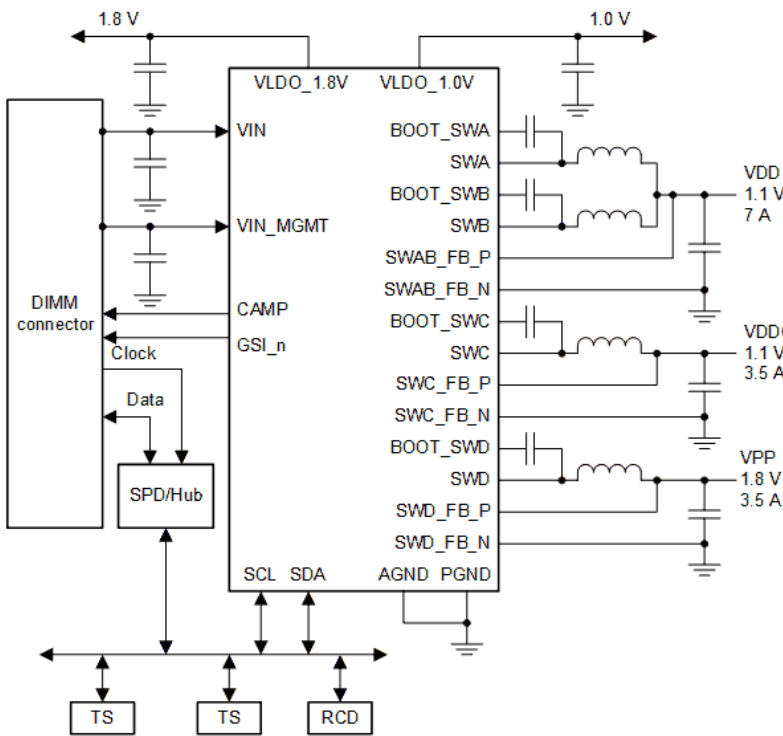
**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>claim 16 of the '918, and a claim analysis is just where I actually look at the claims and compare it to exactly how the products operate, the accused products.”), 381:24–25 (“Q: And what claim did you analyze next? A: Next, I turned to claim 18 . . .”), 344:17–19 (“Q: Let’s turn to claim 5. Have you shown evidence of overvoltage monitoring? A: Yes. Indeed, I have in my analysis of claim 16.”). Netlist previously asserted claim 16 of U.S. Patent No. 11,232,054 against Samsung’s DDR5 DIMMs, which requires among other things, “a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system,” “a voltage conversion circuit [which is part of a PMIC] coupled to the PCB,” and “a controller coupled to the PCB, the controller including a voltage monitor circuit coupled to an input voltage received from the host system via the interface, the voltage monitor circuit configured to detect an amplitude change in the input voltage.” <i>Id.</i> at 345:12–13 (“Q: which claim did you analyze next? A: Next, I turned to claim 16 of the '054 Patent.”).</p> <p>Samsung did not dispute these limitations. <i>Id.</i> at 885:24–886:9 (“With regard to the '918 claims, I have determined that—that these claims are not infringed by the Samsung DDR5 product, at least for the reason that there is a limitation in each of the independent claims directed to the requirement of a converter circuit structure, and that is not found on any of these Samsung DDR5 products. And what’s being pointed to is a linear drop-out regulator, but that’s not a converter circuit. As relates to the '054 claims, the DDR5 product has a normal operating state. The other state is it’s turned off. So it does not have a second operable state as required by the '054 claims.”).</p> |

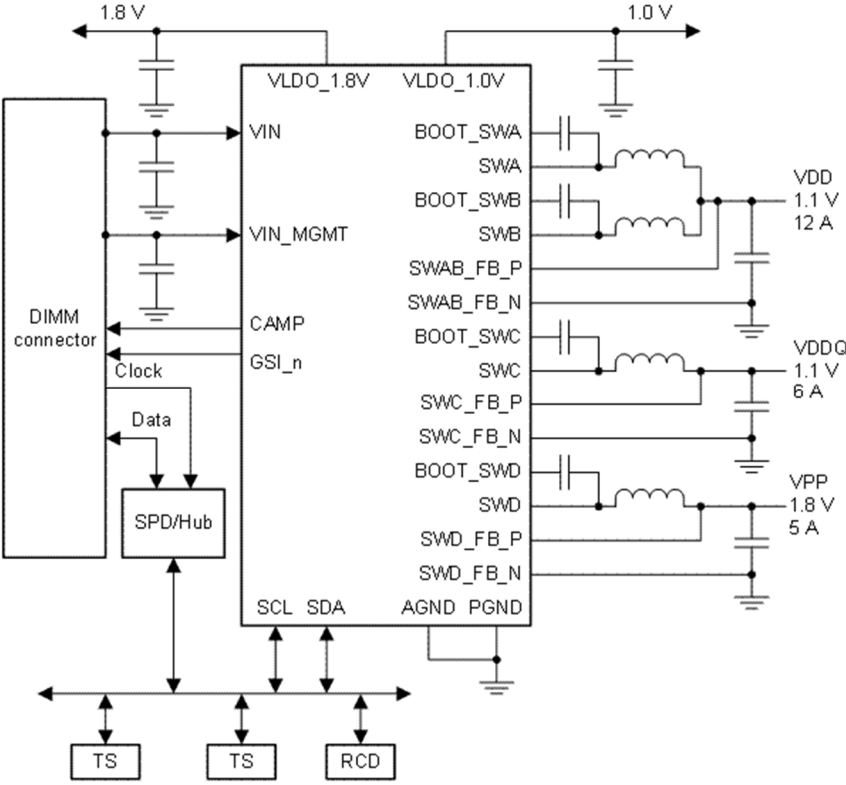
# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |         |                               |             |          |             |                  |                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------|-------------------------------|-------------|----------|-------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>[1.b.2]</b> the voltage monitor circuit configured to: (i) monitor an input voltage supplied via the input voltage supply line; (ii) generate a trigger signal upon detecting a trigger condition; and (iii) transmit the trigger signal to at least one other portion of the controller,</p> <p>wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the nonvolatile memory, and</p> <p>wherein the trigger condition occurs when the input voltage exceeds a first threshold voltage;</p> | <p>As the evidence cited for elements 1b and 1b.1 demonstrates, Samsung’s DDR5 DIMMs includes a PMIC having a voltage monitor that is configured to monitor an input voltage supplied via the input voltage supply line where the trigger occurs when the input voltage exceeds a first threshold voltage (<i>i.e.</i>, when there is an input over-voltage condition) because otherwise, it would not be commercially viable for the voltage monitor circuit to provide the featured input overvoltage lockout (OVLO) or overvoltage protection. <i>See, e.g.</i>,</p> <p>Ex. CC-131 (“S2FPD01 Brochure”) at 2 (annotated):</p> <table border="1"> <thead> <tr> <th></th><th>S2FPD01</th></tr> </thead> <tbody> <tr> <td>VIN_BULK (VINA/INB/VINC/VIND)</td><td>4.25V ~ 15V</td></tr> <tr> <td>VIN_MGMT</td><td>3.0V ~ 3.6V</td></tr> <tr> <td>Power Management</td><td>           Input UVLO/OVLO, Thermal Shut Down function<br/>           Read/Write I2C/I3C for Register Control<br/>           Interface up to 1MHz for I2C and 12.5MHz for I3C<br/>           Error Log Count &amp; Storage<br/>           Built-in Nonvolatile Memory registers<br/>           Dynamic Voltage Scaling controlled by I2C/I3C<br/>           Programmable Soft Start &amp; Soft Stop Time<br/>           Current Limit, Over/Under voltage protection<br/>           Power Meter         </td></tr> </tbody> </table> <p>Ex. CC-75 (Renesas P8900 Short-Form Datasheet) (“VIN_Bulk input supply protection feature: Input overvoltage”).</p> <p>Ex. CC-138 (Renesas P8911 Short-Form Datasheet) at 1 (“VIN_Bulk input supply protection feature: Input overvoltage”).</p> <p>Ex. CC-27 (MPQ8896) (“Full protection features include under-voltage lockout (UVLO), over-current protection (OCP), under-voltage protection (UVP), and thermal shutdown.”); <i>see also</i> Ex. CC-28 (MPQ8895) (same); Ex. CC-123 (MPQ8895F) (same); Ex. CC-29 (MPQ8894) (same).</p> |  | S2FPD01 | VIN_BULK (VINA/INB/VINC/VIND) | 4.25V ~ 15V | VIN_MGMT | 3.0V ~ 3.6V | Power Management | Input UVLO/OVLO, Thermal Shut Down function<br>Read/Write I2C/I3C for Register Control<br>Interface up to 1MHz for I2C and 12.5MHz for I3C<br>Error Log Count & Storage<br>Built-in Nonvolatile Memory registers<br>Dynamic Voltage Scaling controlled by I2C/I3C<br>Programmable Soft Start & Soft Stop Time<br>Current Limit, Over/Under voltage protection<br>Power Meter |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | S2FPD01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |         |                               |             |          |             |                  |                                                                                                                                                                                                                                                                                                                                                                              |
| VIN_BULK (VINA/INB/VINC/VIND)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 4.25V ~ 15V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |         |                               |             |          |             |                  |                                                                                                                                                                                                                                                                                                                                                                              |
| VIN_MGMT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3.0V ~ 3.6V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |         |                               |             |          |             |                  |                                                                                                                                                                                                                                                                                                                                                                              |
| Power Management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Input UVLO/OVLO, Thermal Shut Down function<br>Read/Write I2C/I3C for Register Control<br>Interface up to 1MHz for I2C and 12.5MHz for I3C<br>Error Log Count & Storage<br>Built-in Nonvolatile Memory registers<br>Dynamic Voltage Scaling controlled by I2C/I3C<br>Programmable Soft Start & Soft Stop Time<br>Current Limit, Over/Under voltage protection<br>Power Meter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |         |                               |             |          |             |                  |                                                                                                                                                                                                                                                                                                                                                                              |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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|         | <p>Ex. CC-124 (TPS53832A Product Overview) &amp; Ex. CC-125 (TPS53832A Data Sheet) (“The converter provides a full set of telemetry, including input voltage, output voltage, and output current. [O]vervoltage, undervoltage, overcurrent limit and over-temperature protections are provided as well.”).</p>  <p>Ex. CC-126 (TPS53830A Product Overview) &amp; Ex. CC-127 (TPS53830A Data Sheet) (“The converter provides a full set of telemetry, including input voltage, output voltage, and output current. [O]vervoltage, undervoltage, overcurrent limit and over-temperature protections are provided as</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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|         | <p>well.”).</p>  <p>See also Ex. CC-128 (TPS53830 Product Overview); Ex. CC-129 (TPS53830 Data Sheet); Ex. CC-130 (TPS53832 Product Overview); Ex. CC-26 (TPS53832 Data Sheet).</p> <p>Additionally, the voltage monitor circuit is configured to generate a trigger signal upon detecting a trigger condition and transmit the trigger signal to at least one other portion of the controller. For example, as noted in connection with element 1b, the PMICs used by Samsung are allegedly compliant</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1               | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           |            |                |                     |                 |                     |                 |              |                     |         |         |         |           |    |     |     |                       |         |         |         |         |     |     |     |                       |         |         |         |         |    |      |     |     |     |     |     |     |     |     |     |
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|                       | <p>with JEDEC’s PMIC standards.</p> <p>According to the JEDEC Power Management IC specifications, an over-voltage condition would trigger a VR disable command. <i>See</i> Ex. CC-85 (JESD301-1) at 36 and 37, Table 30; Ex. CC-81 (JESD301-1A) at 35, Table 27; Ex. CC-83 (JESD301-4A) at 35, Table 23; Ex. CC- 137 (JESD301-5) at 45-46, Table 27.</p> <p><b>8.13 Function Interrupt - PWR_GOOD and GSI_n Output Signals</b></p> <p>This section defines the output functionality of GSI_n pin and PWR_GOOD (Table 145, “Register 0x32” [5] = ‘0’) pin.</p> <p>When mask register bits are not set, the PMIC asserts its GSI_n output and PWR_GOOD output signals as shown in Table 30 when any event occurs. The table also highlights 11 events that causes PMIC to generate internally VR Disable command. For remaining events that does not trigger internal VR Disable command, the PMIC continues to operate as normal.</p> <p>Ex. CC-85 (JESD301-1) at 36 (annotated); Ex. CC-81 (JESD301-1A) at 35; <i>see also</i> Ex. CC-83 (JESD301-4A) at 35; Ex. CC-137 (JESD301-5) at 45.</p> <p><b>Table 30 — Events Interrupt Summary</b></p> <table><tr><th>Event</th><th>Status Bit</th><th>Clear Bit</th><th>Mask Bit</th><th>Threshold Bits</th><th>Trigger VR Disable?</th><th>PWR_GOOD Output</th><th>GSI_n Output</th></tr><tr><td>VIN Bulk Power Good</td><td>R08 [7]</td><td>R10 [7]</td><td>R15 [7]</td><td>R1A [7:5]</td><td>No</td><td>Low</td><td>Low</td></tr><tr><td>VIN Bulk Over Voltage</td><td>R08 [0]</td><td>R10 [0]</td><td>R15 [0]</td><td>R1B [7]</td><td>Yes</td><td>Low</td><td>Low</td></tr><tr><td>VIN Mgmt Over Voltage</td><td>R08 [1]</td><td>R10 [1]</td><td>R15 [1]</td><td>R1B [5]</td><td>No</td><td>High</td><td>Low</td></tr><tr><td>...</td><td>...</td><td>...</td><td>...</td><td>...</td><td>...</td><td>...</td><td>...</td></tr></table> <p>Ex. CC-85 (JESD301-1) at 37 (annotated) (showing that input voltage monitoring is performed for both VIN_BULK and VIN_MGMT); Ex. CC-81 (JESD301-1A) at 35 (same); <i>see also</i> Ex. CC-83 (JESD301-4A) at 35 (Table 23); Ex. CC-137 (JESD301-5) at 45-46.</p> <p>The VR Disable Command, which itself can be considered a trigger signal, is generated in response to a trigger signal—such as an input over-voltage detection or alert signals output by the voltage monitor circuit (<i>e.g.</i>, as a result of a comparison by a comparator of the detected input voltage to a reference voltage corresponding to a threshold voltage set by an overvoltage threshold register).</p> | Event     | Status Bit | Clear Bit      | Mask Bit            | Threshold Bits  | Trigger VR Disable? | PWR_GOOD Output | GSI_n Output | VIN Bulk Power Good | R08 [7] | R10 [7] | R15 [7] | R1A [7:5] | No | Low | Low | VIN Bulk Over Voltage | R08 [0] | R10 [0] | R15 [0] | R1B [7] | Yes | Low | Low | VIN Mgmt Over Voltage | R08 [1] | R10 [1] | R15 [1] | R1B [5] | No | High | Low | ... | ... | ... | ... | ... | ... | ... | ... |
| Event                 | Status Bit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Clear Bit | Mask Bit   | Threshold Bits | Trigger VR Disable? | PWR_GOOD Output | GSI_n Output        |                 |              |                     |         |         |         |           |    |     |     |                       |         |         |         |         |     |     |     |                       |         |         |         |         |    |      |     |     |     |     |     |     |     |     |     |
| VIN Bulk Power Good   | R08 [7]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | R10 [7]   | R15 [7]    | R1A [7:5]      | No                  | Low             | Low                 |                 |              |                     |         |         |         |           |    |     |     |                       |         |         |         |         |     |     |     |                       |         |         |         |         |    |      |     |     |     |     |     |     |     |     |     |
| VIN Bulk Over Voltage | R08 [0]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | R10 [0]   | R15 [0]    | R1B [7]        | Yes                 | Low             | Low                 |                 |              |                     |         |         |         |           |    |     |     |                       |         |         |         |         |     |     |     |                       |         |         |         |         |    |      |     |     |     |     |     |     |     |     |     |
| VIN Mgmt Over Voltage | R08 [1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | R10 [1]   | R15 [1]    | R1B [5]        | No                  | High            | Low                 |                 |              |                     |         |         |         |           |    |     |     |                       |         |         |         |         |     |     |     |                       |         |         |         |         |    |      |     |     |     |     |     |     |     |     |     |
| ...                   | ...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ...       | ...        | ...            | ...                 | ...             | ...                 |                 |              |                     |         |         |         |           |    |     |     |                       |         |         |         |         |     |     |     |                       |         |         |         |         |    |      |     |     |     |     |     |     |     |     |     |

**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |                                                                                                                                           |  |  |      |           |         |             |   |    |   |                                                                                                                                           |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------|--|--|------|-----------|---------|-------------|---|----|---|-------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>Table 122 — Register 0x1B</p> <table><tr><th colspan="4">R1B</th></tr><tr><th>Bits</th><th>Attribute</th><th>Default</th><th>Description</th></tr><tr><td>7</td><td>RW</td><td>0</td><td>R1B [7]: VIN_BULK_OVER_VOLTAGE_THRESHOLD<br/>VIN Bulk Input Over Voltage Threshold Setting For GSI_n Assertion<br/>0 = 14.5 V<br/>1 = 16.0 V</td></tr></table> <p>Ex. CC-85 (JESD301-1) at 136 (annotated); Ex. CC-81 (JESD301-1A) at 135; <i>see also</i> Ex. CC-83 (JESD301-4A) at 133, Table 113; Ex. CC-137 (JESD301-5) at 159, Table 132.</p> <p>The trigger signal generated by the voltage monitor circuit is configured to be transmitted by the voltage monitor circuit to other circuitry within the PMIC, e.g., to logic or state machines to determine whether VR Disable Command should issue, and/or to circuitry that would trigger writing to real time status registers (<i>see</i> Ex. CC-81 (JESD301-1A) at 116, Table 98, “Register 0x08”) and/or error log registers. In addition, the VR Disable Command is also transmitted to another portion of the controller, such as the controller circuitry for buck converters to execute a power down sequence in response to the generated VR Disable command. Error log registers would include 0x04 for global error count, and bits 5 and 7 of registers 0x04 would be updated in response to an input over-voltage condition trigger signal. Error log registers would also include 0x05 whose bits [0:2] would be updated to 100 in response to an input over-voltage condition trigger signal.</p> <p>The PMIC is configured to perform, in response to the trigger signal signifying detection of an input over-voltage condition, a write operation to write data into the non-volatile memory, such as duplicating the contents of 0x04 and 0x05 to the corresponding non-volatile memory space.</p> | R1B     |                                                                                                                                           |  |  | Bits | Attribute | Default | Description | 7 | RW | 0 | R1B [7]: VIN_BULK_OVER_VOLTAGE_THRESHOLD<br>VIN Bulk Input Over Voltage Threshold Setting For GSI_n Assertion<br>0 = 14.5 V<br>1 = 16.0 V |
| R1B     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |                                                                                                                                           |  |  |      |           |         |             |   |    |   |                                                                                                                                           |
| Bits    | Attribute                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Default | Description                                                                                                                               |  |  |      |           |         |             |   |    |   |                                                                                                                                           |
| 7       | RW                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0       | R1B [7]: VIN_BULK_OVER_VOLTAGE_THRESHOLD<br>VIN Bulk Input Over Voltage Threshold Setting For GSI_n Assertion<br>0 = 14.5 V<br>1 = 16.0 V |  |  |      |           |         |             |   |    |   |                                                                                                                                           |

## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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|         | <p><b>8.15 Input Over Voltage Protection</b></p> <p>An input over voltage protection mechanism is implemented to limit the voltages to the PMIC. The PMIC actively monitors the input voltage VIN_Bulk and VIN_Mgmt rail.</p> <p>There are two possibilities where PMIC recognizes the input over voltage event.</p> <ol style="list-style-type: none"> <li>1. VIN_Mgmt input goes above the threshold set in register Table 122, "Register 0x1B" [5].</li> <li>2. VIN_Bulk input goes above the threshold set in register Table 122, "Register 0x1B" [7].</li> </ol> <p>When either one or both event occurs for a period longer than tInput_OV_GSI_Assertion time then PMIC sets the register Table 103, "Register 0x08" [1:0] accordingly and drives GSI_n output signal as shown in Table 30 at the same time. Note that at this point, the PMIC does not assert PWR_GOOD output signal. The PMIC allows access to all registers and PMIC continues to operate as normal. The host is responsible for taking any specific action. The host may clear the VIN_Mgmt or VIN_Bulk input over voltage status register by writing '1' to register Table 111, "Register 0x10" [1:0] appropriately or by writing '1' to global status clear register Table 115, "Register 0x14" [0]. If the input over voltage condition is still present then PMIC will continue to assert GSI_n output signal and the status register Table 103, "Register 0x08" [1:0] will remain at '1'.</p> <p>In programmable mode (i.e., Table 142, "Register 0x2F" [2] = '1'), if VIN_Bulk input supply over voltage condition persists greater than tInput_OV_VR_Disable time then PMIC internally generates VR Disable command and disables all of its switching output regulators and asserts PWR_GOOD signal. The PMIC keeps its VOUT_1.8V and</p> <p>Ex. CC-85 (JESD301-1) at 40 (annotated); Ex. CC-81 (JESD301-1A) at 38; Ex. CC-83 (JESD301-4A) at 38-39, Section 1.8.15; Ex. CC-137 (JESD301-5) at 50-51, Section 6.15.</p> <p>For example, when an input over-voltage conditions, at least bits 5 and 7 of 0x04 and bits 0-2 of 0x05 are updated; and the contents of 0x04 and 0x05 are written into non-volatile memory.</p> |

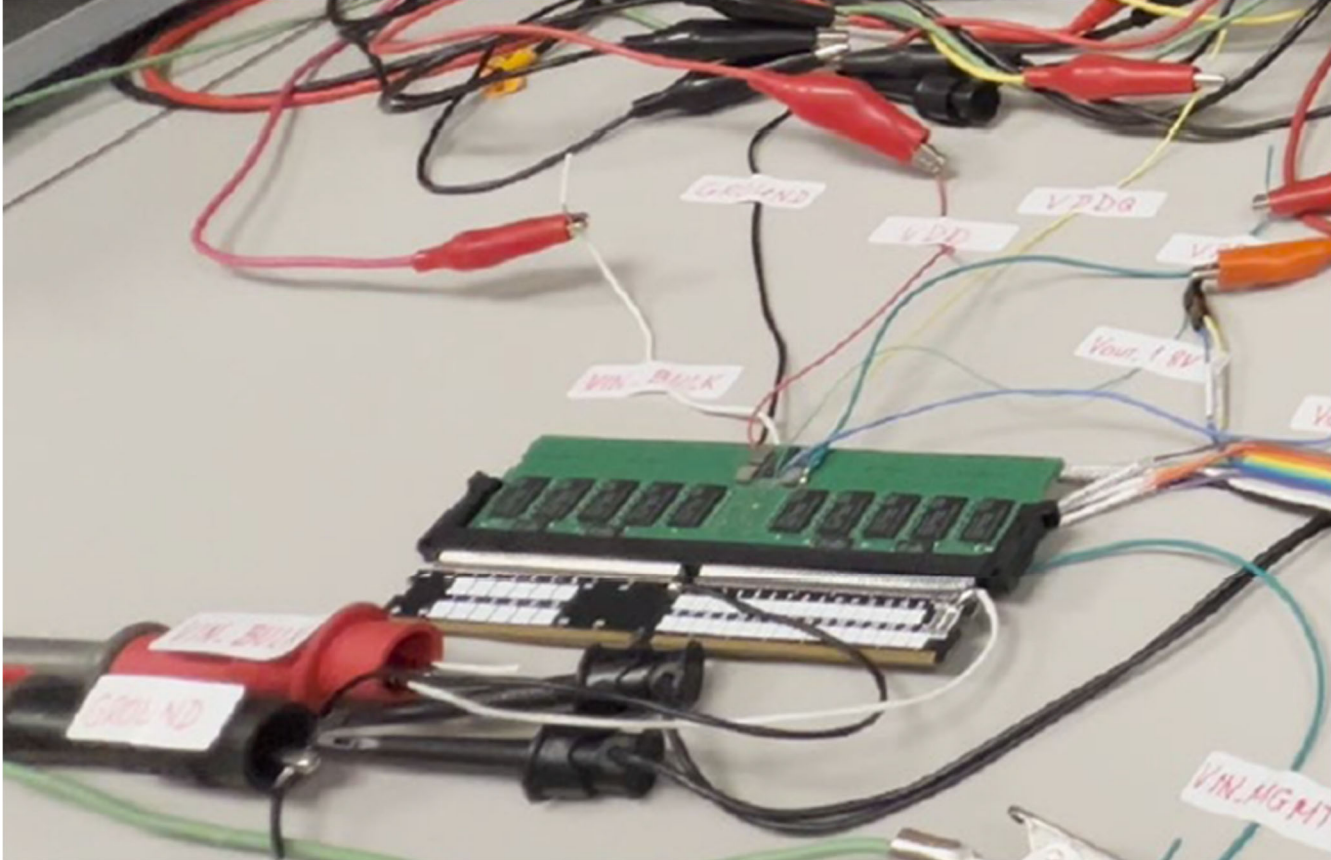
**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |         |                                                                                                                                                                                                               |  |  |      |           |         |                            |   |     |   |                                                                                                                                                                                                               |   |     |   |                                                                                                                                                                                        |   |     |   |                                                                                                                                                                          |   |     |   |                                                                                                                                                                        |     |    |   |                     |
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|         | <p>Table 99 — Register 0x04</p> <table><tr><th colspan="4">R04</th></tr><tr><th>Bits</th><th>Attribute</th><th>Default</th><th>Description<sup>1,2</sup></th></tr><tr><td>7</td><td>ROE</td><td>0</td><td><b>R04 [7]: GLOBAL_ERROR_COUNT</b><br/>Global Error Count Since Last Erase Operation<sup>3</sup><br/>0 = No Error or Only 1 Error since last Erase operation<br/>1 = &gt; 1 Error Count since last Erase operation</td></tr><tr><td>6</td><td>ROE</td><td>0</td><td><b>R04 [6]: GLOBAL_ERROR_LOG_BUCK_OV_OR_UV</b><br/>Global Error Log History for Buck Regulator Output Over or Under Voltage<sup>4</sup><br/>0 = No Error Occurred<br/>1 = Error Occurred</td></tr><tr><td>5</td><td>ROE</td><td>0</td><td><b>R04 [5]: GLOBAL_ERROR_LOG_VIN_BULK_OVER_VOLTAGE</b><br/>Global Error Log History for VIN_Bulk Over Voltage<sup>4</sup><br/>0 = No Error Occurred<br/>1 = Error Occurred</td></tr><tr><td>4</td><td>ROE</td><td>0</td><td><b>R04 [4]: GLOBAL_ERROR_LOG_CRITICAL_TEMPERATURE</b><br/>Global Error Log History for Critical Temperature<sup>4</sup><br/>0 = No Error Occurred<br/>1 = Error Occurred</td></tr><tr><td>3:0</td><td>RV</td><td>0</td><td>R04 [3:0]: Reserved</td></tr></table> <p>1. The PMIC always attempts to write this register into its non-volatile memory. However, it may not be guaranteed depending on how fast the host may shut off the input power to the PMIC. The PMIC needs minimum of 5.0V for VIN_Bulk voltage and 100 ms duration from PWR_GOOD signal assertion to guarantee the write operation into non-volatile memory.</p> <p>2. Host must explicitly perform Erase operation to erase this entire register Table 99, “Register 0x04” [7:0] via Table 151, “Register 0x39”. The PMIC needs minimum of 100 ms for Erase operation.</p> <p>3. PMIC counts the error since last erase operation and if more than one error occurs, it sets this bit to ‘1’. Host must explicitly perform Erase operation to erase this entire register Table 99, “Register 0x04” [7:0].</p> <p>4. PMIC sets the bit when error occurs.</p> <p>Ex. CC-85 (JESD301-1) at 112 (annotated); Ex. CC-81 (JESD301-1A at 112); Ex. CC-83 (JESD301-4A) at 110, Table 90; Ex. CC-137 (JESD301-5) at 137, Table 110.</p> | R04     |                                                                                                                                                                                                               |  |  | Bits | Attribute | Default | Description <sup>1,2</sup> | 7 | ROE | 0 | <b>R04 [7]: GLOBAL_ERROR_COUNT</b><br>Global Error Count Since Last Erase Operation <sup>3</sup><br>0 = No Error or Only 1 Error since last Erase operation<br>1 = > 1 Error Count since last Erase operation | 6 | ROE | 0 | <b>R04 [6]: GLOBAL_ERROR_LOG_BUCK_OV_OR_UV</b><br>Global Error Log History for Buck Regulator Output Over or Under Voltage <sup>4</sup><br>0 = No Error Occurred<br>1 = Error Occurred | 5 | ROE | 0 | <b>R04 [5]: GLOBAL_ERROR_LOG_VIN_BULK_OVER_VOLTAGE</b><br>Global Error Log History for VIN_Bulk Over Voltage <sup>4</sup><br>0 = No Error Occurred<br>1 = Error Occurred | 4 | ROE | 0 | <b>R04 [4]: GLOBAL_ERROR_LOG_CRITICAL_TEMPERATURE</b><br>Global Error Log History for Critical Temperature <sup>4</sup><br>0 = No Error Occurred<br>1 = Error Occurred | 3:0 | RV | 0 | R04 [3:0]: Reserved |
| R04     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |         |                                                                                                                                                                                                               |  |  |      |           |         |                            |   |     |   |                                                                                                                                                                                                               |   |     |   |                                                                                                                                                                                        |   |     |   |                                                                                                                                                                          |   |     |   |                                                                                                                                                                        |     |    |   |                     |
| Bits    | Attribute                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Default | Description <sup>1,2</sup>                                                                                                                                                                                    |  |  |      |           |         |                            |   |     |   |                                                                                                                                                                                                               |   |     |   |                                                                                                                                                                                        |   |     |   |                                                                                                                                                                          |   |     |   |                                                                                                                                                                        |     |    |   |                     |
| 7       | ROE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0       | <b>R04 [7]: GLOBAL_ERROR_COUNT</b><br>Global Error Count Since Last Erase Operation <sup>3</sup><br>0 = No Error or Only 1 Error since last Erase operation<br>1 = > 1 Error Count since last Erase operation |  |  |      |           |         |                            |   |     |   |                                                                                                                                                                                                               |   |     |   |                                                                                                                                                                                        |   |     |   |                                                                                                                                                                          |   |     |   |                                                                                                                                                                        |     |    |   |                     |
| 6       | ROE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0       | <b>R04 [6]: GLOBAL_ERROR_LOG_BUCK_OV_OR_UV</b><br>Global Error Log History for Buck Regulator Output Over or Under Voltage <sup>4</sup><br>0 = No Error Occurred<br>1 = Error Occurred                        |  |  |      |           |         |                            |   |     |   |                                                                                                                                                                                                               |   |     |   |                                                                                                                                                                                        |   |     |   |                                                                                                                                                                          |   |     |   |                                                                                                                                                                        |     |    |   |                     |
| 5       | ROE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0       | <b>R04 [5]: GLOBAL_ERROR_LOG_VIN_BULK_OVER_VOLTAGE</b><br>Global Error Log History for VIN_Bulk Over Voltage <sup>4</sup><br>0 = No Error Occurred<br>1 = Error Occurred                                      |  |  |      |           |         |                            |   |     |   |                                                                                                                                                                                                               |   |     |   |                                                                                                                                                                                        |   |     |   |                                                                                                                                                                          |   |     |   |                                                                                                                                                                        |     |    |   |                     |
| 4       | ROE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0       | <b>R04 [4]: GLOBAL_ERROR_LOG_CRITICAL_TEMPERATURE</b><br>Global Error Log History for Critical Temperature <sup>4</sup><br>0 = No Error Occurred<br>1 = Error Occurred                                        |  |  |      |           |         |                            |   |     |   |                                                                                                                                                                                                               |   |     |   |                                                                                                                                                                                        |   |     |   |                                                                                                                                                                          |   |     |   |                                                                                                                                                                        |     |    |   |                     |
| 3:0     | RV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0       | R04 [3:0]: Reserved                                                                                                                                                                                           |  |  |      |           |         |                            |   |     |   |                                                                                                                                                                                                               |   |     |   |                                                                                                                                                                                        |   |     |   |                                                                                                                                                                          |   |     |   |                                                                                                                                                                        |     |    |   |                     |

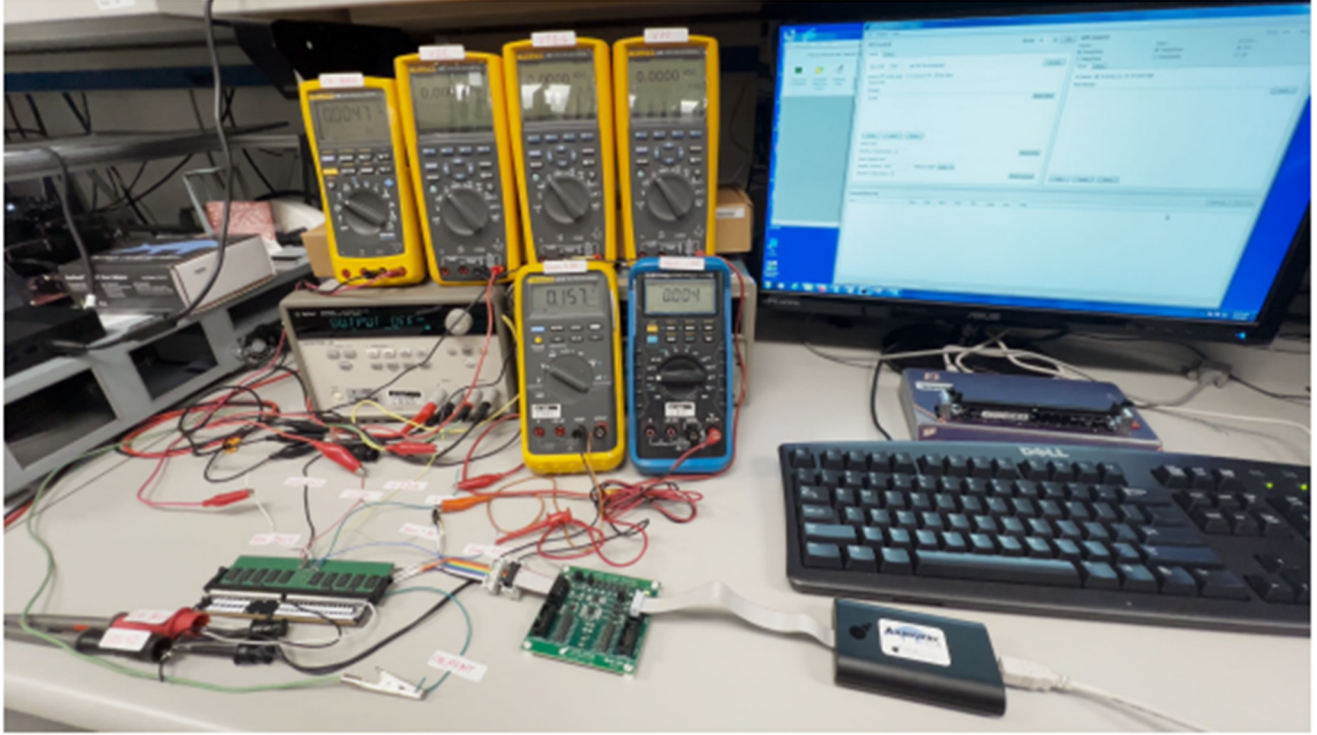
# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |  |      |           |         |                            |     |     |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|------|-----------|---------|----------------------------|-----|-----|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <div>Table 100 — Register 0x05</div> <table><tr><th colspan="4">R05</th></tr><tr><th>Bits</th><th>Attribute</th><th>Default</th><th>Description<sup>1,2</sup></th></tr><tr><td>2:0</td><td>ROE</td><td>0</td><td><div>R05 [2:0]: PMIC_ERROR_LOG<br/>PMIC Power On - High Level Status Bit to Previous Last Known Power Cycle<br/>000 = Normal Power On<br/>001 = Reserved<br/>010 = Buck Regulator Output Over or Under Voltage<sup>5</sup><br/>011 = Critical Temperature<br/>100 = VIN_Bulk Input Over Voltage<br/>101 = Reserved<br/>110 = Reserved<br/>111 = Reserved</div><div>at least three bits are written when Vin_Bulk overvoltage is detected</div></td></tr></table> <div><div>1. The PMIC always attempts to write this register into its non-volatile memory. However, it may not be guaranteed depending on how fast the host may shut off the input power to the PMIC. The PMIC needs minimum of 5.0V for VIN_Bulk voltage and 100 ms duration from PWR_GOOD signal assertion to guarantee the write operation into non-volatile memory.</div><div>2. This entire register status reflects previous power down cycle of the PMIC and is updated by the PMIC on its own at each power cycle, if update is needed. Because this register is updated only if there is an update needed, there is no NVM life time impact. This register is cleared when host issues the erase command via Table 151, “Register 0x39”. The PMIC needs minimum of 100 ms for Erase operation.</div><div>3. This register is set only if PMIC generates internal VR Disable command due to fault condition.</div></div> <div>Ex. CC-85 (JESD301-1) at 113 (annotated); Ex. CC-81 (JESD301-1A at 113); Ex. CC-83 (JESD301-4A) at 111, Table 91; Ex. CC-137 (JESD301-5) at 138, Table 111.</div> <div>Testing shows that various RDIMMs with Samsung PMIC, Renesas PMIC, TI PMIC, Rambus PMIC and MPS PMICs updated the 0x04 and 0x5 registers in the event of an input over-voltage condition, and these changes were retained after a power-down-and-reboot cycle, indicating that the contents in 0x04 and 0x05 registers had been successfully written to non-volatile memory.</div> <div>The below test setup shows a memory module, DIMM extender, and test connections to the various pins on</div> | R05     |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |  | Bits | Attribute | Default | Description <sup>1,2</sup> | 2:0 | ROE | 0 | <div>R05 [2:0]: PMIC_ERROR_LOG<br/>PMIC Power On - High Level Status Bit to Previous Last Known Power Cycle<br/>000 = Normal Power On<br/>001 = Reserved<br/>010 = Buck Regulator Output Over or Under Voltage<sup>5</sup><br/>011 = Critical Temperature<br/>100 = VIN_Bulk Input Over Voltage<br/>101 = Reserved<br/>110 = Reserved<br/>111 = Reserved</div> <div>at least three bits are written when Vin_Bulk overvoltage is detected</div> |
| R05     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |  |      |           |         |                            |     |     |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Bits    | Attribute                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Default | Description <sup>1,2</sup>                                                                                                                                                                                                                                                                                                                                                                                                                      |  |  |      |           |         |                            |     |     |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2:0     | ROE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0       | <div>R05 [2:0]: PMIC_ERROR_LOG<br/>PMIC Power On - High Level Status Bit to Previous Last Known Power Cycle<br/>000 = Normal Power On<br/>001 = Reserved<br/>010 = Buck Regulator Output Over or Under Voltage<sup>5</sup><br/>011 = Critical Temperature<br/>100 = VIN_Bulk Input Over Voltage<br/>101 = Reserved<br/>110 = Reserved<br/>111 = Reserved</div> <div>at least three bits are written when Vin_Bulk overvoltage is detected</div> |  |  |      |           |         |                            |     |     |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>the DIMM (including VIN_BULK and VIN_MGMT, among others):</p>  A photograph showing a DIMM module (green circuit board with black chips) connected to various test wires. The wires are labeled with handwritten text on small white tags. Labels include: "GROUND" (multiple locations), "VDD", "VDDq", "VIN_BULK", "VIN_MGMT", "Vout: 1.8V", and "VIN_MGMT". The wires are connected to the DIMM module and other components, demonstrating the use of the DIMM in a test setup. |

U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                         |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |  <p data-bbox="562 1084 1339 1117">The testing procedures and observations are summarized below:</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Step | Action                                                                                                             | Observation                                                                                                                                                                                                                                                                                                 |
|------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Power Supply output disabled                                                                                       | <div> VIN_Bulk 0.00 V<br/> VIN_Mgmt 0.00 V<br/> VOUT_1.8V (VDDSPD) 0.00 V<br/> VOUT_1.0V (VIO) 0.00 V<br/> SWA/B (VDD) 0.00 V<br/> SWC (VDDQ) 0.00 V<br/> SWD (VPP) 0.00 V </div>                                                                                                                           |
| 2    | Enable the power supplies to output 12.00V to the VIN_Bulk pins and 3.3V to VIN_Mgmt pins of the DIMM              | <div> VIN_Bulk 12.00 V<br/> VIN_Mgmt 3.30 V<br/> VOUT_1.8V (VDDSPD) 1.80 V<br/> VOUT_1.0V (VIO) 1.00 V<br/> SWA/B (VDD) 0.00 V<br/> SWC (VDDQ) 0.00 V<br/> SWD (VPP) 0.00 V </div> <p>PMIC LDO outputs are active.<br/>Switching regulators are disabled.<br/>This is the expected PMIC power on state.</p> |
| 3    | Using the I <sup>2</sup> C Master Read PMIC Registers (0x00 – 0x0F)                                                | Read 16 Bytes (Registers 0x00 – 0x0F):<br>00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00<br>No Error conditions                                                                                                                                                                                            |
| 4    | Using the I <sup>2</sup> C Master Write the VR_ENABLE bit in the PMIC Register 32 Bit 7: VR_ENABLE 1 = PMIC Enable | 0x32 <= 0x80<br><div> VIN_Bulk 12.00 V<br/> VIN_Mgmt 3.30 V<br/> VOUT_1.8V (VDDSPD) 1.80 V<br/> VOUT_1.0V (VIO) 1.00 V<br/> SWA/B (VDD) 1.10 V<br/> SWC (VDDQ) 1.10 V<br/> SWD (VPP) 1.80 V </div> <p>Switching regulators are enabled.</p>                                                                 |
| 5    | Using the I <sup>2</sup> C Master Read PMIC Registers (0x00 – 0x0F)                                                | Read 16 Bytes Registers 0x00 – 0x0F):<br>00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00<br>No error conditions                                                                                                                                                                                             |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Step | Action                                                                                                              | Observation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6    | Increase the Power Supply output voltage connected to the VIN_Bulk pins until an over-voltage condition is detected | <p>VIN_Bulk 12.00 V =&gt; 14.50 V</p> <p>VIN_Mgmt 3.30 V</p> <p>VOUT_1.8V (VDDSPD) 1.80 V</p> <p>VOUT_1.0V (VIO) 1.00 V</p> <p>SWA/B (VDD) 1.10 V =&gt; 0.00 V</p> <p>SWC (VDDQ) 1.10 V =&gt; 0.00 V</p> <p>SWC (VPP) 1.80 V =&gt; 0.00 V</p> <p>Once the overvoltage threshold is reached, the switching regulators (SWA/B, SWB, SWC) turn off.<br/>The LDO outputs are still active.</p>                                                                                                                                             |
| 7    | Using the I <sup>2</sup> C Master Read PMIC Registers (0x00 – 0x0F)                                                 | <p>Read 16 Bytes (Registers 0x00 – 0x0F):<br/>00 00 00 00 20 04 00 00 01 00 02 00 00 00 00 00</p> <p>Register 0x04 = 0x20 =&gt; R04 [5]:<br/>GLOBAL_ERROR_LOG_VIN_BULK_OVER_VOLTAGE<br/>1 = Error Occurred</p> <p>Register 0x05 = 0x04 =&gt; R05 [2:0]: PMIC_ERROR_LOG<br/>100 = VIN_Bulk Input Over Voltage</p> <p>Register 0x08 = 0x01 =&gt; R08 [0]:<br/>VIN_BULK_INPUT_OVER_VOLTAGE_STATUS<br/>1 = Over Voltage</p> <p>Register 0x0A = 0x02 =&gt; R0A [2]:<br/>IBI_AND_GLOBAL_STATUS<br/>1 = Pending IBI or Outstanding Status</p> |
| 8    | Disable the power supply output => 0.00V to the VIN_Bulk and VIN_Mgmt pins of the DIMM                              | <p>VIN_Bulk 0.00 V</p> <p>VIN_Mgmt 0.00 V</p> <p>VOUT_1.8V (VDDSPD) 0.00 V</p> <p>VOUT_1.0V (VIO) 0.00 V</p> <p>SWA/B (VDD) 0.00 V</p> <p>SWC (VDDQ) 0.00 V</p> <p>SWD (VPP) 0.00 V</p>                                                                                                                                                                                                                                                                                                                                                |
| 9    | Wait 30 s – 60 s                                                                                                    | Ensure the UUT is completely discharged                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Step | Action                                                                                                                                                                                              | Observation                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10   | Enable the power supplies to output 12.00V to the VIN_Bulk pins and 3.3V to VIN_Mgmt pins of the DIMM                                                                                               | VIN_Bulk 12.00 V<br>VIN_Mgmt 3.30 V<br>VOUT_1.8V (VDDSPD) 1.80 V<br>VOUT_1.0V (VIO) 1.00 V<br>SWA/B (VDD) 0.00 V<br>SWC (VDDQ) 0.00 V<br>SWD (VPP) 0.00 V<br><br>PMIC LDO outputs are active.<br>Switching regulators are disabled.<br>This is the expected PMIC power on state.                                                                                                                                                                                          |
| 11   | Using the I <sup>2</sup> C Master Read PMIC Registers (0x00 – 0x0F)                                                                                                                                 | Read 16 Bytes (Registers 0x00 – 0x0F):<br>00 00 00 00 20 04 00 00 00 00 00 00 00 00 00 00<br><br>Register 0x04 = 0x20 => R04 [5]:<br>GLOBAL_ERROR_LOG_VIN_BULK_OVER_VOLTAGE<br>1 = Error Occurred<br><br>Register 0x05 = 0x04 => R05 [2:0]: PMIC_ERROR_LOG<br>100 = VIN_Bulk Input Over Voltage<br><br>Notes:<br>Registers 0x04 and 0x05 have saved the over voltage fault condition.<br><br>Registers 0x08 and 0x0A are zero since the fault is not currently occurring. |
| 12   | Using the I <sup>2</sup> C Master Clear the NV Registers by writing 0x74 to Register 39 Bits [7:0]<br><br>Host Region Codes:<br>0x74: Clear Registers R04 to R07, Erase MTP memory for R04 Register | 0x39 <= 0x74                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 13   | Read PMIC Registers (0x00 – 0x0F)                                                                                                                                                                   | Read 16 Bytes (Registers 0x00 – 0x0F):<br>00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00<br><br>No Error conditions                                                                                                                                                                                                                                                                                                                                                      |
| 14   | Screen capture the I2C Transaction Log                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>As noted in Step 5, prior to the over-voltage event, the PMICs registers indicate that there are no error conditions. Step 7 confirms that the PMIC writes to the appropriate status registers following the input over-voltage condition. As noted in Steps 6-7 the error codes stored in the registers were able to be read via the I2C bus. This indicates that at least the I2C bus and the PMIC registers are powered on, which in turn, indicates that the memory module is in an operable state. Steps 8-11 confirm that the error codes written into the status registers maintain the same value upon disabling the power supply and re-powering, indicating that the error codes are stored in nonvolatile memory.</p> <p>The PMIC registers and bits referenced in the test procedure are noted in the following tables from JESD 301-1A:</p> |

## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

### 3.3.5.4 Threshold and Configuration Registers (cont'd)

Table 140 --- Register 0x32

| R32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |         |                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Attribute | Default | Description                                                                                                                                                                                                                                         |
| 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | RW        | 0       | R32 [7]: VR_ENABLE<br>PMIC Enable <sup>1,2,3,4</sup><br>0 = PMIC Disable<br>1 = PMIC Enable                                                                                                                                                         |
| 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | RO        | 0       | R32 [6]: MANAGEMENT_INTERFACE_SELECTION<br>PMIC Management Bus Interface Protocol Selection <sup>5</sup><br>0 = I <sup>2</sup> C Protocol (Max speed 1 MHz)<br>1 = I3C Basic Protocol                                                               |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | RW        | 1       | R32 [5]: EXECUTE_VR_ENABLE_CONTROL_<br>PMIC VR Enable Command Execution Control over I <sup>2</sup> C/I3C Bus <sup>4</sup><br>0 = Do Not Execute VR Enable Command; i.e., ignore bit [7] = '1' and keep it as '0'.<br>1 = Execute VR Enable Command |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | RW        | 0       | R32 [4]: EXECUTE_CAMP_FAIL_N_FUNCTION_CONTROL<br>PMIC CAMP Fail_n function (Transition from High to Low) Control<br>0 = Execute VR Disable Command<br>1 = Do Not Execute VR Disable Command                                                         |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | RW        | 0       | R32 [3]: CAMP_PWR_GOOD_OUTPUT_SIGNAL_CONTROL<br>PMIC CAMP PWR_GOOD Output Signal Control<br>0 = PMIC controls PWR_GOOD output on its own based on internal status<br>1 = PWR_GOOD Output Float <sup>6</sup>                                         |
| 2:0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | RV        | 00      | R32 [2:0]: Reserved                                                                                                                                                                                                                                 |
| <p>NOTE 1 Host sets this bit at first power on. After this bit is set, the PMIC executes Power On Sequence configuration 0 (Table 152, "Register 0x40") to Power On Sequence configuration 3 (Table 155, "Register 0x43") registers. At least one bit in Table 152, "Register 0x40" [6:3] must be set to '1' to issue VR Enable command.</p> <p>NOTE 2 The host shall ensure that prior to issuing VR Enable command, there is no pending IBI interrupt (i.e., Table 100, "Register 0x0A" [1] = '1') status. After host issues VR Enable command, the PMIC may NACK any I<sup>2</sup>C or I3C Basic bus transaction by host until tPMIC_PWR_GOOD_OUT timing parameter is satisfied. The host shall not access any device specific registers or issue any CCCs until tPMIC_PWR_GOOD_OUT parameter is satisfied. The PMIC device may request for an IBI during power up sequence (i.e., during tPMIC_PWR_GOOD_OUT time) if there is any event.</p> <p>NOTE 3 After host issues VR Enable command, the PMIC may NACK any I2C or I3C Basic bus transactions by host until tPMIC_PWR_GOOD_OUT timing parameter is satisfied.</p> <p>NOTE 4 Once Table 140, "Register 0x32" [7] is set to '1' via VR Enable command, the subsequent write to register Table 140, "Register 0x32" [5] = '0' is ignored by the PMIC. If there is a simultaneous write to register Table 140, "Register 0x32" [7,5] = '10', the PMIC prioritizes bit [5] and does not execute VR Enable command.</p> <p>NOTE 5 This register is automatically updated when SETAASA CCC or RSTDAAC CCC is registered by the PMIC device or when PMIC device goes through bus reset as described in clause 2.10.15 regardless of whether PMIC is in write protect mode or non write protect mode of operation. This register can be read by the Host through normal Read operation but it cannot be written with normal write operation either in I<sup>2</sup>C mode or I3C Basic mode of operation. When this register is updated, it takes in effect where there is a next START operation (i.e., after STOP operation).</p> <p>NOTE 6 When this encoding is set, the PMIC always floats the PWR_GOOD output signal even when there is an internal VR Disable command due to fault condition.</p> |           |         |                                                                                                                                                                                                                                                     |

## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

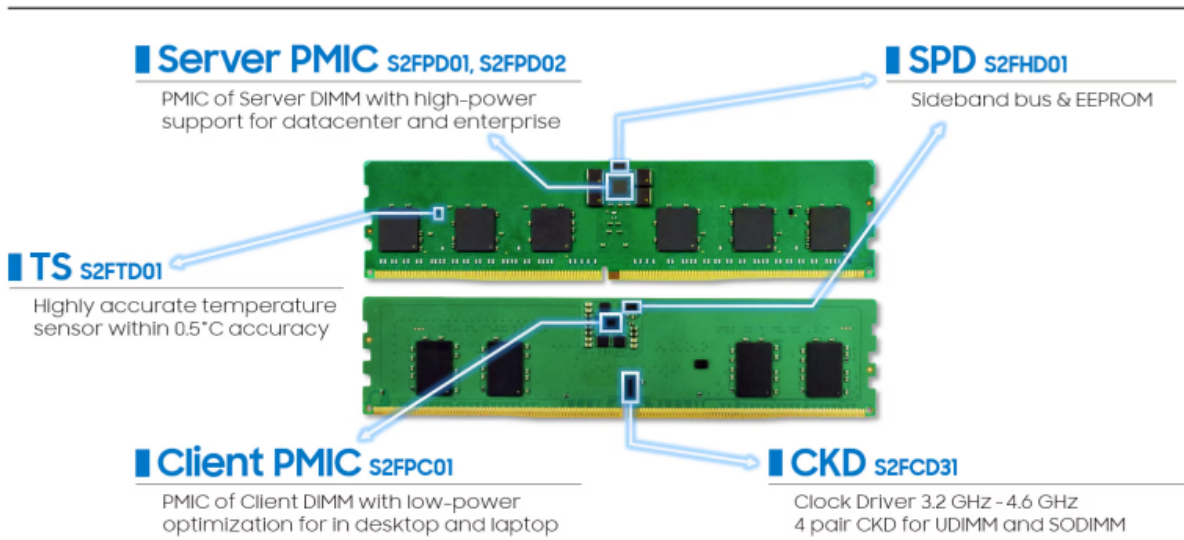
| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>Ex. CC-25 at 159; <i>see also id.</i> at 112-113, 116, 118, 135, 163 (Tables 94, 95, 98, 100, 117, 146).</p> <p>Moreover, Netlist previously asserted claims 16, and 18–19 of U.S. Patent No. 11,016,918 against Samsung’s DDR5 DIMMs, which recites, among other limitations, “<b>a voltage monitor circuit</b> configured to monitor an input voltage received via a first portion of the plurality of edge connections, <b>the voltage monitor circuit configured to produce a [trigger] signal in response to the input voltage having a voltage amplitude that is greater than a first threshold voltage,</b>” “<b>a controller [which is the other portion of the controller in the 366 patent] coupled to the voltage monitor circuit and configured to receive the signal,</b> wherein <b>the controller executes a write operation in response to the [trigger] signal,</b>” and “wherein the write operation includes <b>writing data information into non-volatile memory.</b>” ’918 Patent, claims 16, 18–19 (emphases added). <i>See, e.g.,</i> Ex. CC-21 (EDTX Trial Tr. Day 2, 4 (2023)) at 333:9–12 (“Q. And can you tell us about claim 16 of the ’918 Patent? A. So here’s claim 16 of the ’918, and a claim analysis is just where I actually look at the claims and compare it to exactly how the products operate, the accused products.”), 381:24–25 (“Q: And what claim did you analyze next? A: Next, I turned to claim 18 . . . .”), 341:3 (“Q: Let’s turn to ’918, claim 19.”). Netlist previously asserted claim 16 of U.S. Patent No. 11,232,054 against Samsung’s DDR5 DIMMs, which requires among other things, “a controller coupled to the PCB, the controller including a voltage monitor circuit coupled to an input voltage received from the host system via the interface, the voltage monitor circuit configured to detect an amplitude change in the input voltage, wherein, in response to the voltage monitor detecting an amplitude change in the input voltage, the memory module transitions from a first operable state to a second operable state.” <i>Id.</i> at 345:12–13 (“Q: which claim did you analyze next? A: Next, I turned to claim 16 of the ’054 Patent.”).</p> <p>Samsung did not dispute these limitations. <i>Id.</i> at 885:24–886:9 (“With regard to the ’918 claims, I have determined that—that these claims are not infringed by the Samsung DDR5 product, at least for the reason that there is a limitation in each of the independent claims directed to the requirement of a converter circuit structure, and that is not found on any of these Samsung DDR5 products. And what’s being pointed to is a linear drop-out regulator, but that’s not a converter circuit. As relates to the ’054 claims, the DDR5 product has a normal operating state. The other state is it’s turned off. So it does not have a second operable state as required by the ’054 claims.”).</p> |

**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

| Claim 1                                              | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>[1.c]</b> a set of components coupled to the PCB, | <p>The Accused Instrumentalities each include a set of components coupled to the PCB, such as DDR5 SDRAMs, RCD/MRCD (for RDIMMs and MRDIMMs), and data buffers (for MRDIMMs) as further explained in element 1d.2.</p> <p>Analysis below uses Samsung’s RDIMM format as an example, but MRDIMMs would operate in materially the same way in all relevant part. Ex. CC-13 (JEDEC Press Release) (JEDEC noting that other planned features for MRDIMMs include “Utilization of standard DDR5 DIMM components including DRAM, DIMM Form Factor &amp; Pinout, SPD, PMIC, and TS for ease of adoption”).</p> <p><i>See, e.g.</i>, Ex. CC-104 (Optimized DDR5 DIMM Solutions):</p> |

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>The diagram illustrates a system architecture with the following components and connections:</p> <ul style="list-style-type: none"><li><b>Legend:</b><ul style="list-style-type: none"><li>I2C/I3C line: Black double-headed arrow</li><li>Power line: Red arrow</li><li>DDR Clock line: Blue arrow</li></ul></li><li><b>Components:</b><ul style="list-style-type: none"><li><b>DRAM</b> (Teal box)</li><li><b>S2FCD31</b> (Yellow box inside a teal box)</li><li><b>PMIC</b> (Teal box containing <b>S2FPC01</b>, <b>S2FPD01</b>, and <b>S2FPD02</b> yellow boxes)</li><li><b>SPD/TS</b> (Teal box containing <b>S2FHD01</b> and <b>S2FTD01</b> yellow boxes)</li><li><b>CPU</b> (Teal box)</li></ul></li><li><b>Connections:</b><ul style="list-style-type: none"><li><b>Power lines (Red):</b> From an external source to S2FPC01, S2FPD01, S2FPD02, and DRAM. A CKD line also connects S2FPD01 to S2FCD31.</li><li><b>DDR Clock lines (Blue):</b> From CPU to S2FCD31, and from S2FCD31 to DRAM.</li><li><b>I2C/I3C lines (Black):</b> Bidirectional connections between S2FCD31 and DRAM, S2FCD31 and SPD/TS, and between S2FHD01 and S2FTD01.</li><li><b>Other connections:</b> A bidirectional I2C/I3C line between S2FPC01 and S2FHD01, and a bidirectional I2C/I3C line between S2FTD01 and CPU.</li></ul></li></ul> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p style="text-align: center;"><b>System LSI's DDR5 Solution</b></p>  <p><b>Server PMIC S2FPD01, S2FPD02</b><br/>PMIC of Server DIMM with high-power support for datacenter and enterprise</p> <p><b>SPD S2FHD01</b><br/>Sideband bus &amp; EEPROM</p> <p><b>TS S2FTD01</b><br/>Highly accurate temperature sensor within 0.5°C accuracy</p> <p><b>Client PMIC S2FPC01</b><br/>PMIC of Client DIMM with low-power optimization for in desktop and laptop</p> <p><b>CKD S2FCD31</b><br/>Clock Driver 3.2 GHz - 4.6 GHz<br/>4 pair CKD for UDIMM and SODIMM</p> <p><i>See also images of Samsung DDR5 RDIMM products:</i></p>  <p>Samsung 64GB DDR5 5600MHz PC5-44800 ECC RDIMM 2Rx4 (EC8 10x4) Dual Rank 1.1V Registered DIMM 288-Pin Server RAM Memory M321R8GA0PB0-CWM. <i>See Ex. CC-117.</i></p> |

U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <div data-bbox="560 337 1881 787">A photograph of a Samsung 64GB DDR5 PC5-4800 EC8 RDIMM 2Rx4 M321R8GA0BB0-CQK memory module. The module is green with gold-plated pins. It features a white label with the Samsung logo, part number 80CE03251213251911, and other specifications. The module is populated with 16 black memory chips, each labeled with 'SEC513' and 'K4R4004 6V8 BCC'.</div> <p data-bbox="560 797 1757 829">Ex. CC-118 (Samsung 64GB DDR5 PC5-4800 EC8 RDIMM 2Rx4 M321R8GA0BB0-CQK).</p> <div data-bbox="560 875 1881 1279">A photograph of a Samsung 64GB DDR5 PC5-6400 EC8 RDIMM 2Rx4 M321R8GA0PB2-CCP memory module. The module is green with gold-plated pins. It features a white label with the Samsung logo, part number 80CE01251501799891, and other specifications. The module is populated with 16 black memory chips, each labeled with 'SEC513' and 'K4R4004 6V8 BCC'.</div> <p data-bbox="560 1289 1747 1321">Ex. CC-119 (Samsung 64GB DDR5 PC5-6400 EC8 RDIMM 2Rx4 M321R8GA0PB2-CCP).</p> |

U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p data-bbox="558 310 1274 342"><i>See also images of Samsung DDR5 MRDIMM products:</i></p> <div data-bbox="558 375 1791 1356"><p>The top image shows a Samsung DDR5 MRDIMM module with a label that reads: SAMSUNG, DDR5 ECU MRDIMM, B0CE01251101663069, W068040, 64GB 2Rx4 PC5-80/560-MA0-1211-XT, Korea, M327R8GA0EB0-CLVXB, MB1M11G. The middle image shows the underside of a module with various components. The bottom image is a close-up of the label, showing the same information as the top image.</p></div> <p data-bbox="558 1365 1843 1398">An image of a Samsung MRDIMM M327R8GA0EB0-CLVXB. <i>See</i> Exs. CC-111, CC-112, CC-113.</p> |

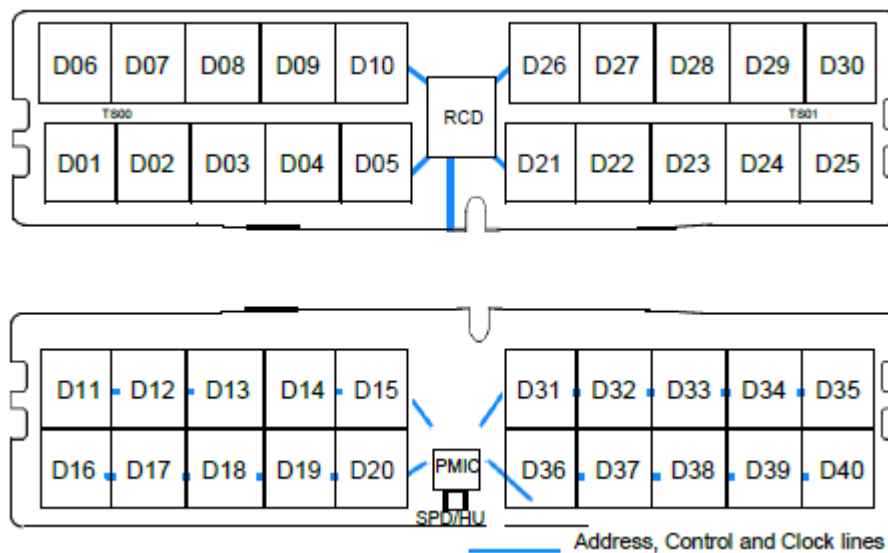
U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p data-bbox="558 342 1297 378"><i>See also</i> images of other Samsung DDR5 DIMM products:</p> <div data-bbox="558 412 1749 596"></div> <p data-bbox="558 695 1377 730"><i>See also</i> Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 33:</p> |

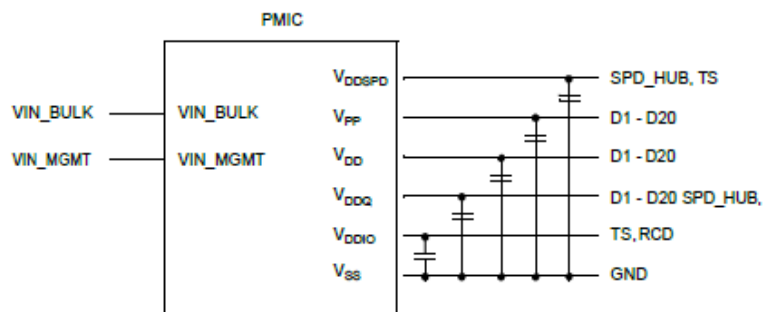
# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

## Claim 1

## Evidence of Use



See also *id.* at 29, 25, 21; *id.* at 10 (showing SPD Hub, TS, D1-D20 (SDRAMs) and RCD as components coupled to the output of the PMIC):



**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

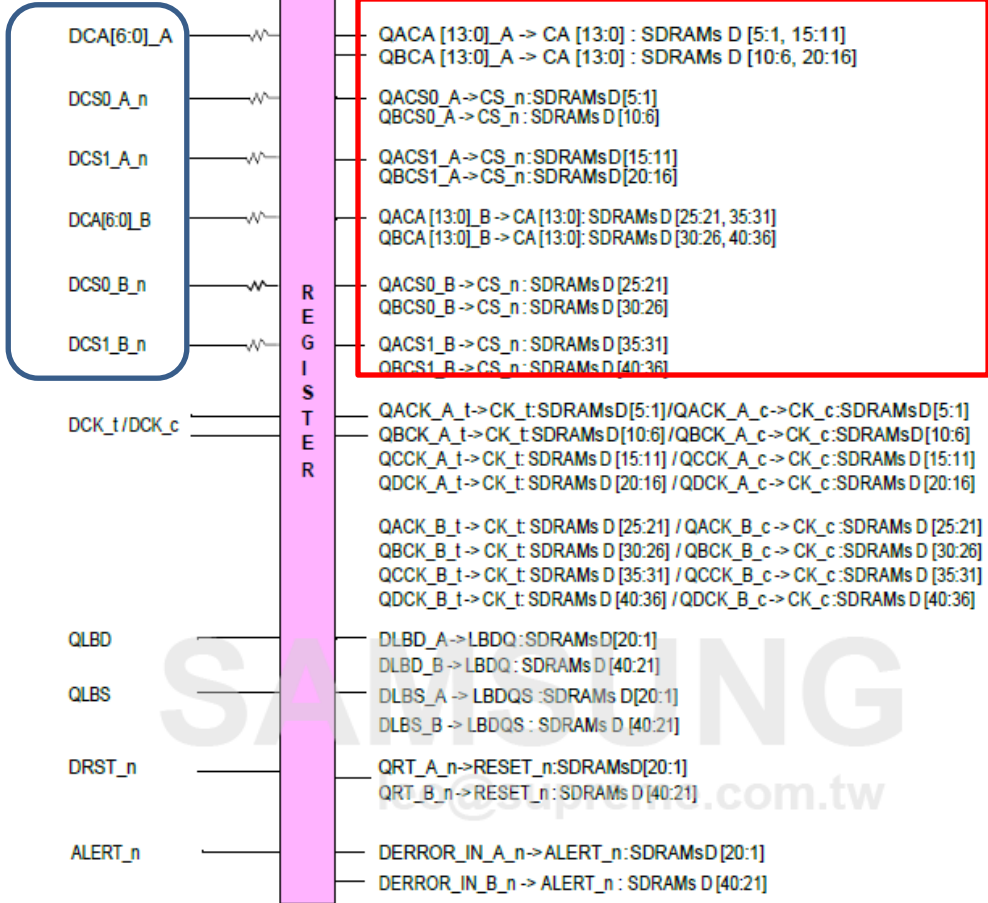
| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>Moreover, Netlist previously asserted claim 16 of U.S. Patent No. 11,016,918 against Samsung's DDR5 DIMMs, which recites, among other limitations, "a plurality of components coupled to the PCB." <i>See, e.g.</i>, Ex. CC-21 (EDTX Trial Tr. Day 2, 4 (2023)) at 333:9–12 ("Q. And can you tell us about claim 16 of the '918 Patent? A. So here's claim 16 of the '918, and a claim analysis is just where I actually look at the claims and compare it to exactly how the products operate, the accused products.").</p> <p>Samsung did not dispute this limitation. <i>Id.</i> at 885:24–886:9 ("With regard to the '918 claims, I have determined that—that these claims are not infringed by the Samsung DDR5 product, at least for the reason that there is a limitation in each of the independent claims directed to the requirement of a converter circuit structure, and that is not found on any of these Samsung DDR5 products. And what's being pointed to is a linear drop-out regulator, but that's not a converter circuit. As relates to the '054 claims, the DDR5 product has a normal operating state. The other state is it's turned off. So it does not have a second operable state as required by the '054 claims.").</p> |

**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

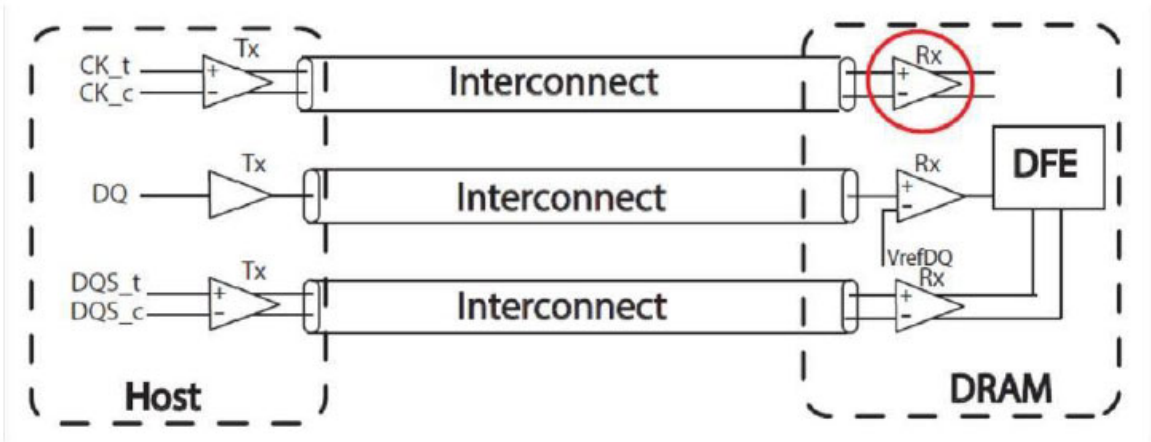
| Claim 1                                                                                                                                                                                                                            | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>[1.c.1]</b> the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines,</p> | <p>The set of components coupled to the PCB includes a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, <i>e.g.</i>, the DDR5 SDRAMs coupled to the C/A and data lines.</p> <p>Analysis below uses Samsung’s RDIMM format as an example, but MRDIMMs would operate in materially the same way in all relevant part. Ex. CC-13 (JEDEC Press Release) (JEDEC noting that other planned features for MRDIMMs include “Utilization of standard DDR5 DIMM components including DRAM, DIMM Form Factor &amp; Pinout, SPD, PMIC, and TS for ease of adoption”).</p> <p>For example, as shown in the Samsung DDR5 RDIMM Datasheet, there are 10 DDR5 DRAMs per rank for RDIMMs composed of x8 SDRAMs and 20 DRAMs per rank for RDIMMs composed of x4 SDRAMs. The SDRAMs are coupled to the C/A lines connected to the C/A edge connections, via the RCD. <i>See, e.g.</i>, Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 33:</p> |

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>The diagram illustrates a memory array architecture. It features two main memory blocks, TS00 and TS01, each containing a 2x5 grid of data cells (D01-D10 for TS00, D21-D30 for TS01). These blocks are connected to a central RCD (Row Driver/Column Decoder) block. Below this, a second memory block is shown, containing a 2x5 grid of data cells (D11-D20 and D31-D40). This block is connected to a PMIC (Power Management IC) block. A legend indicates that blue lines represent Address, Control and Clock lines. The PMIC block is also connected to a SPD/HV block.</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                            |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |  <p><i>Id.</i> (annotated) for 2Rx4; <i>see also id.</i> at 21 (1Rx8), 25 (2Rx8) and 29 (1Rx4).</p> <p>QACA[13:0]_A/B, QBCA[13:0]_A/B, QACS0_A/B, QACS1_A/B, QBCS0_A/B and QBCS1_A/B are C/A signal inputs to DRAMs, and these signals are generated in response to DCA[6:0]_A, DCS0_A_n, DCS1_A_n,</p> |

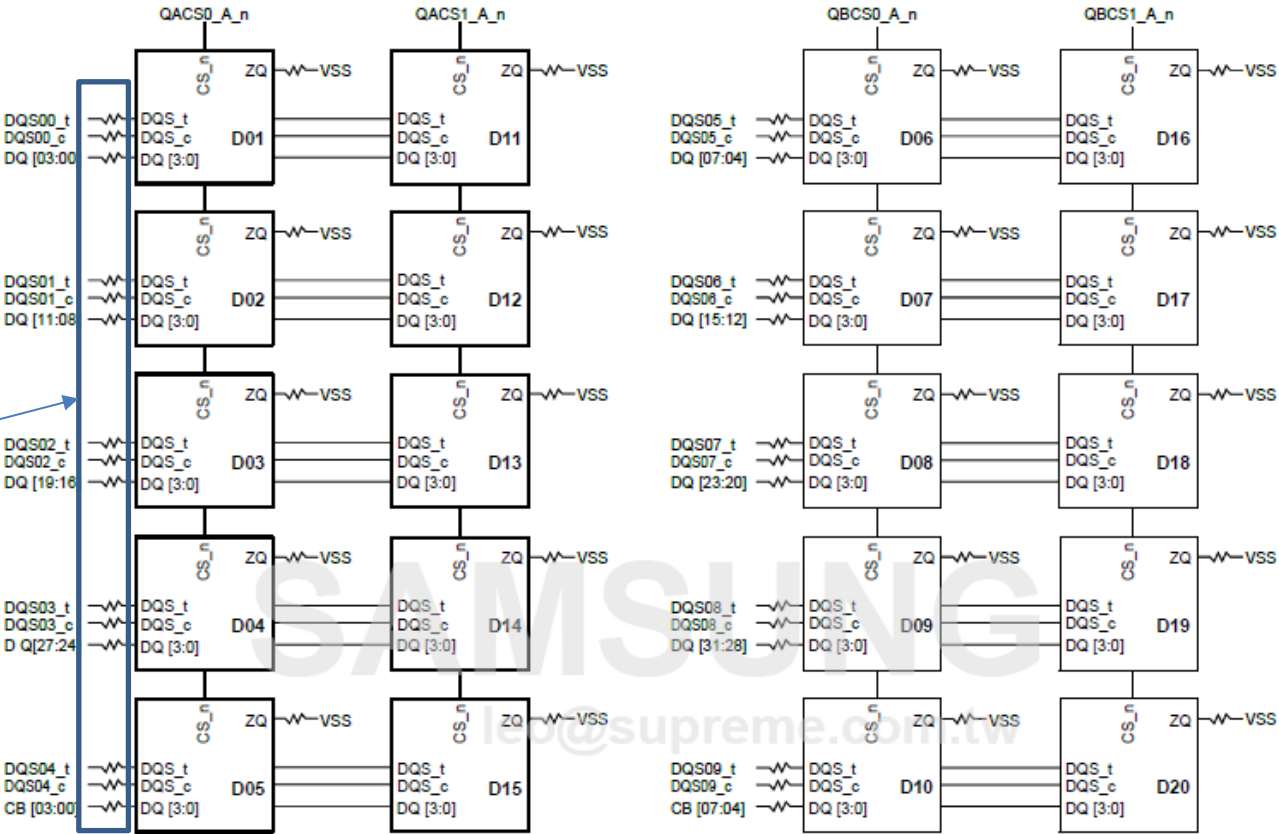
**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>DCA[6:0]_B received by the RCD, which signals are carried over the set of C/A signal lines recited in 1a.6. QACA and QBCA signals are copies of the corresponding DCA signals; and QACS and QBCS signals are copies of the DCS signals.</p> <p>Additionally, the plurality of SDRAMs are coupled to the set of data (DQ) lines recited in 1a.5:</p>  <p align="center"><b>Figure 10. Example of DDR5 Memory Interconnect</b></p> |

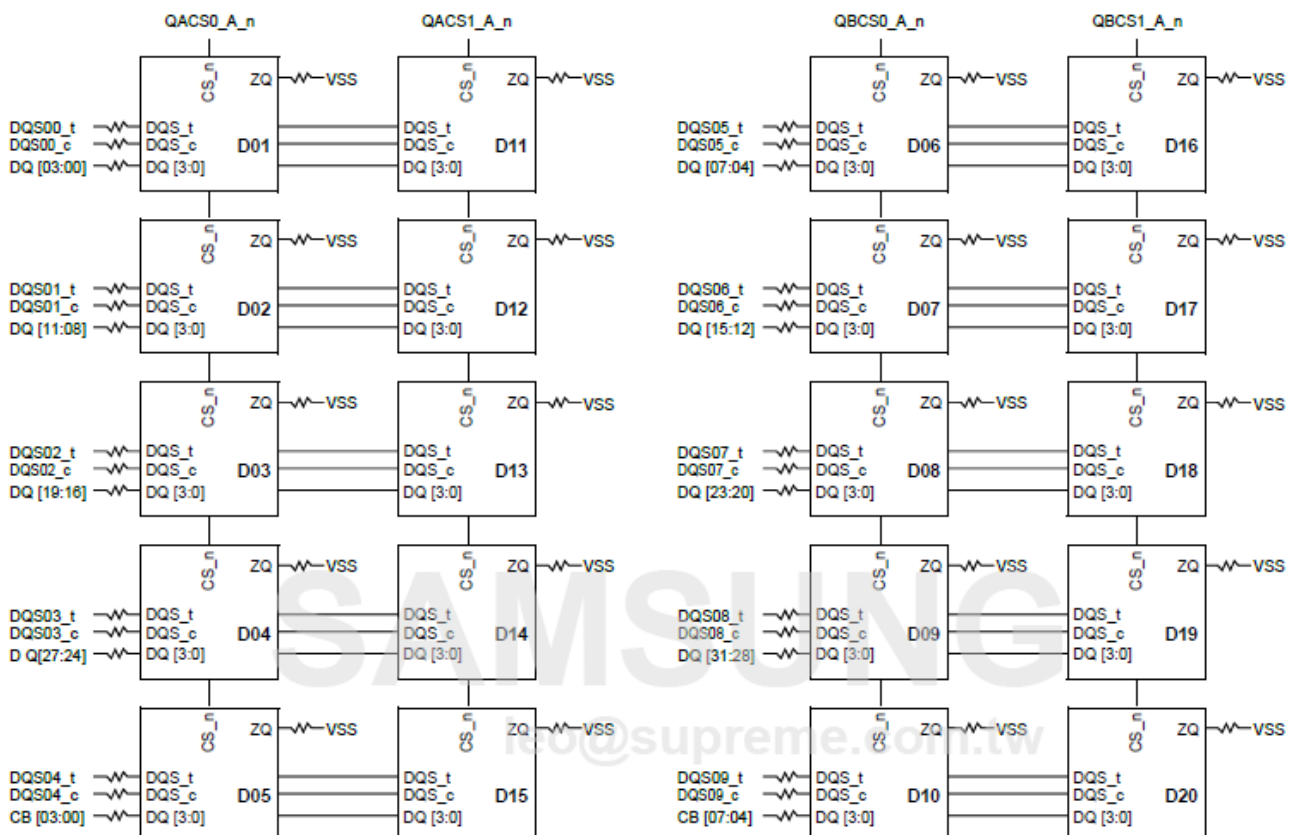
U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <div data-bbox="604 321 1816 673"> </div> <p data-bbox="947 690 1478 716">Figure 17. Example of DDR5 Memory Interconnect</p> <p data-bbox="558 732 1730 768">Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at Figs. 10, 17; <i>see also id.</i> at Figs. 13–14.</p> <p data-bbox="558 805 1848 878"><i>See also id.</i> at 34–35 (showing data lines electrically connected to the second set of edge connections (e.g., lines connecting DQ pins to DRAMs)):</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                                                                                                                                                                                       | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p data-bbox="247 751 451 1117">Examples of data lines coupled between the 2<sup>nd</sup> set of edge connections (data pins) and the SDRAMs; for simplicity, only a subset is annotated.</p> | <p data-bbox="579 313 787 354"><b>Channel A</b></p>  <p>The diagram illustrates the internal structure of Channel A, showing two banks of SDRAMs. Each bank consists of five SDRAMs (D01-D05 on the left, D11-D15 on the right, and D06-D10 on the right, D16-D20 on the right). Each SDRAM is connected to a common data bus (DQ [3:0]) and has its own address and data pins. The diagram shows the coupling between the 2<sup>nd</sup> set of edge connections (data pins) and the SDRAMs. A blue box highlights the data lines for D01-D05, with an arrow pointing to the text in the left column.</p> |

U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                            |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p><b>Channel B</b></p>  <p>See also <i>id.</i> at 30–31, 26–27, 22–23 (annotated).</p> |

**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>Moreover, Netlist previously asserted claim 16 of U.S. Patent No. 11,016,918 against Samsung's DDR5 DIMMs, which recites, among other limitations, "a plurality of components coupled to the PCB, the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices." <i>See, e.g.,</i> Ex. CC-21 (EDTX Trial Tr. Day 2, 4 (2023)) at 333:9–12 ("Q. And can you tell us about claim 16 of the '918 Patent? A. So here's claim 16 of the '918, and a claim analysis is just where I actually look at the claims and compare it to exactly how the products operate, the accused products."). Netlist also asserted claim 1 of the '918 patent, which recites, among other limitations, "at least one circuit coupled between a first portion of the plurality of edge connections and the plurality of SDRAM devices, the at least one circuit operable to (i) receive a first plurality of address and control signals via the first portion of the plurality of edge connections, and (ii) output a second plurality of address and control signals to the plurality of SDRAM devices, . . . ." <i>Id.</i> at 342:8–9 ("Next, I turned to claim 1 of the '918. Now, as we'll see, claim 1 only applies to the RDIMM products.").</p> <p>Samsung did not dispute this limitation. <i>Id.</i> at 885:24–886:9 ("With regard to the '918 claims, I have determined that—that these claims are not infringed by the Samsung DDR5 product, at least for the reason that there is a limitation in each of the independent claims directed to the requirement of a converter circuit structure, and that is not found on any of these Samsung DDR5 products. And what's being pointed to is a linear drop-out regulator, but that's not a converter circuit. As relates to the '054 claims, the DDR5 product has a normal operating state. The other state is it's turned off. So it does not have a second operable state as required by the '054 claims.").</p> |

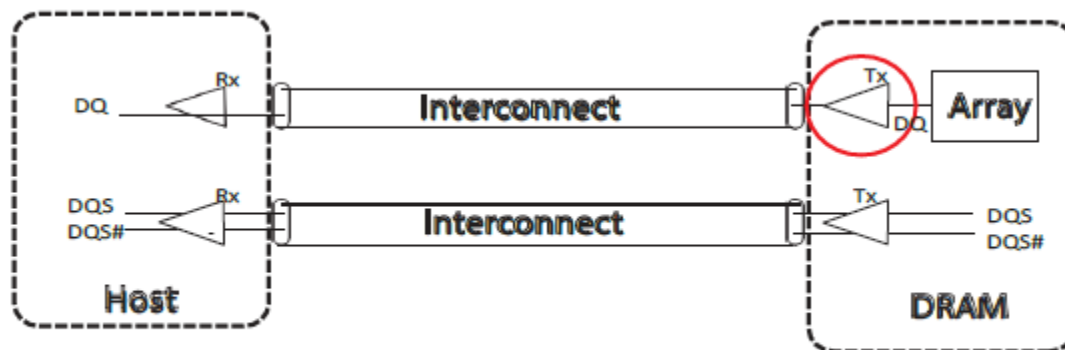
# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                                                                                                                                                                                                                                  | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>[1.c.2]</b> the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines; and</p> | <p>In each of Samsung's DDR5 DIMMs, the DDR5 SDRAM devices are operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines.</p> <p>For example, as illustrated in the Samsung DDR5 RDIMM Datasheet, each DRAM is coupled to a corresponding group of DQ lines for receiving (as shown in Figure 17) or outputting (as shown in Figure 34) data signals. The plurality of SDRAMs are therefore configured to receive or output data via the set of data lines recited in 1a.5.</p> <div data-bbox="625 678 1711 998"> </div> <p style="text-align: center;">Figure 17. Example of DDR5 Memory Interconnect</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

## Claim 1

## Evidence of Use



**Figure 34. Example of DDR5 Memory Interconnect**

Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at Figs. 17 and 34.

See also *id.* at 11:

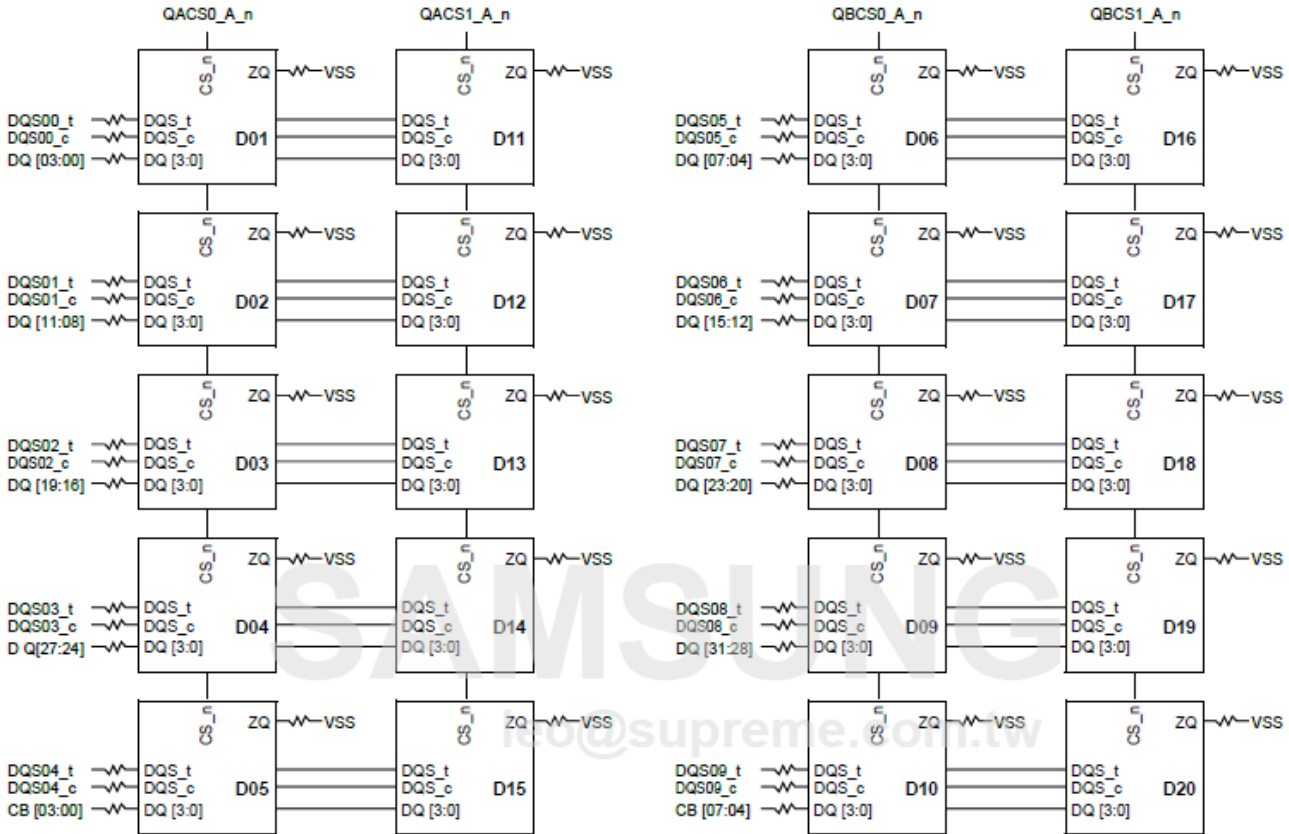
|                          |                  |     |                                                                                                                                                                                                                                                                                                                   |
|--------------------------|------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DQ[31:0]_A<br>DQ[31:0]_B | Input/<br>Output | VDD | Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used. |
| CB[7:0]_A<br>CB[7:0]_B   | Input/<br>Output | VDD | ECC Check Bits Input/ Output: Bi-directional data bus.                                                                                                                                                                                                                                                            |

The input and output of the data by SDRAMs are in response to C/A signals input to the SDRAMs. For example, as illustrated below, one input to the SDRAM is chip select (CS) signals. Ex. CC-89 (Samsung DDR5 RDIMM datasheet) at 34–35:

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p><b>Channel A</b></p> <p>The diagram illustrates the internal structure of Channel A, showing two columns of data path blocks. The left column contains blocks D01-D05, and the right column contains blocks D11-D15. To the right of these are blocks D06-D10 and D16-D20. Each block has inputs for DQS (data strobe) and DQ (data) signals, and outputs for ZQ (write enable) and VSS (ground). The diagram shows the internal structure of these blocks, including the DQS and DQ signals and their connections to the ZQ and VSS lines.</p> <p><b>Block D01:</b> Inputs: DQS00_t, DQS00_c, DQ [03:00]. Outputs: DQS_t, DQS_c, DQ [3:0].</p> <p><b>Block D02:</b> Inputs: DQS01_t, DQS01_c, DQ [11:08]. Outputs: DQS_t, DQS_c, DQ [3:0].</p> <p><b>Block D03:</b> Inputs: DQS02_t, DQS02_c, DQ [19:16]. Outputs: DQS_t, DQS_c, DQ [3:0].</p> <p><b>Block D04:</b> Inputs: DQS03_t, DQS03_c, DQ [27:24]. Outputs: DQS_t, DQS_c, DQ [3:0].</p> <p><b>Block D05:</b> Inputs: DQS04_t, DQS04_c, CB [03:00]. Outputs: DQS_t, DQS_c, DQ [3:0].</p> <p><b>Block D11:</b> Inputs: DQS05_t, DQS05_c, DQ [07:04]. Outputs: DQS_t, DQS_c, DQ [3:0].</p> <p><b>Block D12:</b> Inputs: DQS06_t, DQS06_c, DQ [15:12]. Outputs: DQS_t, DQS_c, DQ [3:0].</p> <p><b>Block D13:</b> Inputs: DQS07_t, DQS07_c, DQ [23:20]. Outputs: DQS_t, DQS_c, DQ [3:0].</p> <p><b>Block D14:</b> Inputs: DQS08_t, DQS08_c, DQ [31:28]. Outputs: DQS_t, DQS_c, DQ [3:0].</p> <p><b>Block D15:</b> Inputs: DQS09_t, DQS09_c, CB [07:04]. Outputs: DQS_t, DQS_c, DQ [3:0].</p> <p><b>Block D06:</b> Inputs: DQS05_t, DQS05_c, DQ [07:04]. Outputs: DQS_t, DQS_c, DQ [3:0].</p> <p><b>Block D07:</b> Inputs: DQS06_t, DQS06_c, DQ [15:12]. Outputs: DQS_t, DQS_c, DQ [3:0].</p> <p><b>Block D08:</b> Inputs: DQS07_t, DQS07_c, DQ [23:20]. Outputs: DQS_t, DQS_c, DQ [3:0].</p> <p><b>Block D09:</b> Inputs: DQS08_t, DQS08_c, DQ [31:28]. Outputs: DQS_t, DQS_c, DQ [3:0].</p> <p><b>Block D10:</b> Inputs: DQS09_t, DQS09_c, CB [07:04]. Outputs: DQS_t, DQS_c, DQ [3:0].</p> <p><b>Block D16:</b> Inputs: DQS05_t, DQS05_c, DQ [07:04]. Outputs: DQS_t, DQS_c, DQ [3:0].</p> <p><b>Block D17:</b> Inputs: DQS06_t, DQS06_c, DQ [15:12]. Outputs: DQS_t, DQS_c, DQ [3:0].</p> <p><b>Block D18:</b> Inputs: DQS07_t, DQS07_c, DQ [23:20]. Outputs: DQS_t, DQS_c, DQ [3:0].</p> <p><b>Block D19:</b> Inputs: DQS08_t, DQS08_c, DQ [31:28]. Outputs: DQS_t, DQS_c, DQ [3:0].</p> <p><b>Block D20:</b> Inputs: DQS09_t, DQS09_c, CB [07:04]. Outputs: DQS_t, DQS_c, DQ [3:0].</p> |

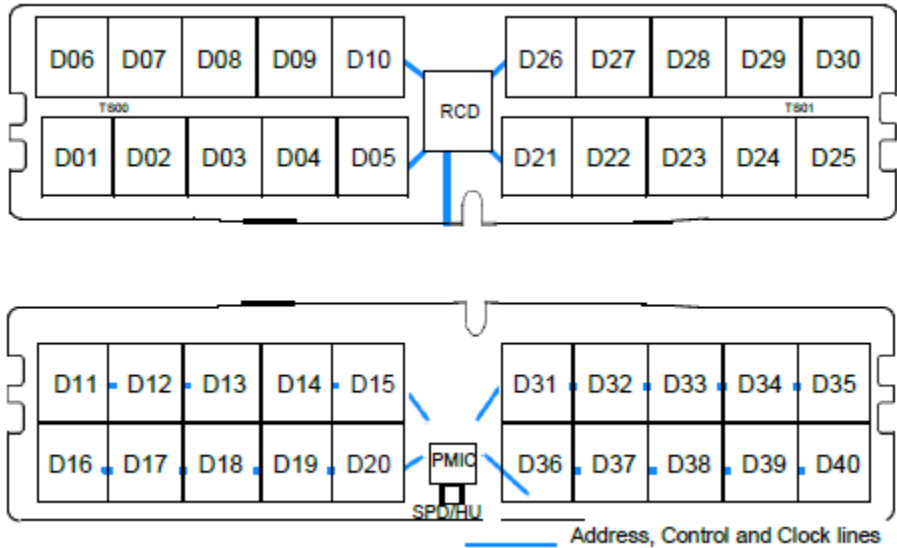
# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                              |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p><b>Channel B</b></p>  <p>See also <i>id.</i> at 30–31, 26–27, 22–23. Each SDRAM should also receive C/A signals (indicated as QACA or QBCA signals). See <i>id.</i> at 33, 29, 25, 21 (annotated):</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>QACA[13:0]_A/B, QBCA[13:0]_A/B, QACS0_A/B, QACS1_A/B, QBCS0_A/B and QBCS1_A/B are C/A signal input to DRAMs, and these signals are generated in response to DCA[6:0]_A,</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>received from the C/A signals recited in 1a.6:</p>  <p>The diagram illustrates a memory architecture with two memory banks, T800 and T801, connected to a central RCD (Row Control Driver) and a PMIC (Power Management IC) via Address, Control and Clock lines. The diagram includes data paths D01-D10, D11-D20, D21-D30, and D31-D40.</p> <p>See <i>id.</i> at 33, 29, 25, 21.</p> <p>See also <i>id.</i> at 11:</p> |

## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                                                                                                                                                                                                                        | Evidence of Use        |       |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                | Symbol                 | Type  | I/O Levels | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                                                                                                                                                                                                | CS[1:0]_A<br>CS[1:0]_B | Input | VDD        | Chip Select : All commands are masked when CS_n is registered HIGH. CS_n provides for external Rank selection on systems with multiple Ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and VREF parameters as configured by the CA_ODT strap setting or mode register. When in self refresh the CS_n is a CMOS rail to rail signal with DC high and low at 80% and 20% of VDD. |
|                                                                                                                                                                                                                                | PAR_A<br>PAR_B         | Input | VDD        | Command and Address Parity Input: DDR5 Supports Even Parity check in DRAMs with MR setting. Once it's enabled via Register in MR5, then DRAM calculates Parity. Input parity should be maintained at the rising edge of the clock and at the same time with command & address with CSx_x_n LOW                                                                                                                                                                                                                                                               |
| <p>The data signals are input and output by the SDRAMs in accordance with so-called command truth table, which shows that read and write operations are indicated by specific combinations of command and address signals:</p> |                        |       |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

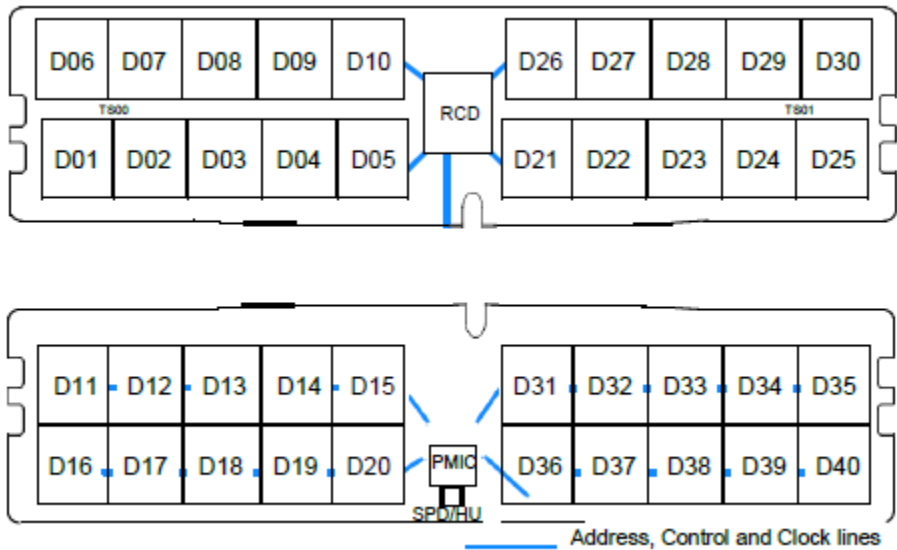
# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                         | Evidence of Use |                   |      |         |     |     |     |       |      |      |      |      |      |                 |      |                |                             |      |
|---------------------------------|-----------------|-------------------|------|---------|-----|-----|-----|-------|------|------|------|------|------|-----------------|------|----------------|-----------------------------|------|
|                                 | Function        | Abbrevia-<br>tion | CS_n | CA Pins |     |     |     |       |      |      |      |      |      |                 |      |                | NOTES                       |      |
|                                 |                 |                   |      | CA0     | CA1 | CA2 | CA3 | CA4   | CA5  | CA6  | CA7  | CA8  | CA9  | CA10            | CA11 | CA12           |                             | CA13 |
| Activate                        | ACT             | L                 | L    | L       | R0  | R1  | R2  | R3    | BA0  | BA1  | BG0  | BG1  | BG2  | CID0            | CID1 | CID2/<br>DOPIO | 11,17,20,21,<br>23,24       |      |
|                                 |                 | H                 | R4   | R5      | R6  | R7  | R8  | R9    | R10  | R11  | R12  | R13  | R14  | R15             | R16  | CID3/<br>R17   |                             |      |
| RFU                             | RFU             | L                 | H    | L       | L   | L   | L   | V     | V    | V    | V    | V    | V    | V               | V    | V              |                             |      |
|                                 |                 | H                 | V    | V       | V   | V   | V   | V     | V    | V    | V    | V    | V    | V               | V    | V              |                             |      |
| RFU                             | RFU             | L                 | H    | L       | L   | L   | H   | V     | V    | V    | V    | V    | V    | V               | V    | V              |                             |      |
|                                 |                 | H                 | V    | V       | V   | V   | V   | V     | V    | V    | V    | V    | V    | V               | V    | V              |                             |      |
| Write Pattern                   | WRP             | L                 | H    | L       | L   | H   | L   | H     | BA0  | BA1  | BG0  | BG1  | BG2  | CID0            | CID1 | CID2/<br>DOPIO | 11,15,18,19,<br>20,21,23,24 |      |
|                                 |                 | H                 | V    | C3      | C4  | C5  | C6  | C7    | C8   | C9   | C10  | V    | H    | H               | V    | CID3           |                             |      |
| Write Pattern w/ Auto Precharge | WRPA            | L                 | H    | L       | L   | H   | L   | H     | BA0  | BA1  | BG0  | BG1  | BG2  | CID0            | CID1 | CID2/<br>DOPIO | 11,15,18,19,<br>20,21,23,24 |      |
|                                 |                 | H                 | V    | C3      | C4  | C5  | C6  | C7    | C8   | C9   | C10  | V    | AP=L | H               | V    | CID3           |                             |      |
| RFU                             | RFU             | L                 | H    | L       | L   | H   | H   | V     | V    | V    | V    | V    | V    | V               | V    | V              |                             |      |
|                                 |                 | H                 | V    | V       | V   | V   | V   | V     | V    | V    | V    | V    | V    | V               | V    | V              |                             |      |
| Mode Register Write             | MRW             | L                 | H    | L       | H   | L   | L   | MRA0  | MRA1 | MRA2 | MRA3 | MRA4 | MRA5 | MRA6            | MRA7 | V/<br>DOPIO    | 8,11,13,20,<br>22,23,24     |      |
|                                 |                 | H                 | OP0  | OP1     | OP2 | OP3 | OP4 | OP5   | OP6  | OP7  | V    | V    | CW   | V               | V    | V              |                             |      |
| Mode Register Read              | MRR             | L                 | H    | L       | H   | L   | H   | MRA0  | MRA1 | MRA2 | MRA3 | MRA4 | MRA5 | MRA6            | MRA7 | V/<br>DOPIO    | 8,13,20,22,<br>23,24,25,26  |      |
|                                 |                 | H                 | L    | L       | V   | V   | V   | V     | V    | V    | V    | V    | CW   | V               | V    | V              |                             |      |
| Write                           | WR              | L                 | H    | L       | H   | H   | L   | BL*=L | BA0  | BA1  | BG0  | BG1  | BG2  | CID0            | CID1 | CID2/<br>DOPIO | 8,12,15,19,<br>20,21,23,24  |      |
|                                 |                 | H                 | V    | C3      | C4  | C5  | C6  | C7    | C8   | C9   | C10  | V    | H    | WR<br>Partial=L | V    | CID3           |                             |      |
| Write w/Auto Precharge          | WRA             | L                 | H    | L       | H   | H   | L   | BL*=L | BA0  | BA1  | BG0  | BG1  | BG2  | CID0            | CID1 | CID2/<br>DOPIO | 8,12,15,19,<br>20,21,23,24  |      |
|                                 |                 | H                 | V    | C3      | C4  | C5  | C6  | C7    | C8   | C9   | C10  | V    | AP=L | WR<br>Partial=L | V    | CID3           |                             |      |
| Read                            | RD              | L                 | H    | L       | H   | H   | H   | BL*=L | BA0  | BA1  | BG0  | BG1  | BG2  | CID0            | CID1 | CID2/<br>DOPIO | 8,15,19,20,<br>21,23,24     |      |
|                                 |                 | H                 | C2   | C3      | C4  | C5  | C6  | C7    | C8   | C9   | C10  | V    | H    | V               | V    | CID3           |                             |      |
| Read w/Auto Precharge           | RDA             | L                 | H    | L       | H   | H   | H   | BL*=L | BA0  | BA1  | BG0  | BG1  | BG2  | CID0            | CID1 | CID2/<br>DOPIO | 8,15,19,20,<br>21,23,24     |      |
|                                 |                 | H                 | C2   | C3      | C4  | C5  | C6  | C7    | C8   | C9   | C10  | V    | AP=L | V               | V    | CID3           |                             |      |

See also, e.g., Ex. CC-46 (DDR5 Command Truth Table) (DDR5 SDRAM command truth table, showing that write or read operations which result in receiving or outputting data along the set of data lines are in response to combination of C/A signals).

See also, e.g., Ex. CC-46 (DDR5 Command Truth Table) (DDR5 SDRAM command truth table, showing that write or read operations which result in receiving or outputting data along the set of data lines are in response to combination of C/A signals).

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                                                                                                                                                                                                                                                                         | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>[1.d] first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line, and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,</p> | <p>In each of Samsung's DDR5 DIMMs, there are multiple buck converters and LDOs contained in a PMIC coupled to the PCB. Analysis below uses Samsung's RDIMM format as an example, but MRDIMMs would operate in materially the same way in all relevant part. Ex. CC-13 (JEDEC Press Release) (JEDEC noting that other planned features for MRDIMMs include "Utilization of standard DDR5 DIMM components including DRAM, DIMM Form Factor &amp; Pinout, SPD, PMIC, and TS for ease of adoption").</p>  <p><i>E.g.</i>, Ex. CC-89 (Samsung DDR5 RDIMM datasheet) at 33, 29, 25, 21.</p> <p>For example, as noted above, each of Samsung's DDR5 DIMMs includes a PMIC, which in turn includes at least four voltage regulators or converters. <i>See, e.g.</i>, Ex. CC-104 (Optimized DDR5 DIMM Solutions) (annotated):</p> |

## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |         |                                                                                              |     |                                                                                              |     |                                                                                              |      |                                                                             |      |                                                                                 |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------------------------------------------------------------------------------------------|-----|----------------------------------------------------------------------------------------------|-----|----------------------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------|------|---------------------------------------------------------------------------------|
|         | <p>As PMICs have evolved, the number and type of power channels has increased accordingly, with server products generally requiring more channels and higher power capacity than client products. This is why server products use a 4-channel buck converter and a dual low-dropout regulator (2LDO)<sup>1</sup> — and client products use a 3-channel buck converter and 2LDO.<sup>2</sup> Initial plans were to have only one product each for servers and clients, but as DRAM modules increased in capacity and speed, it became clear that necessary power would have to be supplied in different ways, leading to the diversification of products.</p> <p><sup>1</sup>) Server Models: PMIC5000 5010 5020 5030.</p> <p><sup>2</sup>) Client Models: PMIC 5100 5120 5200.</p> <p>In DDR5 RDIMMs and MRDIMMs, the VDD voltage supplied to the SDRAMs is provided a combination of <i>e.g.</i>, two buck converters (SWA and SWB), which together forms a dual-phase buck converter. The SWC and SWD buck converters each operate as a single-phase buck converter. The two LDOs also each serve as a voltage converter circuit. Thus, for example, SWA+SWB, SWC, SWD and LD0_1.0V can be the claimed first, second, third and fourth converter circuits.</p> <p><i>See also, e.g., Ex. CC-139 (S2FPC01 Brochure) at 2:</i></p> <table border="1" data-bbox="562 1114 1873 1416"> <tbody> <tr> <td data-bbox="562 1114 999 1174">SWA+SWB</td><td data-bbox="999 1114 1873 1174">Programmable output with 5mV step (0.6V ~ 1.435V)<br/>Switching Frequency : 0.5MHz to 1.25MHz</td></tr> <tr> <td data-bbox="562 1174 999 1234">SWC</td><td data-bbox="999 1174 1873 1234">Programmable output with 5mV step (0.6V ~ 1.435V)<br/>Switching Frequency : 0.5MHz to 1.25MHz</td></tr> <tr> <td data-bbox="562 1234 999 1294">SWD</td><td data-bbox="999 1234 1873 1294">Programmable output with 5mV step (1.5V ~ 2.835V)<br/>Switching Frequency : 0.5MHz to 1.25MHz</td></tr> <tr> <td data-bbox="562 1294 999 1354">LD01</td><td data-bbox="999 1294 1873 1354">Programmable output voltage 1.7/1.8/1.9V<br/>Output current capacity of 25mA</td></tr> <tr> <td data-bbox="562 1354 999 1416">LD02</td><td data-bbox="999 1354 1873 1416">Programmable output voltage 0.9/1.0/1.1/1.2V<br/>Output current capacity of 20mA</td></tr> </tbody> </table> | SWA+SWB | Programmable output with 5mV step (0.6V ~ 1.435V)<br>Switching Frequency : 0.5MHz to 1.25MHz | SWC | Programmable output with 5mV step (0.6V ~ 1.435V)<br>Switching Frequency : 0.5MHz to 1.25MHz | SWD | Programmable output with 5mV step (1.5V ~ 2.835V)<br>Switching Frequency : 0.5MHz to 1.25MHz | LD01 | Programmable output voltage 1.7/1.8/1.9V<br>Output current capacity of 25mA | LD02 | Programmable output voltage 0.9/1.0/1.1/1.2V<br>Output current capacity of 20mA |
| SWA+SWB | Programmable output with 5mV step (0.6V ~ 1.435V)<br>Switching Frequency : 0.5MHz to 1.25MHz                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |                                                                                              |     |                                                                                              |     |                                                                                              |      |                                                                             |      |                                                                                 |
| SWC     | Programmable output with 5mV step (0.6V ~ 1.435V)<br>Switching Frequency : 0.5MHz to 1.25MHz                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |                                                                                              |     |                                                                                              |     |                                                                                              |      |                                                                             |      |                                                                                 |
| SWD     | Programmable output with 5mV step (1.5V ~ 2.835V)<br>Switching Frequency : 0.5MHz to 1.25MHz                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |                                                                                              |     |                                                                                              |     |                                                                                              |      |                                                                             |      |                                                                                 |
| LD01    | Programmable output voltage 1.7/1.8/1.9V<br>Output current capacity of 25mA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |         |                                                                                              |     |                                                                                              |     |                                                                                              |      |                                                                             |      |                                                                                 |
| LD02    | Programmable output voltage 0.9/1.0/1.1/1.2V<br>Output current capacity of 20mA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |         |                                                                                              |     |                                                                                              |     |                                                                                              |      |                                                                             |      |                                                                                 |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                     |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>Ex. CC-74 (Renesas P8910 Short-Form Datasheet) (annotated):</p> <p>(showing output of SWA and SWB combined to supply VDD, SWC supplying VDDQ, SWD supplying VPP, and one LDO supplying 1.0V)</p> |

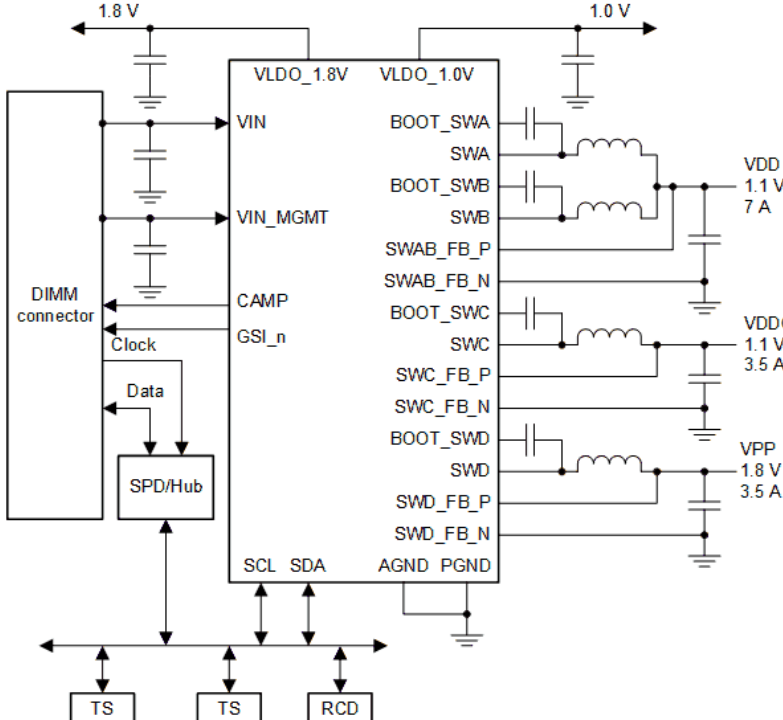
# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>(showing output of SWA and SWB combined to supply VDD, SWC supplying VDDQ, SWD supplying VPP and LDO supplying VOUT_1.0V)</p> <p>Ex. CC-27 (MPQ8896) (“The MPQ8896 provides four channel buck outputs. Buck A and buck B can be configured as a dual-phase interleaving buck converter with current-sharing performance. Buck C and buck D can be configured as two single-phase buck converters that are designed to support the specific DDR5 power requirements.”; “Two Low-Dropout (LDO) Regulators”).</p> <p>See also Ex. CC-28 (MPQ8895) (“The MPQ8895 provides four channel buck outputs. Buck A and</p> |

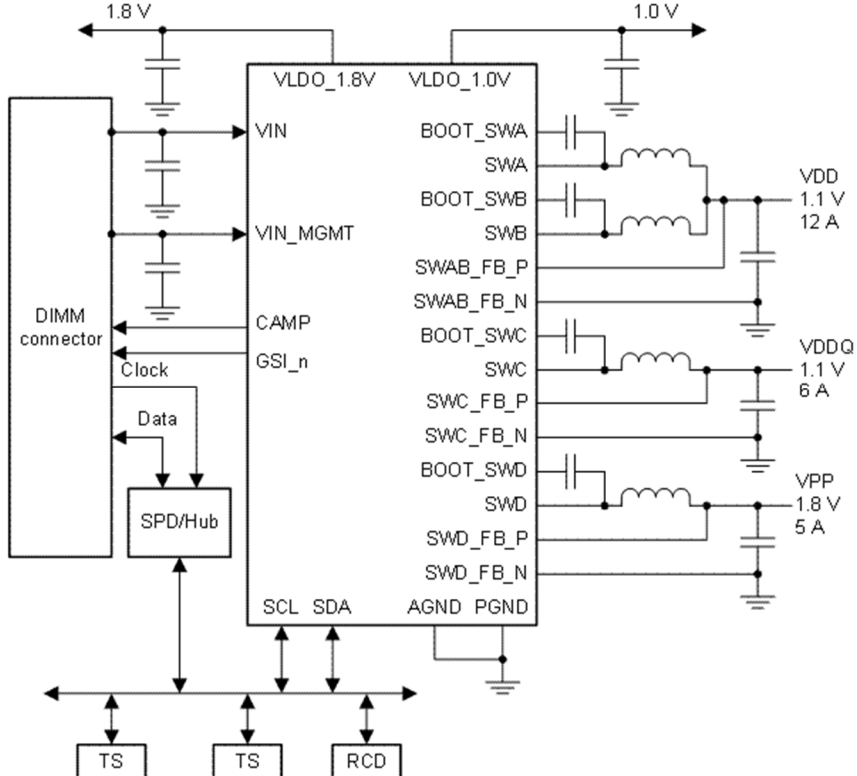
**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>buck B are configured to a dual-phase, interleaving buck converter with current-sharing performance. Buck C and buck D [are] configured to two single-phase buck converters that are designed to support the specific power requirement of DDR5.”; “Two Low-Dropout (LDO) Regulators”).</p> <p>Ex. CC-123 (MPQ8895F) (“The MPQ8895F provides four channel buck outputs. Buck A and buck B can be configured as a dual-phase interleaving buck converter with current-sharing performance. Buck C and buck D can be configured as two single-phase buck converters that are designed to support the specific DDR5 power requirements.”; “Two Low-Dropout (LDO) Regulators”).</p> <p>Ex. CC-29 (MPQ8894) (“The MPQ8894 provides four channel buck outputs. Buck A and buck B are configured to a dual-phase, interleaving buck converter with current-sharing performance. Buck C and buck D are configured to two single-phase buck converters that are designed to support the specific power requirements of DDR5.”; “Two Low-Dropout (LDO) Regulators”).</p> <p>Ex. CC-124 (TPS53832A Product Overview) &amp; Ex CC-125 (TPS53832A Data Sheet) (“The converter provides a full set of telemetry, including input voltage, output voltage, and output current. [O]vervoltage, undervoltage, overcurrent limit and over-temperature protections are provided as well.”).</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |  <p>(showing output of SWA and SWB combined to supply VDD, SWC supplying VDDQ, SWD supplying VPP and VLDO_1.0V supplying 1.0V)</p> <p>Ex. CC-126 (TPS53830A Product Overview) &amp; Ex. CC-127 (TPS53830A Data Sheet) (“The converter provides a full set of telemetry, including input voltage, output voltage, and output current. [O]vervoltage, undervoltage, overcurrent limit and over-temperature protections are provided as well.”).</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |  <p>(showing output of SWA and SWB combined to supply VDD, SWC supplying VDDQ, SWD supplying VPP and VLDO_1.0V supplying VOUT_1.0V)</p> <p>See also Ex. CC-128 (TPS53830 Product Overview); Ex. CC-129 (TPS53830 Data Sheet); Ex. CC-130 (TPS53832 Product Overview); Ex. CC-26 (TPS53832 Data Sheet).</p> <p>Additionally, as noted above, the PMICs used in Samsung's DDR5 DIMMs allegedly comply with the appropriate JEDEC standards:</p> |

## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>As PMICs have evolved, the number and type of power channels has increased accordingly, with server products generally requiring more channels and higher power capacity than client products. This is why server products use a 4-channel buck converter and a dual low-dropout regulator (2LDO)<sup>1</sup> — and client products use a 3-channel buck converter and 2LDO.<sup>2</sup> Initial plans were to have only one product each for servers and clients, but as DRAM modules increased in capacity and speed, it became clear that necessary power would have to be supplied in different ways, leading to the diversification of products.</p> <p><sup>1</sup>) Server Models: PMIC5000 5010 5020 5030.</p> <p><sup>2</sup>) Client Models: PMIC 5100 5120 5200.</p> <p>Ex. CC-104 (Optimized DDR5 DIMM Solutions).</p> <p>According to the JEDEC PMIC standards, PMICs for server applications have four buck converters and two LDOs for external output, and SWA and SWB can be combined to form a dual phase buck converter.</p> <p>The four converter circuits are coupled to the one or more input voltage supply lines, and configured to receive power from the input voltage supply lines and to deliver power via first, second, third, and fourth regulated voltage lines. <i>See, e.g.</i>, Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 10 (showing the PMIC receiving power input from VIN_BULK and VIN_MGMT lines and outputting various regulated voltages to associated components). For instance, when VIN_MGMT is operating normally, VDD, VDDQ and VPP are generated from VIN_BULK while VDDIO (1.0V) and VDDSPD (1.8V) are generated from VIN_MGMT. However, when VIN_MGMT drops below a threshold, VDDIO and VDDSPD are generated from the 12V VIN_BULK in a switch-over event.</p> |

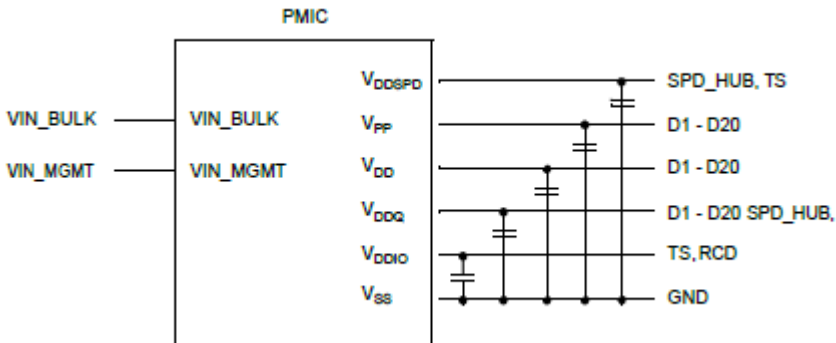
# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <div data-bbox="787 321 1617 662"> <p>The diagram shows a PMIC block with two input pins on the left: VIN_BULK and VIN_MGMT. On the right, there are six output pins: VDDSPD, VPP, VDD, VDDQ, VDDIO, and VSS. Each output pin is connected to a horizontal line representing a voltage rail. From top to bottom, these rails are: SPD_HUB, TS (connected to VDDSPD); D1 - D20 (connected to VPP); D1 - D20 (connected to VDD); D1 - D20 SPD_HUB, (connected to VDDQ); TS, RCD (connected to VDDIO); and GND (connected to VSS). Capacitors are shown at various points along these rails.</p> </div> <p>Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 24, 28, 32, 36.</p> <p>Testing as referenced above in limitation [1.b.2] confirms the outputs for each voltage rail.</p> <table border="1" data-bbox="562 795 1722 1174"> <tr> <td data-bbox="562 795 646 1174">4</td><td data-bbox="646 795 966 1174">           Using the I<sup>2</sup>C Master<br/>           Write the VR_ENABLE<br/>           bit in the PMIC<br/>           Register 32 Bit 7:<br/>           VR_ENABLE<br/>           1 = PMIC Enable         </td><td data-bbox="966 795 1722 1174">           0x32 &lt;= 0x80<br/><br/> <div> <div>VIN_Bulk</div> <div>12.00 V</div> </div> <div> <div>VIN_Mgmt</div> <div>3.30 V</div> </div> <div> <div>VOUT_1.8V (VDDSPD)</div> <div>1.80 V</div> </div> <div> <div>VOUT_1.0V (VIO)</div> <div>1.00 V</div> </div> <div> <div>SWA/B (VDD)</div> <div>1.10 V</div> </div> <div> <div>SWC (VDDQ)</div> <div>1.10 V</div> </div> <div> <div>SWD (VPP)</div> <div>1.80 V</div> </div> <br/>           Switching regulators are enabled.         </td></tr> </table> <p>Moreover, Netlist previously asserted claim 16 of U.S. Patent No. 11,016,918 against Samsung's DDR5 DIMMs, which recites, among other limitations, "first, second, and third buck converters configured to receive a pre-regulated input voltage and to produce first, second and third regulated voltages, respectively; a converter circuit configured to reduce the pre-regulated input voltage to provide a fourth regulated voltage, wherein the first, second, third and fourth regulated voltages have first, second, third, and fourth voltage</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 4 | Using the I <sup>2</sup> C Master<br>Write the VR_ENABLE<br>bit in the PMIC<br>Register 32 Bit 7:<br>VR_ENABLE<br>1 = PMIC Enable | 0x32 <= 0x80<br><br><div> <div>VIN_Bulk</div> <div>12.00 V</div> </div> <div> <div>VIN_Mgmt</div> <div>3.30 V</div> </div> <div> <div>VOUT_1.8V (VDDSPD)</div> <div>1.80 V</div> </div> <div> <div>VOUT_1.0V (VIO)</div> <div>1.00 V</div> </div> <div> <div>SWA/B (VDD)</div> <div>1.10 V</div> </div> <div> <div>SWC (VDDQ)</div> <div>1.10 V</div> </div> <div> <div>SWD (VPP)</div> <div>1.80 V</div> </div><br>Switching regulators are enabled. |
| 4       | Using the I <sup>2</sup> C Master<br>Write the VR_ENABLE<br>bit in the PMIC<br>Register 32 Bit 7:<br>VR_ENABLE<br>1 = PMIC Enable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0x32 <= 0x80<br><br><div> <div>VIN_Bulk</div> <div>12.00 V</div> </div> <div> <div>VIN_Mgmt</div> <div>3.30 V</div> </div> <div> <div>VOUT_1.8V (VDDSPD)</div> <div>1.80 V</div> </div> <div> <div>VOUT_1.0V (VIO)</div> <div>1.00 V</div> </div> <div> <div>SWA/B (VDD)</div> <div>1.10 V</div> </div> <div> <div>SWC (VDDQ)</div> <div>1.10 V</div> </div> <div> <div>SWD (VPP)</div> <div>1.80 V</div> </div><br>Switching regulators are enabled. |   |                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>amplitudes, respectively.” <i>See, e.g.</i>, Ex. CC-21 (EDTX Trial Tr. Day 2, 4 (2023)) at 333:9–12 (“Q. And can you tell us about claim 16 of the ’918 Patent? A. So here’s claim 16 of the ’918, and a claim analysis is just where I actually look at the claims and compare it to exactly how the products operate, the accused products.”). Netlist also asserted claim 1 of the ’918 patent, which recites, among other limitations, “a first buck converter configured to provide a first regulated voltage having a first voltage amplitude; a second buck converter configured to provide a second regulated voltage having a second voltage amplitude; a third buck converter configured to provide a third regulated voltage having a third voltage amplitude; a converter circuit configured to provide a fourth regulated voltage having a fourth voltage amplitude.” <i>Id.</i> at 342:8–9 (“Next, I turned to claim 1 of the ’918. Now, as we’ll see, claim 1 only applies to the RDIMM products.”).</p> <p>Samsung only disputed whether an LDO is a claimed converter circuit. <i>Id.</i> at 885:24–886:9 (“With regard to the ’918 claims, I have determined that—that these claims are not infringed by the Samsung DDR5 product, at least for the reason that there is a limitation in each of the independent claims directed to the requirement of a converter circuit structure, and that is not found on any of these Samsung DDR5 products. And what’s being pointed to is a linear drop-out regulator, but that’s not a converter circuit. As relates to the ’054 claims, the DDR5 product has a normal operating state. The other state is it’s turned off. So it does not have a second operable state as required by the ’054 claims.”).</p> |

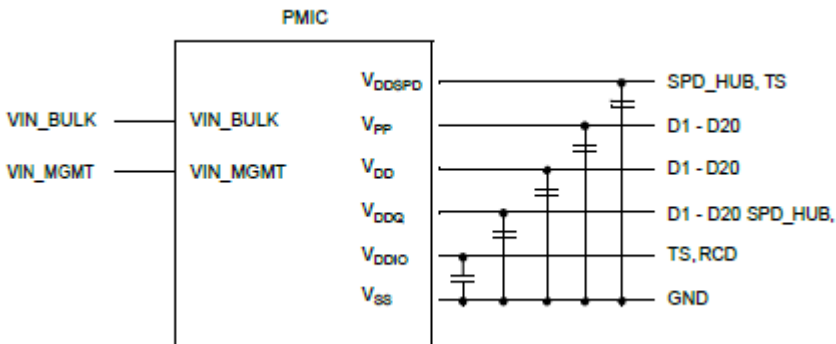
# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>[1.d.1]</b> wherein the set of components includes:<br/> a first component configured to receive power via at least the first regulated voltage line;<br/> a second component configured to receive power via at least the second regulated voltage line;<br/> a third component configured to receive power via at least the third regulated voltage line; and<br/> a fourth component configured to receive power via at least the fourth regulated voltage line;</p> | <p>In each of Samsung's DDR5 DIMMs, the set of components includes:</p> <ul style="list-style-type: none"> <li>• a first component configured to receive power via at least the first regulated voltage line;</li> <li>• a second component configured to receive power via at least the second regulated voltage line;</li> <li>• a third component configured to receive power via at least the third regulated voltage line; and</li> <li>• a fourth component configured to receive power via at least the fourth regulated voltage line.</li> </ul> <p>For example, as shown by Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 10, any of SDRAMs D1-D20 are designed to receive VDD, VPP and VDDQ output by the buck converters; and RCD is designed to receive 1.0V VDDIO from 1.0V LDO and although not shown, also VDDQ (*).</p>  <p>*As noted above, although not shown, RCD also receives a 1.1V voltage. This would be understood as for I/O communication with SDRAM for C/A signal transmissions. The 1.0V VDDIO is for operating the sideband bus (I2C/I3C). For example, Rambus, one DDR5 RCD supplier to Samsung states on its website that one feature of the DDR5 RCD is “Low Power 1.1V VDD.” See Ex. CC-96 (Rambus DDR5 RCD). Likewise, Montage, another DDR5 RCD supplier to Samsung, shows that its RCDs</p> |

**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>receive “1.1V VDD and 1.0V VDDIO voltages.” Ex. CC-48 (Montage M88DR5RCD01 (Gen1 DDR5 RCD)); Ex. CC-87 (Montage M88DR5RCD02 (Gen2 DDR5 RCD)); Ex. CC-97 (Montage M88DR5RCD03 (Gen3 DDR5 RCD)); Ex. CC-98 (Montage M88DR5RCD04 (Gen4 DDR5 RCD)).</p> <p>*On the other hand, SPD—being required to be on before the buck converters are ready to operate and generate the 1.1V VDDQ power output—should not be required to operate off the VDDQ voltage rail. And this is confirmed by SPD suppliers’ web sites. For example, Montage shows that its SPD_Hub operates off “1.8 V VDDSPD and 1.0 V VDDIO power supply.” Ex. CC-140 (Montage M88SPD5118 (DDR5 SPD Hub)). Likewise, Renesas’s SPD5118 block diagram shows only input of VDDSPD and VIO, but nothing about VDDQ (or VDD). <i>See</i> Ex. CC-141 (Renesas SPD5118 – SPD HUB for DDR5):</p> <div data-bbox="911 740 1541 1172" data-label="Diagram"> <pre> graph LR     VDDSPD --&gt; Hub     VIO --&gt; Hub     Hub --&gt; LSDA     Hub --&gt; LSCL     Hub &lt;--&gt; HSDA     Hub &lt;--&gt; HSCL     Hub --- LDO     Hub --- TempSensor[Temperature Sensor]     Hub --- EEPROM     Hub --- HSA_Sense[HSA Sense]     HSA_Sense --&gt; HSA[HSA I3C/I2C Address Pin]     GND --&gt; Hub     </pre> </div> <p>Thus, it is believed that in the figure below, VDDQ is to be supplied to D1-D20 and RCD, not D1-D20 and SPD.</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                        | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |                   |                                |       |                            |     |                           |     |                            |      |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------|--------------------------------|-------|----------------------------|-----|---------------------------|-----|----------------------------|------|
|                                |  <p>The diagram shows a PMIC block with inputs VIN_BULK and VIN_MGMT. On the right, it lists output rails: VDDSPD, VPP, VDD, VDDQ, VDDIO, and VSS. These rails are connected to various components: VDDSPD to SPD_HUB, TS; VPP to D1 - D20; VDD to D1 - D20; VDDQ to D1 - D20 SPD_HUB,; VDDIO to TS, RCD; and VSS to GND. Capacitors are shown at several output nodes.</p> <p>Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 10.</p> <p>In Samsung MRDIMMs, the voltage rails for the SDRAMs would be the same because the same DDR5 SDRAMs are used. Ex. CC-13 (JEDEC Press Release) (one planned feature for MRDIMM is “Utilization of standard DDR5 DIMM components including DRAM, DIMM Form Factor &amp; Pinout, SPD, PMIC, and TS for ease of adoption”). Samsung’s MRDIMMs also use Montage’s MRCD01. <i>See</i> images of Samsung MRDIMM in 1pre. Montage states that MRCD receives a 1.1V VDD and a 1.0V VDDIO input. <i>See</i> Ex. CC-12 (Montage M88MR5RCD01 (Gen 1 DDR5 MRCD)). Because the 1.1V is to support I/O communication with host and DRAM, it is more likely that the 1.1V is connected to the VDDQ rail than the VDD rail. But that would not affect the analysis.</p> <p>A combination that would satisfy this limitation and the next limitation is a combination of three SDRAMs and an RCD, with the three SDRAMs receiving respectively VDD, VDDQ and VPP and the RCD receiving VDDIO. Some specific combinations are as below:</p> <table border="1"> <thead> <tr> <th>Components</th><th>Regulated Voltage</th></tr> </thead> <tbody> <tr> <td>First Component: RCD (or MRCD)</td><td>VDDIO</td></tr> <tr> <td>Second Component: D1 SDRAM</td><td>VDD</td></tr> <tr> <td>Third Component: D2 SDRAM</td><td>VPP</td></tr> <tr> <td>Fourth Component: D3 SDRAM</td><td>VDDQ</td></tr> </tbody> </table> | Components | Regulated Voltage | First Component: RCD (or MRCD) | VDDIO | Second Component: D1 SDRAM | VDD | Third Component: D2 SDRAM | VPP | Fourth Component: D3 SDRAM | VDDQ |
| Components                     | Regulated Voltage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |                   |                                |       |                            |     |                           |     |                            |      |
| First Component: RCD (or MRCD) | VDDIO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |                   |                                |       |                            |     |                           |     |                            |      |
| Second Component: D1 SDRAM     | VDD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |                   |                                |       |                            |     |                           |     |                            |      |
| Third Component: D2 SDRAM      | VPP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |                   |                                |       |                            |     |                           |     |                            |      |
| Fourth Component: D3 SDRAM     | VDDQ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |                   |                                |       |                            |     |                           |     |                            |      |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                         | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |                   |                          |               |                                 |       |                           |     |                            |               |            |                   |                           |     |                            |     |                           |      |                                 |       |            |                   |                           |     |                            |      |                           |     |                                 |       |                           |     |                            |      |                           |     |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------|--------------------------|---------------|---------------------------------|-------|---------------------------|-----|----------------------------|---------------|------------|-------------------|---------------------------|-----|----------------------------|-----|---------------------------|------|---------------------------------|-------|------------|-------------------|---------------------------|-----|----------------------------|------|---------------------------|-----|---------------------------------|-------|---------------------------|-----|----------------------------|------|---------------------------|-----|
|                                 | <p>or</p> <table> <tr> <th>Components</th><th>Regulated Voltage</th></tr> <tr> <td>First Component: D1SDRAM</td><td>VDDQ (or VDD)</td></tr> <tr> <td>Second Component: RCD (or MRCD)</td><td>VDDIO</td></tr> <tr> <td>Third Component: D2 SDRAM</td><td>VPP</td></tr> <tr> <td>Fourth Component: D3 SDRAM</td><td>VDD (or VDDQ)</td></tr> </table><br><table> <tr> <th>Components</th><th>Regulated Voltage</th></tr> <tr> <td>First Component: D1 SDRAM</td><td>VDD</td></tr> <tr> <td>Second Component: D2 SDRAM</td><td>VPP</td></tr> <tr> <td>Third Component: D3 SDRAM</td><td>VDDQ</td></tr> <tr> <td>Fourth Component: RCD (or MRCD)</td><td>VDDIO</td></tr> </table> <p>or</p> <table> <tr> <th>Components</th><th>Regulated Voltage</th></tr> <tr> <td>First Component: D1 SDRAM</td><td>VDD</td></tr> <tr> <td>Second Component: D2 SDRAM</td><td>VDDQ</td></tr> <tr> <td>Third Component: D3 SDRAM</td><td>VPP</td></tr> <tr> <td>Fourth Component: RCD (or MRCD)</td><td>VDDIO</td></tr> </table> <p>or</p> <table> <tr> <td>First Component: D1 SDRAM</td><td>VPP</td></tr> <tr> <td>Second Component: D2 SDRAM</td><td>VDDQ</td></tr> <tr> <td>Third Component: D3 SDRAM</td><td>VDD</td></tr> </table> | Components | Regulated Voltage | First Component: D1SDRAM | VDDQ (or VDD) | Second Component: RCD (or MRCD) | VDDIO | Third Component: D2 SDRAM | VPP | Fourth Component: D3 SDRAM | VDD (or VDDQ) | Components | Regulated Voltage | First Component: D1 SDRAM | VDD | Second Component: D2 SDRAM | VPP | Third Component: D3 SDRAM | VDDQ | Fourth Component: RCD (or MRCD) | VDDIO | Components | Regulated Voltage | First Component: D1 SDRAM | VDD | Second Component: D2 SDRAM | VDDQ | Third Component: D3 SDRAM | VPP | Fourth Component: RCD (or MRCD) | VDDIO | First Component: D1 SDRAM | VPP | Second Component: D2 SDRAM | VDDQ | Third Component: D3 SDRAM | VDD |
| Components                      | Regulated Voltage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |                   |                          |               |                                 |       |                           |     |                            |               |            |                   |                           |     |                            |     |                           |      |                                 |       |            |                   |                           |     |                            |      |                           |     |                                 |       |                           |     |                            |      |                           |     |
| First Component: D1SDRAM        | VDDQ (or VDD)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |                   |                          |               |                                 |       |                           |     |                            |               |            |                   |                           |     |                            |     |                           |      |                                 |       |            |                   |                           |     |                            |      |                           |     |                                 |       |                           |     |                            |      |                           |     |
| Second Component: RCD (or MRCD) | VDDIO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |                   |                          |               |                                 |       |                           |     |                            |               |            |                   |                           |     |                            |     |                           |      |                                 |       |            |                   |                           |     |                            |      |                           |     |                                 |       |                           |     |                            |      |                           |     |
| Third Component: D2 SDRAM       | VPP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |                   |                          |               |                                 |       |                           |     |                            |               |            |                   |                           |     |                            |     |                           |      |                                 |       |            |                   |                           |     |                            |      |                           |     |                                 |       |                           |     |                            |      |                           |     |
| Fourth Component: D3 SDRAM      | VDD (or VDDQ)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |                   |                          |               |                                 |       |                           |     |                            |               |            |                   |                           |     |                            |     |                           |      |                                 |       |            |                   |                           |     |                            |      |                           |     |                                 |       |                           |     |                            |      |                           |     |
| Components                      | Regulated Voltage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |                   |                          |               |                                 |       |                           |     |                            |               |            |                   |                           |     |                            |     |                           |      |                                 |       |            |                   |                           |     |                            |      |                           |     |                                 |       |                           |     |                            |      |                           |     |
| First Component: D1 SDRAM       | VDD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |                   |                          |               |                                 |       |                           |     |                            |               |            |                   |                           |     |                            |     |                           |      |                                 |       |            |                   |                           |     |                            |      |                           |     |                                 |       |                           |     |                            |      |                           |     |
| Second Component: D2 SDRAM      | VPP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |                   |                          |               |                                 |       |                           |     |                            |               |            |                   |                           |     |                            |     |                           |      |                                 |       |            |                   |                           |     |                            |      |                           |     |                                 |       |                           |     |                            |      |                           |     |
| Third Component: D3 SDRAM       | VDDQ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |                   |                          |               |                                 |       |                           |     |                            |               |            |                   |                           |     |                            |     |                           |      |                                 |       |            |                   |                           |     |                            |      |                           |     |                                 |       |                           |     |                            |      |                           |     |
| Fourth Component: RCD (or MRCD) | VDDIO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |                   |                          |               |                                 |       |                           |     |                            |               |            |                   |                           |     |                            |     |                           |      |                                 |       |            |                   |                           |     |                            |      |                           |     |                                 |       |                           |     |                            |      |                           |     |
| Components                      | Regulated Voltage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |                   |                          |               |                                 |       |                           |     |                            |               |            |                   |                           |     |                            |     |                           |      |                                 |       |            |                   |                           |     |                            |      |                           |     |                                 |       |                           |     |                            |      |                           |     |
| First Component: D1 SDRAM       | VDD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |                   |                          |               |                                 |       |                           |     |                            |               |            |                   |                           |     |                            |     |                           |      |                                 |       |            |                   |                           |     |                            |      |                           |     |                                 |       |                           |     |                            |      |                           |     |
| Second Component: D2 SDRAM      | VDDQ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |                   |                          |               |                                 |       |                           |     |                            |               |            |                   |                           |     |                            |     |                           |      |                                 |       |            |                   |                           |     |                            |      |                           |     |                                 |       |                           |     |                            |      |                           |     |
| Third Component: D3 SDRAM       | VPP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |                   |                          |               |                                 |       |                           |     |                            |               |            |                   |                           |     |                            |     |                           |      |                                 |       |            |                   |                           |     |                            |      |                           |     |                                 |       |                           |     |                            |      |                           |     |
| Fourth Component: RCD (or MRCD) | VDDIO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |                   |                          |               |                                 |       |                           |     |                            |               |            |                   |                           |     |                            |     |                           |      |                                 |       |            |                   |                           |     |                            |      |                           |     |                                 |       |                           |     |                            |      |                           |     |
| First Component: D1 SDRAM       | VPP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |                   |                          |               |                                 |       |                           |     |                            |               |            |                   |                           |     |                            |     |                           |      |                                 |       |            |                   |                           |     |                            |      |                           |     |                                 |       |                           |     |                            |      |                           |     |
| Second Component: D2 SDRAM      | VDDQ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |                   |                          |               |                                 |       |                           |     |                            |               |            |                   |                           |     |                            |     |                           |      |                                 |       |            |                   |                           |     |                            |      |                           |     |                                 |       |                           |     |                            |      |                           |     |
| Third Component: D3 SDRAM       | VDD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |                   |                          |               |                                 |       |                           |     |                            |               |            |                   |                           |     |                            |     |                           |      |                                 |       |            |                   |                           |     |                            |      |                           |     |                                 |       |                           |     |                            |      |                           |     |

**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

| Claim 1 | Evidence of Use                 |       |
|---------|---------------------------------|-------|
|         | Fourth Component: RCD (or MRCD) | VDDIO |
|         | or                              |       |
|         | First Component: D1 SDRAM       | VPP   |
|         | Second Component: D2 SDRAM      | VDD   |
|         | Third Component: D3 SDRAM       | VDDQ  |
|         | Fourth Component: RCD (or MRCD) | VDDIO |
|         | or                              |       |
|         | First Component: D1 SDRAM       | VDDQ  |
|         | Second Component: D2 SDRAM      | VPP   |
|         | Third Component: D3 SDRAM       | VDD   |
|         | Fourth Component: RCD (or MRCD) | VDDIO |
|         | or                              |       |
|         | First Component: D1 SDRAM       | VDDQ  |
|         | Second Component: D2 SDRAM      | VDD   |
|         | Third Component: D3 SDRAM       | VPP   |
|         | Fourth Component: RCD (or MRCD) | VDDIO |

**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

| Claim 1                                                                                                                                                                                                                                                                                                | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                   |                          |                |        |                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------|--------------------------|----------------|--------|----------------|
| <p><b>[1.d.2]</b> wherein each component of the set of components: (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines,</p> | <p>As also shown in the datasheet, each component of the set of components is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines, as explained above and below.</p> <p>For example, to the extent that the four regulated voltages are respectively VDD, VDDQ, VPP and VDDIO, the set of components and the associated regulated voltages are as follows:</p> <table border="1" data-bbox="558 591 1877 706"> <tr> <th>Components</th><th>Regulated Voltage</th></tr> <tr> <td>RCD (or MRCD in MRDIMMs)</td><td>VDDQ and VDDIO</td></tr> <tr> <td>SDRAMs</td><td>VDD, VDDQ, VPP</td></tr> </table> <p>Each component of the set of components (RCD and SDRAMs) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines as explained above.</p> <p>Note that in this combination, SPD and TS are not part of the set of the component, as they also derive power from VDDSPD, which is not one of the named four regulated voltages.</p> | Components | Regulated Voltage | RCD (or MRCD in MRDIMMs) | VDDQ and VDDIO | SDRAMs | VDD, VDDQ, VPP |
| Components                                                                                                                                                                                                                                                                                             | Regulated Voltage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |                   |                          |                |        |                |
| RCD (or MRCD in MRDIMMs)                                                                                                                                                                                                                                                                               | VDDQ and VDDIO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |                   |                          |                |        |                |
| SDRAMs                                                                                                                                                                                                                                                                                                 | VDD, VDDQ, VPP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |                   |                          |                |        |                |

**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

| Claim 1                                                                                                                                                                                                                                                                                                                                                                                                                           | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>[1.e]</b> wherein the plurality of DDR SDRAM devices includes:</p> <p>a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal;</p> <p>a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal;</p> | <p>The plurality of DDR SDRAM devices includes a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal (<i>e.g.</i>, those SDRAMs in channel A, rank 0 or channel A, rank 1) and a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal (<i>e.g.</i>, those SDRAMs in channel B, rank 0 or channel B, rank 1). For example, an ECC DIMM would include five x8 SDRAMs and ten x4 SDRAMs in each rank associated with each of channels A and B.</p> <p>Analysis below uses Samsung’s RDIMM format as an example, but MRDIMMs would operate in materially the same way in all relevant part. Ex. CC-13 (JEDEC Press Release) (JEDEC noting that other planned features for MRDIMMs include “Utilization of standard DDR5 DIMM components including DRAM, DIMM Form Factor &amp; Pinout, SPD, PMIC, and TS for ease of adoption”).</p> <p><i>E.g.</i>, Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 33–35 (annotated):</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>Two examples of a first group of SDRAMs, each shown as electrically connected to a respective DCS signal</p> <p>Two examples of a second group of SDRAMs, each shown as electrically connected to a respective DCS signal</p> <p>REGISTER</p> <p>DCA[6:0]_A → QACA [13:0]_A → CA [13:0] : SDRAMs D [5:1, 15:11]<br/> QBCA [13:0]_A → CA [13:0] : SDRAMs D [10:6, 20:16]</p> <p>DCS0_A_n → QACS0_A → CS_n : SDRAMs D [5:1]<br/> QBCS0_A → CS_n : SDRAMs D [10:6]</p> <p>DCS1_A_n → QACS1_A → CS_n : SDRAMs D [15:11]<br/> QBCS1_A → CS_n : SDRAMs D [20:16]</p> <p>DCA[6:0]_B → QACA [13:0]_B → CA [13:0] : SDRAMs D [25:21, 35:31]<br/> QBCA [13:0]_B → CA [13:0] : SDRAMs D [30:26, 40:36]</p> <p>DCS0_B_n → QACS0_B → CS_n : SDRAMs D [25:21]<br/> QBCS0_B → CS_n : SDRAMs D [30:26]</p> <p>DCS1_B_n → QACS1_B → CS_n : SDRAMs D [35:31]<br/> QBCS1_B → CS_n : SDRAMs D [40:36]</p> <p>DCK_t/DCK_c → QACK_A_t → CK_t : SDRAMs D [5:1] / QACK_A_c → CK_c : SDRAMs D [5:1]<br/> QBCK_A_t → CK_t : SDRAMs D [10:6] / QBCK_A_c → CK_c : SDRAMs D [10:6]<br/> QCCK_A_t → CK_t : SDRAMs D [15:11] / QCCK_A_c → CK_c : SDRAMs D [15:11]<br/> QDCK_A_t → CK_t : SDRAMs D [20:16] / QDCK_A_c → CK_c : SDRAMs D [20:16]</p> <p>QACK_B_t → CK_t : SDRAMs D [25:21] / QACK_B_c → CK_c : SDRAMs D [25:21]<br/> QBCK_B_t → CK_t : SDRAMs D [30:26] / QBCK_B_c → CK_c : SDRAMs D [30:26]<br/> QCCK_B_t → CK_t : SDRAMs D [35:31] / QCCK_B_c → CK_c : SDRAMs D [35:31]<br/> QDCK_B_t → CK_t : SDRAMs D [40:36] / QDCK_B_c → CK_c : SDRAMs D [40:36]</p> <p>QLBD → DLBD_A → LBDQ : SDRAMs D [20:1]<br/> DLBD_B → LBDQ : SDRAMs D [40:21]</p> <p>QLBS → DLBS_A → LBDQS : SDRAMs D [20:1]<br/> DLBS_B → LBDQS : SDRAMs D [40:21]</p> <p>DRST_n → QRT_A_n → RESET_n : SDRAMs D [20:1]<br/> QRT_B_n → RESET_n : SDRAMs D [40:21]</p> <p>ALERT_n → DERROR_IN_A_n → ALERT_n : SDRAMs D [20:1]<br/> DERROR_IN_B_n → ALERT_n : SDRAMs D [40:21]</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p><b>Channel A</b></p> <p>The diagram illustrates the internal structure of Channel A, organized into two main sections. Each section contains five rows of data storage elements, labeled D01 through D05 on the left and D11 through D15 on the right, and D06 through D10 on the left and D16 through D20 on the right. Each storage element is represented by a square block with internal labels 'CS' and 'Q'. The word lines (QACS0_A_n, QACS1_A_n, QBCS0_A_n, QBCS1_A_n) are connected to the top of each element. The bit lines (ZQ) are connected to the right side of each element, which is then connected to VSS. The data inputs/outputs are labeled with DQS (data strobe), DQS_c (data strobe complement), and DQ (data) with specific address ranges. For example, D01 is connected to DQS00_t, DQS00_c, and DQ [03:00]. D11 is connected to DQS_t, DQS_c, and DQ [3:0]. D06 is connected to DQS05_t, DQS05_c, and DQ [07:04]. D16 is connected to DQS_t, DQS_c, and DQ [3:0]. D09 is connected to DQS08_t, DQS08_c, and DQ [31:28]. D19 is connected to DQS_t, DQS_c, and DQ [3:0]. D10 is connected to DQS09_t, DQS09_c, and DQ [3:0]. D20 is connected to DQS_t, DQS_c, and DQ [3:0].</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

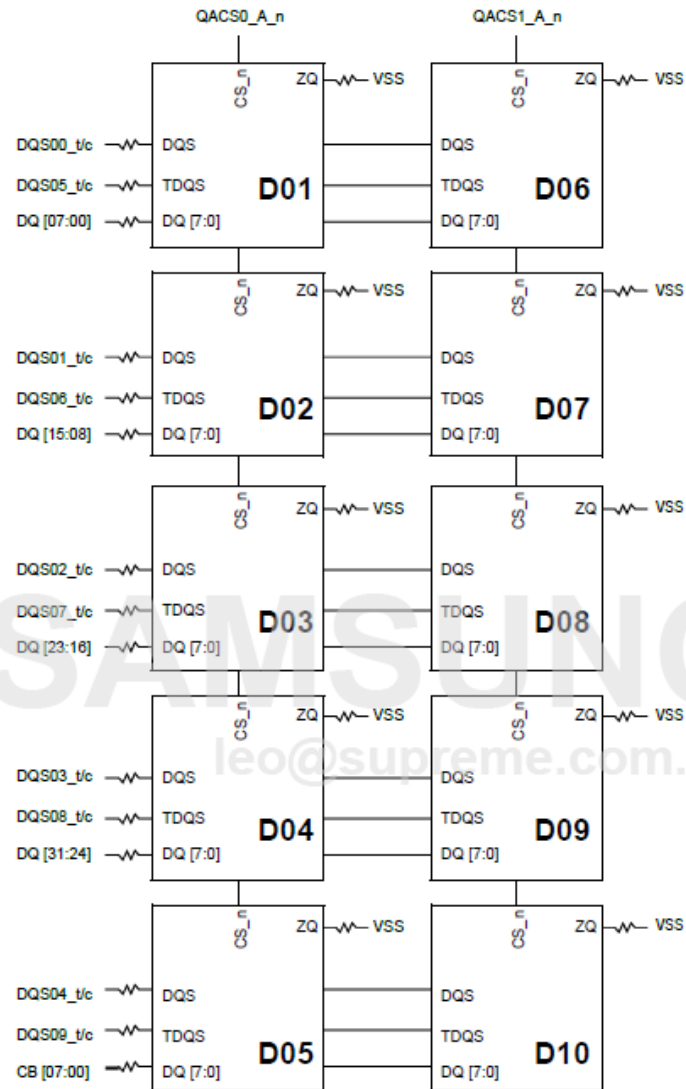
| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p><b>Channel B</b></p> <p>The diagram illustrates the internal structure of Channel B, showing 20 data buffers (D01-D20) organized into two columns. Each buffer is connected to a common data bus (DQ [3:0]) and has its own DQS_t and DQS_c inputs. The ZQ output of each buffer is connected to VSS. The buffers are grouped by address ranges: D01-D05 (03:00), D06-D10 (07:04), D11-D15 (11:08), and D16-D20 (15:12). The diagram also shows the connection of the DQS_t and DQS_c signals to the buffers.</p> <p><i>E.g., Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 25–27 (annotated):</i></p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

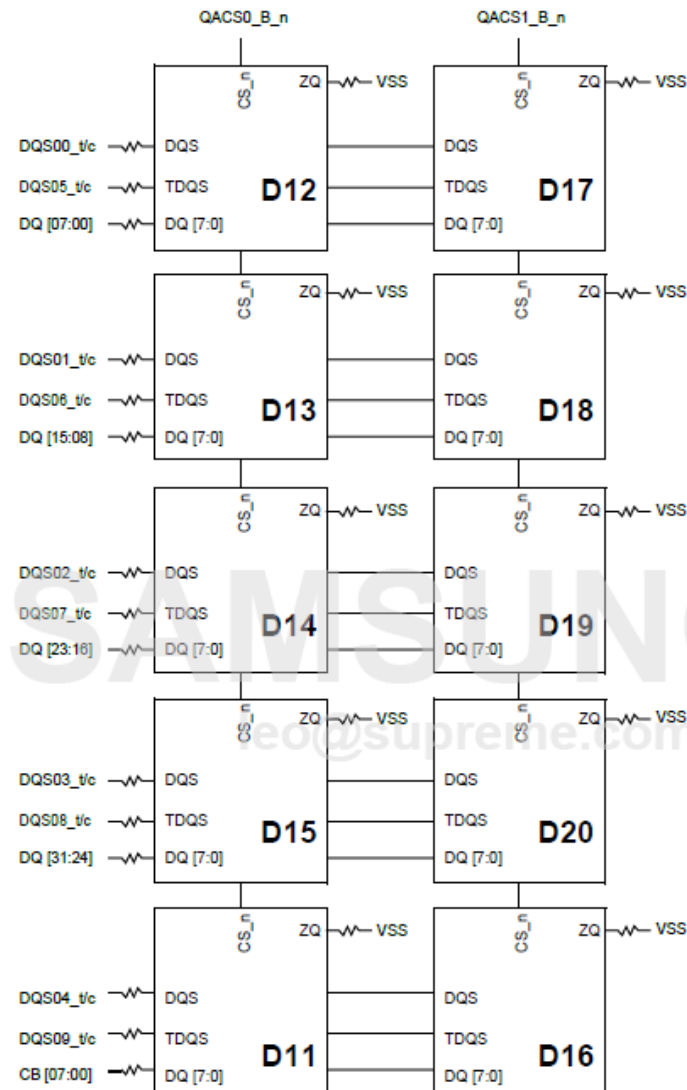
| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p> DCA[06:0]_A ——— QACA[13:00]_A -&gt; CA [13:00] : SDRAMs D [10:01]<br/> DCS0_A_n ——— QACS0_A -&gt; CS_n : SDRAMs D [05:01]<br/> DCS1_A_n ——— QACS1_A -&gt; CS_n : SDRAMs D [10:06]<br/> DCA [06:00]_B ——— QACA [13:00]_B -&gt; CA [13:00] : SDRAMs D [20:11]<br/> DCS0_B_n ——— QACS0_B -&gt; CS_n : SDRAMs D [15:11]<br/> DCS1_B_n ——— QACS1_B -&gt; CS_n : SDRAMs D [20:16]<br/> DCK_t/DCK_c ——— QACK_A_t -&gt; CK_t : SDRAMs D [05:01] / QACK_A_c -&gt; CK_c : SDRAMs D [05:01]<br/> — QCCCK_A_t -&gt; CK_t : SDRAMs D [10:06] / QCCCK_A_c -&gt; CK_c : SDRAMs D [10:06]<br/> — QACK_B_t -&gt; CK_t : SDRAMs D [15:11] / QACK_B_c -&gt; CK_c : SDRAMs D [15:11]<br/> — QCCCK_B_t -&gt; CK_t : SDRAMs D [20:16] / QCCCK_B_c -&gt; CK_c : SDRAMs D [20:16]<br/> QLBD ——— DLBD_A -&gt; LBDQ : SDRAMs D [10:01]<br/> — DLBD_B -&gt; LBDQ : SDRAMs D [20:11]<br/> QLBS ——— DLBS_A -&gt; LBDQS : SDRAMs D [10:01]<br/> — DLBS_B -&gt; LBDQS : SDRAMs D [20:11]<br/> DRST_n ——— QRT_A_n -&gt; RESET_n : SDRAMs D [10:01]<br/> — QRT_B_n -&gt; RESET_n : SDRAMs D [20:11]<br/> ALERT_n ——— ERROR_IN_A_n -&gt; ALERT_n : SDRAMs D [10:01]<br/> — ERROR_IN_B_n -&gt; ALERT_n : SDRAMs D [20:11] </p> <p>REGISTER</p> <p>Two examples of a first group of SDRAMs, each shown as electrically connected to a respective DCS signal</p> <p>Two examples of a second group of SDRAMs, each shown as electrically connected to a respective DCS</p> |

U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

**Channel A**



## Channel B



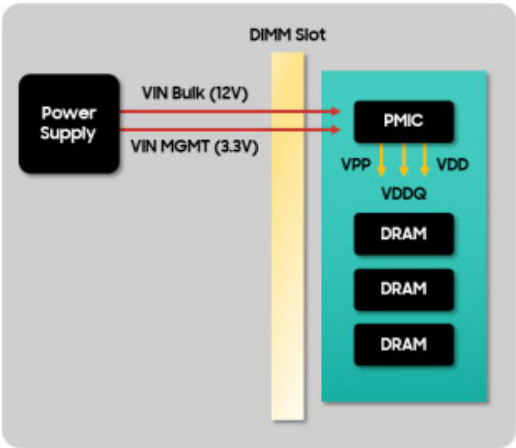
**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p><i>See also, e.g., id.</i> at 21–23, 29–31. As seen above, there are separate and independent C/A and CS signals for SDRAMs in channel A (DCA[6:0]_A and DCS0_A_n, DCS1_A_n) and SDRAMs in channel B (DCA[6:0]_B and DCS0_B_n, DCS1_B_n). The first group of SDRAMs are therefore each connected to a first chip select line (signal line connected to DCS0_A_n or DCS1_A_n) configured to electrically conduct a first chip select signal (DCS0_A_n or DCS1_A_n). The second group of SDRAMs are therefore each connected to a second chip select line (signal line connected to DCS0_B_n or DCS1_B_n) configured to electrically conduct a first chip select signal (DCS0_B_n or DCS1_B_n).</p> |

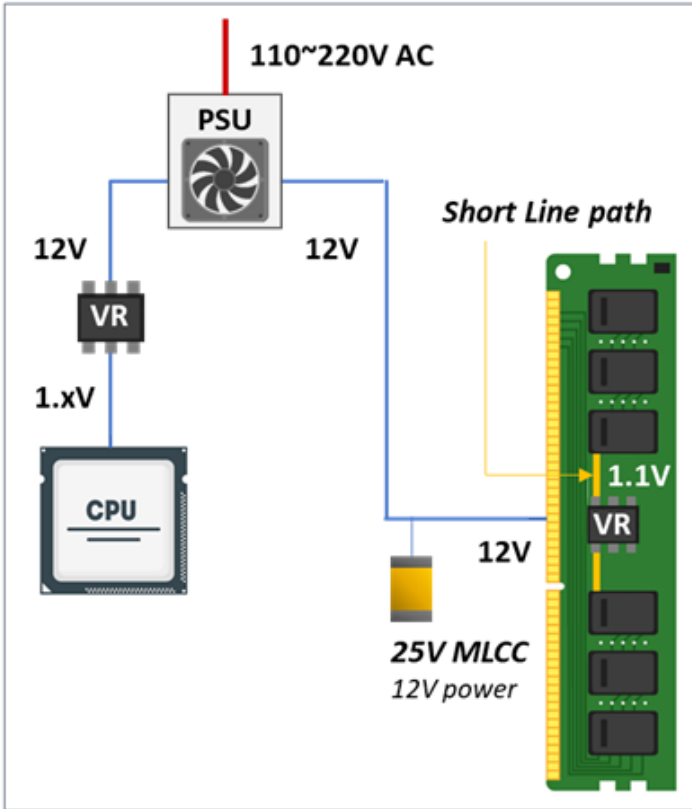
**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

| Claim 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>[1.e.1]</b> wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode:</p> <p>the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines;</p> | <p>Analysis below primarily uses Samsung’s RDIMM format as an example, but MRDIMMs would operate in materially the same way in all relevant part. Ex. CC-13 (JEDEC Press Release) (JEDEC noting that other planned features for MRDIMMs include “Utilization of standard DDR5 DIMM components including DRAM, DIMM Form Factor &amp; Pinout, SPD, PMIC, and TS for ease of adoption”).</p> <p>Samsung’s DDR5 DIMMs are designed to input or output data when it receives power from the power rails of the system board and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode. For example, as shown in Ex. CC-104 (Optimized DDR5 DIMM Solutions), DRAMs require VPP, VDDQ and VDD generated in response to power from the power rails of the system board via the DIMM slot:</p> |

## U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                                                                                                                                                                  | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>the at least 40 1-bit DDR data signals include at least 32 1-bit wide DDR data signals and at least eight 1-bit DDR error-correcting code (ECC) data signals; and</p> | <div style="text-align: center;">  <p>The diagram, titled "DDR5 DIMM", illustrates the power architecture. A "Power Supply" block on the left is connected to a "DIMM Slot" (represented by a yellow vertical bar) via two red lines labeled "VIN Bulk (12V)" and "VIN MGMT (3.3V)". The "DIMM Slot" is connected to a teal-colored board containing a "PMIC" (Power Management IC) and three "DRAM" modules. The PMIC is connected to the DIMM Slot and provides power to the DRAM modules via three yellow lines labeled "VPP", "VDDQ", and "VDD".</p> </div> <p><i>See also</i> Ex. CC-120 (Samsung, “Memory VR changes in the system by DDR5 adoption”) (the off-DIMM 12V line is an example of power rails of the system board, which would then be fed to the DIMM slot and from the DIMM slot to the power pins of the DIMM’s edge connections).</p> |

U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p data-bbox="1010 331 1373 370"><b>[ DDR5 Based-System ]</b></p>  <p data-bbox="558 1219 1608 1255">Ex. CC-120 (Samsung, “Memory VR changes in the system by DDR5 adoption”).</p> <p data-bbox="558 1292 1829 1398">A self-refresh mode is a low-power mode where the DRAM periodically refreshes its content without commands from the memory controller. Thus, when the SDRAMs are performing read or write operations, they are not in self-refresh mode.</p> |

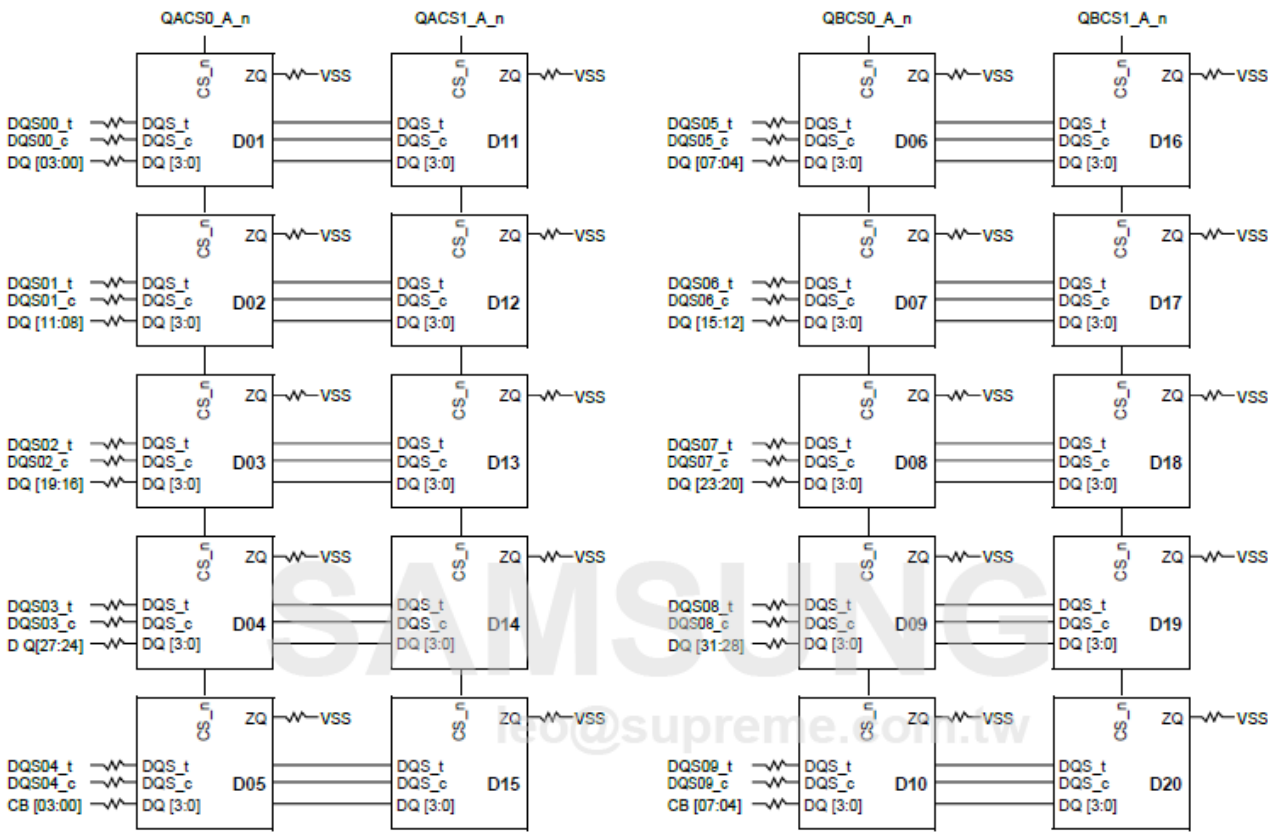
**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

| Claim 1                  | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                          |                                                                                                                                                                                                                                                                                                                   |     |                                                                                                                                                                                                                                                                                                                   |                        |                  |     |                                                        |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------|-----|--------------------------------------------------------|
|                          | <p>In a read or write operation, the first group of DDR SDRAM devices (<i>e.g.</i>, Rank0 or Rank1 SDRAMs in channel A) is configured to be enabled, in response to the first chip select signal (<i>e.g.</i>, DCS0_A_n or DCS1_A_n), to receive or output 40 1-bit DDR data signals in parallel via 40 data lines of the set of data lines (<i>e.g.</i>, via data lines associated with DQ[31:0]_A and CB[7:0]_A), independently of whether the second group of DDR SDRAM devices (Rank0 or Rank1 SDRAMs in channel B) is enabled, in response to the second chip select signal (<i>e.g.</i>, DCS0_B_n or DCS1_B_n), to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines (<i>e.g.</i>, via data lines associated with DQ[31:0]_B and CB[7:0]_B), because channel A and channel B receive independent C/A and CS signals. <i>See</i> evidence in 1.e.</p> <p>The at least 40 1-bit DDR data signals include at least 32 1-bit wide DDR data signals and at least eight 1-bit DDR error-correcting code (ECC) data signals as explained above. <i>See also</i>:</p> <table><tr><td>DQ[31:0]_A<br/>DQ[31:0]_B</td><td>Input/<br/>Output</td><td>VDD</td><td>Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.</td></tr><tr><td>CB[7:0]_A<br/>CB[7:0]_B</td><td>Input/<br/>Output</td><td>VDD</td><td>ECC Check Bits Input/ Output: Bi-directional data bus.</td></tr></table> <p>Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 11.</p> <p><i>See also id.</i> at 34–35 (showing separate DQ[00:31] and CB[7:0] for channel A and channel B respectively):</p> | DQ[31:0]_A<br>DQ[31:0]_B | Input/<br>Output                                                                                                                                                                                                                                                                                                  | VDD | Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used. | CB[7:0]_A<br>CB[7:0]_B | Input/<br>Output | VDD | ECC Check Bits Input/ Output: Bi-directional data bus. |
| DQ[31:0]_A<br>DQ[31:0]_B | Input/<br>Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | VDD                      | Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used. |     |                                                                                                                                                                                                                                                                                                                   |                        |                  |     |                                                        |
| CB[7:0]_A<br>CB[7:0]_B   | Input/<br>Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | VDD                      | ECC Check Bits Input/ Output: Bi-directional data bus.                                                                                                                                                                                                                                                            |     |                                                                                                                                                                                                                                                                                                                   |                        |                  |     |                                                        |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

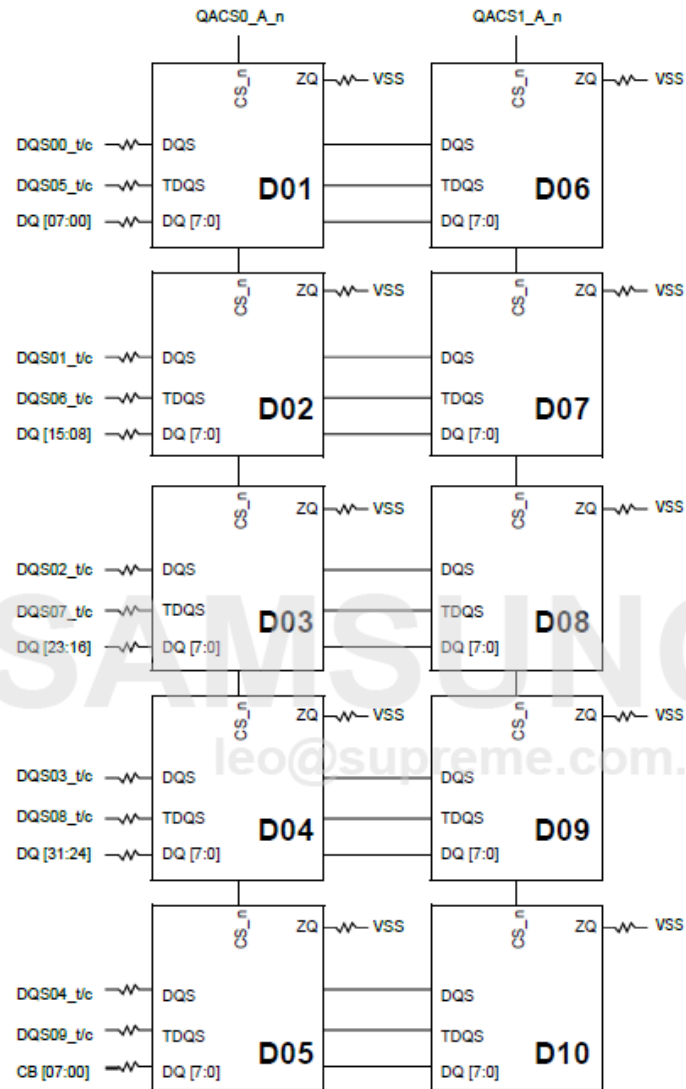
| Claim 1 | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p><b>Channel A</b></p> <p>The diagram illustrates the internal structure of Channel A, organized into two main sections. Each section contains five rows of data path blocks. The left section, connected to QACS0_A_n and QACS1_A_n, includes blocks D01-D05 and D11-D15. The right section, connected to QBCS0_A_n and QBCS1_A_n, includes blocks D06-D10 and D16-D20. Each block (D01-D20) is a square component with a capacitor symbol (C<sub>S</sub>) and a ZQ input connected to VSS. The blocks are interconnected via DQS (data strobe) and DQ (data) lines. For example, DQS00_t, DQS00_c, and DQ [03:00] are inputs to D01, which is connected to D11. Similarly, DQS05_t, DQS05_c, and DQ [07:04] are inputs to D06, which is connected to D16. The diagram shows a complex interconnection of these data paths across the two sections.</p> |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

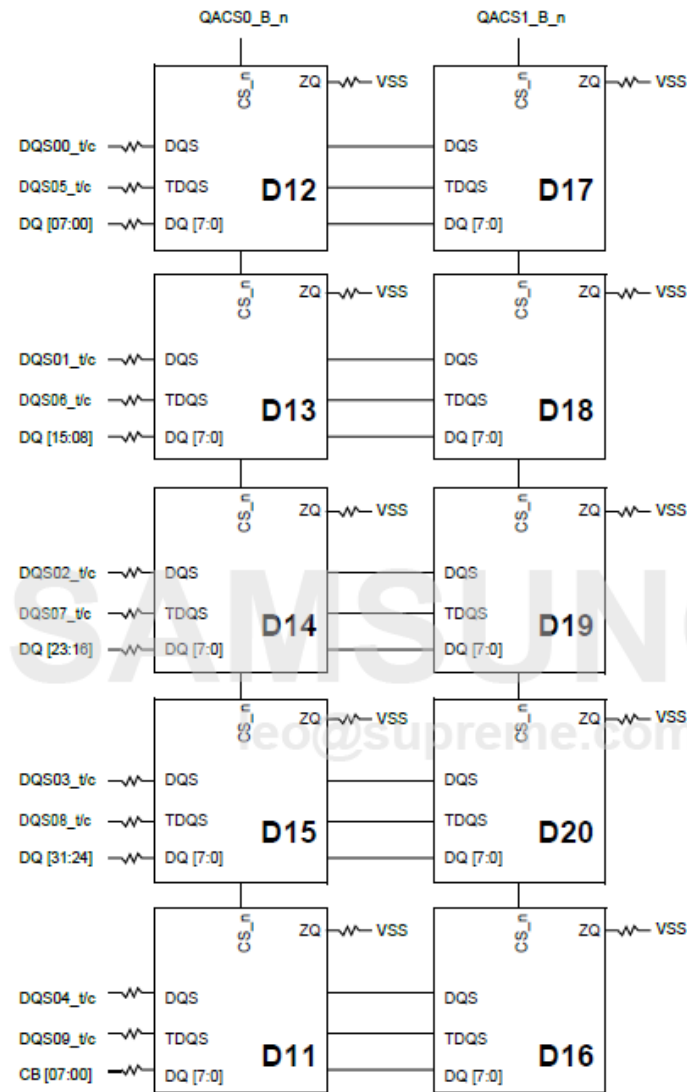
| Claim 1 | Evidence of Use                                                                                                                                                                                                                    |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p><b>Channel B</b></p>  <p>Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 26–27 (separate DQ[00:31] and CB[7:0] for channel A and channel B):</p> |

U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

**Channel A**



## Channel B



**U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)**

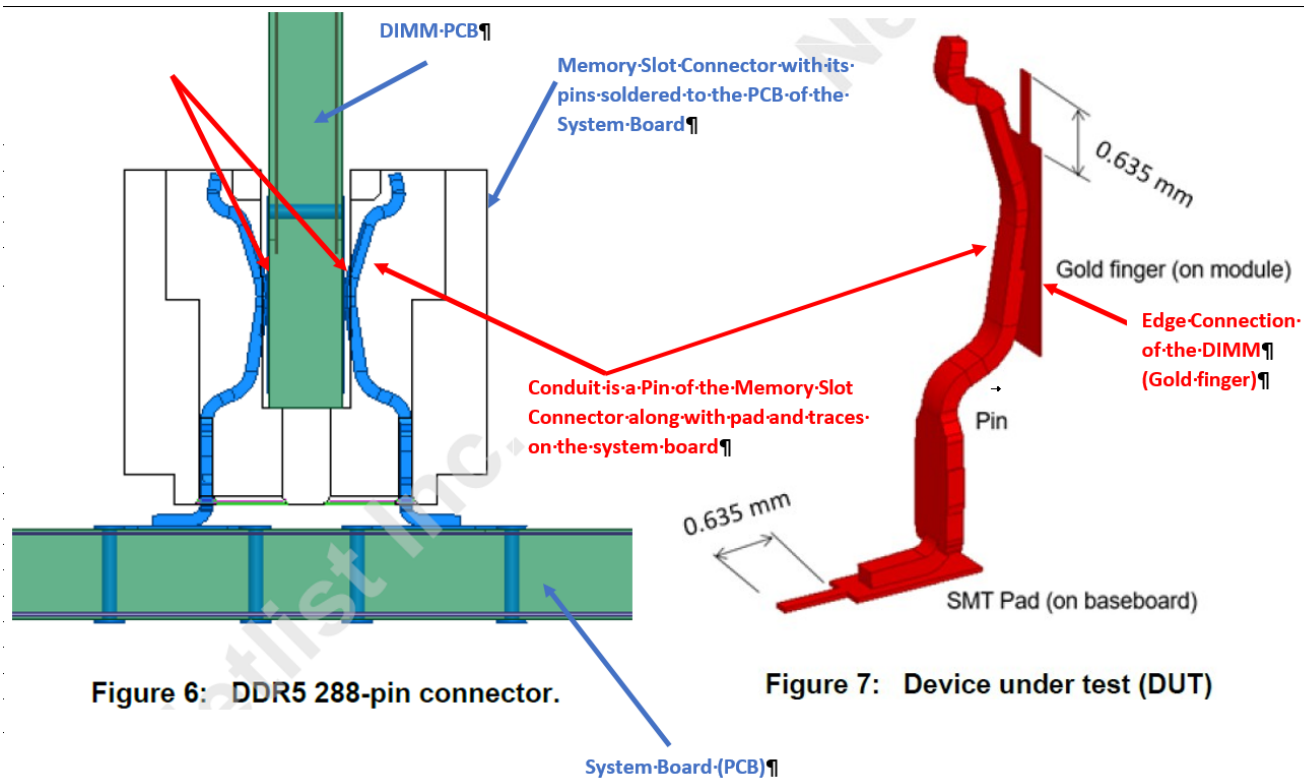
| Claim 1 | Evidence of Use                                    |
|---------|----------------------------------------------------|
|         | <i>See also, e.g., id.</i> at 21–23, 29–31 (same). |

# U.S. Patent No. 12,373,366 Domestic Industry Claim Chart (Samsung)

| Claim 1                                                                                                                                                                                                                                                                       | Evidence of Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                          |                                                        |                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                   |                        |                  |     |                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------|-----|--------------------------------------------------------|
| <p><b>[1.e.2]</b> wherein the sum of the at least 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.</p> | <p>Analysis below uses Samsung’s RDIMM format as an example, but MRDIMMs would operate in materially the same way in all relevant part. Ex. CC-13 (JEDEC Press Release) (JEDEC noting that other planned features for MRDIMMs include “Utilization of standard DDR5 DIMM components including DRAM, DIMM Form Factor &amp; Pinout, SPD, PMIC, and TS for ease of adoption”).</p>                                                                                                                                                                                         |                          |                                                        |                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                   |                        |                  |     |                                                        |
|                                                                                                                                                                                                                                                                               | <p>The sum (80 in EC8 DDR5 DIMMs) of the at least 40 data lines and the integer number of data lines (40 in the EC8 DIMMs) equals a total number of data conduits of the set of data conduits of the memory slot connector (40), and the total number of data conduits (80) is at least 72 data conduits.</p>                                                                                                                                                                                                                                                            |                          |                                                        |                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                   |                        |                  |     |                                                        |
|                                                                                                                                                                                                                                                                               | <p>As shown above, each of channel A and channel B in an EC8 DDR5 DIMM has 40 data lines associated with DQ and CB pins in the edge connections (DQ[31:0] and CB[7:0]).</p>                                                                                                                                                                                                                                                                                                                                                                                              |                          |                                                        |                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                   |                        |                  |     |                                                        |
|                                                                                                                                                                                                                                                                               | <table><tr><td>DQ[31:0]_A<br/>DQ[31:0]_B</td><td>Input/<br/>Output</td><td>VDD</td><td>Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.</td></tr><tr><td>CB[7:0]_A<br/>CB[7:0]_B</td><td>Input/<br/>Output</td><td>VDD</td><td>ECC Check Bits Input/ Output: Bi-directional data bus.</td></tr></table> | DQ[31:0]_A<br>DQ[31:0]_B | Input/<br>Output                                       | VDD                                                                                                                                                                                                                                                                                                               | Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used. | CB[7:0]_A<br>CB[7:0]_B | Input/<br>Output | VDD | ECC Check Bits Input/ Output: Bi-directional data bus. |
|                                                                                                                                                                                                                                                                               | DQ[31:0]_A<br>DQ[31:0]_B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Input/<br>Output         | VDD                                                    | Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used. |                                                                                                                                                                                                                                                                                                                   |                        |                  |     |                                                        |
| CB[7:0]_A<br>CB[7:0]_B                                                                                                                                                                                                                                                        | Input/<br>Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | VDD                      | ECC Check Bits Input/ Output: Bi-directional data bus. |                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                   |                        |                  |     |                                                        |
| <p>Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 11.</p>                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                          |                                                        |                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                   |                        |                  |     |                                                        |
|                                                                                                                                                                                                                                                                               | <p>The same pinouts apply to MRDIMMs. <i>See</i> Ex. CC-13 (JEDEC Press Release); Ex. CC-145 (DDR5 MRDIMM: A Transformational Evolution For DDR5 DIMM) (“DDR5 MRDIMM is pin compatible with traditional DIMM, and it can be used in the same DIMM slots that are used for DDR5 RDIMMs”).</p>                                                                                                                                                                                                                                                                             |                          |                                                        |                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                   |                        |                  |     |                                                        |
|                                                                                                                                                                                                                                                                               | <p>Each edge connection is electrically coupled to a corresponding conduit in the memory slot, as illustrated below:</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                          |                                                        |                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                   |                        |                  |     |                                                        |

Claim 1

Evidence of Use



Ex. CC-73 (JEDEC PS-005B) at 13, Figs. 6 and 7 (annotated) (showing an electrical connection between a connector in a memory slot and a corresponding edge connection of the DIMM).

Thus, the sum of the data lines equals a total number of data conduits of the set of data conduits of the memory slot connector, and given that there are 80 data lines per EC8 DDR5 DIMM, that total number of data conduits (80) is at least 72 data conduits.